

The Information Channel of Asset Purchases*

Gaetano Gaballo

Carlo Galli

HEC Paris and CEPR

UC3M and CEPR

July 8, 2026

Abstract

We study central bank asset purchases (APs) in bond markets where rational investors face position limits and learn from private signals and market prices. APs affect information aggregation, altering the informativeness of prices and their sensitivity to fundamentals through an information channel distinct from risk-premium effects. Consistent with this mechanism, we find that U.S. Federal Reserve APs strengthened the pass-through of labor market news to Treasury yields. Dispersed beliefs and position limits also provide a microfoundation for the optimality of APs.

*We thank the editor and five anonymous referees for their insightful comments and constructive suggestions, which have substantially improved the paper. We thank Vladimir Asriyan, Chiara Felli, Christian Hellwig, Luigi Iovino, Christiaan van der Kwaak, Vytautas Valaitis, and Cathy Zhang for insightful discussions of the paper. For helpful comments and suggestions, we are indebted to George-Marios Angeletos, Manuel Amador, Marco Bassetto, Anmol Bhandari, Pierpaolo Benigno, Wei Cui, Sebastián Fanelli, William Fuchs, Refet Gürkaynak, Alberto Martin, Eric Mengus, Luigi Paciello, Evi Pappa, Ricardo Reis, Giovanni Ricco, Edouard Schaal, Georg Strasser, Victoria Vanasco, Mirko Wiederholt, and participants at various conferences and seminars. An earlier version of this work circulated under the title “Asset Purchases and Heterogeneous Beliefs.” The present paper subsumes and substantially extends that version. Gaballo acknowledges funding from the European Research Council (ERC) under the European Union’s Horizon Europe research and innovation program (Grant agreement No 101088411). Galli gratefully acknowledges financial support from Fundación Ramón Areces, and from MICIN/AEI/10.13039/501100011033 and ERDF/UE with grants RYC2023-044688-I, CEX2021-001181-M, PID2021-122931NB-I00, and PID2024-158085NB-I00.

1 Introduction

The widespread adoption of large-scale asset purchase (AP) programs arguably marks the most significant shift in monetary policy over the past decades. While robust evidence confirms that announced APs increase asset prices in targeted markets, their general-equilibrium effects remain debated.

Pinning down the microfoundations of the elasticity of asset prices to APs is crucial for assessing their effectiveness. A long tradition focuses on risk and term premia: by shrinking the risk the private sector must absorb, APs lower the compensation required by investors and push prices up. This logic, however, relies on frictions in risk allocation. In the absence of such frictions, risk is efficiently priced, and reallocating it from the private to the public sector does not affect asset prices, as in Wallace (1981).

In this paper, we study the information channel of APs, which arises from frictions in information rather than in risk allocation. The contrast is instructive: in the risk tradition, APs redistribute aggregate risk between sectors, leaving its total amount unchanged. In our framework, APs alter how markets aggregate information, changing the precision of what investors collectively know. Dispersed beliefs, unlike risk preferences, respond endogenously to the intervention, shifting investors' valuations irrespective of term premia.

Importantly, the channel neither depends on uncertainty about the intervention nor on signaling through the intervention itself. We show that the fixed and observable nature of APs, as in central bank practice, does not guarantee informational neutrality unless aggregate asset demand satisfies a separability condition that we characterize.

The information channel of APs leads to two main implications. First, the impact of APs on asset prices is *conditional* on the state of the economy. To fix ideas, consider a large public debt purchase program implemented during a period of high yields. If the intervention fails to sufficiently inflate prices, each investor interprets this as evidence that others hold pessimistic information about fundamentals, driving prices down. APs can thus accelerate fundamentally-driven crises rather than tame them. Consistent with this intuition, we show that even a simple, uncontingent AP program can increase the amount of information that investors learn from the market, and the sensitivity of prices to fundamentals. We provide empirical support for this prediction, showing that the flows of AP operations under the QE programs of the U.S. Federal Reserve strengthened the pass-through of labor market news to Treasury yields in daily data.

Second, APs also change asset prices *unconditionally*. This occurs because learning from prices generates a wedge between the equilibrium price and the fundamental value of the

asset, which changes as APs affect the distribution of prices. By affecting this wedge, APs move the average price. We further show that, since the intervention is non-contingent, an increase in the average price leads to expected losses for the central bank and an implicit financing of the government.

To make our theoretical mechanism as transparent as possible, we study two models: a risk-neutral model in which investors are subject to position bounds, and a more traditional CARA model with and without position bounds. While only the former admits simple closed-form solutions, the latter embeds our mechanism within the class of traditional asset pricing models.

In the first model, a continuum of rational, risk-neutral investors trade government bonds alongside unobservable noise trader demand. Bond returns are stochastic, investors face position limits, and they form expectations based on private signals and the equilibrium price. The latter is determined by the beliefs of the marginal investor—the investor making zero expected gains. We consider AP policies whereby the central bank commits to purchasing a fixed, publicly known quantity of bonds at any price.

APs mechanically crowd out relatively pessimistic investors, allowing more optimistic investors to become marginal for a given realization of fundamentals. Consequently, low prices can only result from a combination of high supply and sufficiently negative private signals, making them informative about poor fundamentals. APs thus create a revelation region in which prices reveal low payoffs (that is, returns) and investors revise beliefs accordingly. As this region expands with APs, both the informativeness of prices and their sensitivity to fundamentals increase.

APs not only reshuffle probability mass across states, but also raise the unconditional mean of prices, consistent with the empirical evidence on AP announcements. This occurs because learning from prices generates a wedge between the market price of the asset and its fundamental value. In equilibrium, a higher price plays two roles at once: it conveys favorable public information, which raises investors' valuations, and makes the asset more expensive, which lowers demand at any given valuation. Prices therefore respond more strongly to market signals than investors' valuations, generating a wedge between the two. APs increase the average value of this wedge by ruling out bad market signals that would otherwise arise from large asset supply. As a by-product, they raise average prices above fundamental valuations, generating expected losses for the central bank.

We present a second model to uncover the conditions under which the same mechanism may arise in a more standard CARA environment. We assume the same payoff and information structure as before. In contrast, investors choose asset positions by balancing expected

returns against risk exposure, in the spirit of workhorse finance models. We first show that, in the version without position bounds, uncontingent APs are informationally neutral: price effects arise only because the central bank removes risky asset supply through the classical risk- or term-premium channel. With position bounds, however, neutrality breaks down and APs generate informational effects. We resort to numerical simulations to show that the CARA model with position bounds can generate the same main predictions as the simpler risk-neutral economy.

More generally, we show that non-contingent APs are informationally neutral in any model where aggregate asset demand is additively separable in asset fundamentals and market prices. What unifies the risk-neutral model and the CARA model with position bounds is precisely the non-separability of aggregate demand, which makes the distribution of market signals depend on APs. When aggregate demand is non-separable, APs can amplify as well as dampen the sensitivity of prices depending on the details of the environment. In both models, we deliberately focus our analysis on the amplification case for two reasons. First, it is the most policy-relevant because it implies that AP programs may amplify, rather than buffer, fundamentally driven crises. Second, amplification is the case supported by our empirical evidence.

To assess the empirical relevance of our channel, we test the prediction that APs increase the sensitivity of asset prices to fundamentals using U.S. data. We focus on daily changes in the risk-neutral component of Treasury yields—constructed by the NY Fed following Adrian et al. (2013)—to capture revisions in expectations about future short rates, net of term premia, consistent with our channel. We estimate an interaction model in which the sensitivity of yields to macro labor news—measured by Bloomberg surprises in the Employment Situation report—varies linearly with the flow of Fed purchases on the same day. We find a robust, positive, and significant interaction coefficient across specifications, maturities, and subsamples, supporting our models’ prediction that APs heighten the sensitivity of asset prices to fundamental shocks.

Finally, we extend our framework to a consumption-saving model to address what frictions justify central bank intervention through APs. We show that the same assumptions—limited arbitrage and dispersed information—that make APs non-neutral also microfound their optimality, without resorting to additional frictions such as price rigidities or frictions in the allocation of risk. In this economy, households save by depositing funds with investors who act as financial intermediaries. The combination of position bounds and dispersed beliefs generates positive expected gains for investors, which are passed through to households as above-market deposit returns. This induces excessive saving relative to the

social optimum, since households take these returns as given while they are financed through lump-sum taxes. By unconditionally raising asset prices, AP policies correct this distortion by compressing the expected return on deposits and stimulating current consumption.

The rest of the paper is organized as follows. After a review of the related literature, Section 2 presents the risk-neutral model and characterizes the effects of APs on price informativeness, sensitivity, average prices, and the distribution of gains and losses across market participants. Section 3 develops the CARA model with and without position bounds and provides a general characterization of when APs are informationally neutral. Section 4 presents our empirical evidence from the U.S. Treasury market. Section 5 develops the welfare analysis, and a final section concludes. The appendices collect proofs, several model extensions, and empirical robustness checks.

Related Literature. Our paper relates to the literature on central bank APs in finance and macroeconomics. Workhorse finance models of APs—notably Vayanos and Vila (2021) and Greenwood and Vayanos (2014)—combine limits to arbitrage (Shleifer and Vishny (1997)) with structural heterogeneity in investors’ preferences for risk, maturity, or liquidity, and focus on the local impact of APs across maturities, as documented by a large empirical literature (see Bhattarai and Neely (2022) for a survey). Evidence of heterogeneity in demand elasticity has been found in the context of AP programs by Koijen et al. (2021) and Jiang et al. (2024) for the Eurozone and the U.S., respectively, although there is a debate on the level of asset demand elasticity (Gabaix and Koijen (2021), Koijen and Yogo (2019)). Similarly, a vast literature in macroeconomics emphasizes market segmentation and structural heterogeneity as the basis for departing from the neutrality benchmark of Wallace (1981). Portfolio rebalancing effects have been studied with quantitative models by Cúrdia and Woodford (2011) and Gertler and Karadi (2015), among others; Ray et al. (2024) combines a full macroeconomic framework with preferred-habitat investors; Cui and Sterk (2021) studies how APs affect risk sharing across households in the presence of incomplete markets. A shared feature of all these works is some form of structural heterogeneity and limited arbitrage. We share the limited arbitrage assumption but characterize a novel channel grounded in dispersed information, which—unlike structural heterogeneity—responds endogenously to central bank policy. Our theory implies that demand elasticities are not policy-invariant structural objects, but change with the informational regime.

Our paper also relates to the literature on trading in financial markets in the presence of information frictions. A tradition dating back to Miller (1977) and Harrison and Kreps (1978) studies markets where investors hold *exogenous* heterogeneous beliefs and face short-sale or credit constraints (e.g., Geanakoplos (2003) and Geanakoplos (2010)). Hong et al.

(2006) show that a lower asset float raises prices by increasing the resale option value of future optimistic investors. In a related setting, Ottaviani and Sørensen (2015) show how wealth effects shape the reaction of prices to information, generating underreaction and momentum. In all these models, a reduction in supply shifts the unconditional price level monotonically through the identity of the marginal investor, without affecting the amount of dispersed information that markets aggregate into prices and from which investors learn. In contrast, in our model investors have dispersed information and learn from prices, as in models of noisy rational expectations equilibria following the seminal contributions by Grossman (1976), Grossman and Stiglitz (1980), and Hellwig (1980). Albagli et al. (2024), formalizing a concept known at least since Vives (2010), demonstrate that the existence of a wedge between prices and fundamental valuations is a general implication of learning from prices in this class of models. They show that this is independent of risk aversion or position bounds, and that the average level of such a wedge depends on the asset’s payoff asymmetry.¹ We characterize how APs affect the distribution of the wedge through price informativeness, and show that our results are independent of the asymmetry of payoffs. A related body of work (see Diamond and Verrecchia (1987), Goldstein and Guembel (2008), Marin and Olivier (2008), and Lim (2011)) studies the informational content of prices in the presence of trading constraints, within dynamic models in the tradition of Glosten and Milgrom (1985) and Kyle (1985). Our paper is, to our knowledge, the first to analyze the effect of a common-knowledge, non-random reduction in asset supply on price informativeness and sensitivity to fundamentals, within a class of static noisy rational expectations models.

The interaction between APs and information frictions has been studied by several papers, but through mechanisms distinct from ours.² Iovino and Sergeyev (2023) uses level-k thinking—a form of bounded rationality—as the source of AP non-neutrality in an otherwise frictionless model. Fontanier (2023) shows that APs reduce the incentives of informed investors to acquire information. In Candian et al. (2023), investors overreact to the information content of exchange rates. Mussa (1981), Jeanne and Svensson (2007), Christensen and Rudebusch (2012), and Bhattarai et al. (2022) study the signaling role of APs about future policy objectives. In all of this work, investors either do not have rational expectations or do not learn from prices—the key channel affected by APs in our model.

¹Albagli et al. (2023) and Bassetto and Galli (2019) study the implications of this wedge on investment distortions and the sovereign debt market, respectively.

²While they do not study APs, Cole et al. (2022) and Cole et al. (2024) show that sovereign debt markets are pervaded by asymmetric information, which generates adverse selection in the primary market.

2 Risk-Neutral Model

2.1 The Financial Market

The financial market consists of a continuum of risk-neutral investors facing a random supply of government bonds, given by a fixed unitary amount issued by the government, net of a stochastic demand by noise traders, and a given asset purchase rule by the central bank. Investors share the same preferences but differ in their beliefs about bond payoffs.³ Central bank interventions affect bond prices and the expected gains or losses of investors, noise traders, the government, and the central bank.

Asset payoff and supply. At the beginning of the period, the government issues a unit quantity of bonds. The bonds are sold at the market-clearing price of Q , and at the end of the period pay a stochastic real payoff θ , which potentially captures uncertainty relative to inflation or default. We assume θ is distributed according to the exogenous lottery

$$\theta = \begin{cases} \theta_H & \text{with probability } \gamma, \\ \theta_L & \text{with probability } 1 - \gamma, \end{cases} \quad (1)$$

where γ and $1 - \gamma$ respectively denote the probability of a high and low payoff. The fundamental shock θ could represent any variation in the real payoff of the bond, for example, due to fluctuations in the resale price, inflation, or default. Despite its simplicity, this binary distribution allows for the asymmetry of bonds' payoffs, which are left-skewed if and only if $\gamma > 1/2$. The lottery (1) captures all publicly available information at the time of the intervention, potentially including any disclosure that the central bank may have made in conjunction with it.⁴

Following a tradition dating back to Grossman and Stiglitz (1980), we assume a stochastic, non-fundamental demand of $1 - \tilde{S}$ by noise traders, where $\tilde{S}$ is uniformly distributed between 0 and 1. Thus,

$$\tilde{S} \sim \text{Uniform}[0, 1] \quad (2)$$

represents the stochastic net supply of government bonds available to investors and the central bank. More generally, $\tilde{S}$ can be interpreted as any aggregate shock orthogonal to bonds' payoffs; this source of noise is standard in models of learning so that prices are not

³We compare our channel to a mechanism relying on heterogeneous preferences in Online Appendix J.

⁴We study an extension with explicit public signals in Online Appendix E.2.

immediately revealing of the fundamental.⁵

Investors. There is a continuum of measure one of investors that are risk neutral. They do not observe the realization of the aggregate shocks $(\theta, \tilde{S})$, but receive noisy information about it. Specifically, we assume that each investor receives a noisy private signal on the payoff, given by

$$x_i = \theta + \sigma_x \xi_i, \quad (3)$$

where $\xi_i \sim N(0, 1)$ is independently and identically distributed across agents. The marginal distribution of private signal x_i is thus a normal mixture, which we denote with $\mathcal{N}$. Investors also receive two pieces of public information: they observe the quantity of central bank APs b_{cb} , and the bond price Q . We thus denote the information set of each investor $i \in [0, 1]$ with $\Omega_i := \{x_i, b_{cb}, Q\}$.⁶

Each investor i chooses her bond position b_i , subject to position bounds $[0, 1]$, to maximize the expected present value of net trading gains, which are given by

$$\max_{b_i \in [0, 1]} \mathbb{E}[\pi_{inv,i} | \Omega_i], \quad (4)$$

where $\pi_{inv,i} := b_i(\theta - Q)$. As we will show later, the solution to the investor problem is a price-contingent demand function $b_i = b(\Omega_i)$ taking the form of a limit order, i.e., the investor buys up to the portfolio upper bound if the price is below her reservation value, and nothing otherwise.

Methodologically, our model belongs to the limits-to-arbitrage tradition (see Gromb and Vayanos (2010) for a review). The presence of exogenous position bounds is essential to generate a lack of individual arbitrage and for APs to have an impact on prices in our model.⁷ This is equivalent to assuming the absence of free entry in the market, and allows investors to earn strictly positive rents in equilibrium (as we will show later) in analogy to the rents sustained by capital constraints in Shleifer and Vishny (1997) and Gromb and Vayanos (2002), and by information acquisition costs in Grossman and Stiglitz (1980).

⁵For example, $\tilde{S}$ could be interpreted directly as the government supply in cases where such information is not publicly available.

⁶The distinction between the prior in (1) and the private signals in (3) is made for expositional simplicity but is not essential. One could instead treat the beliefs resulting from updating (1) with (3) as a primitive distribution of dispersed priors without affecting any of the results.

⁷Online Appendix G.1 considers a version of this model with generic position bounds $[-\bar{b}, \bar{b}] \supseteq [0, 1]$, allowing for short selling. Online Appendix C.1 shows that, in models where investors do not face position bounds and the gains and losses from APs are passed on to investors in a lump-sum way, APs affect neither prices nor allocations, as in Wallace (1981).

Central Bank. We are interested in studying the impact of central bank open market operations in financial markets. We consider the simplest possible rule, a quantity-targeting policy, which consists of the central bank buying bonds up to a fixed quantity target,⁸ independent of the market price, and subject to its demand being no larger than the supply. Online Appendix H considers price-targeting policies, where the central bank intervenes only if the bond price is below a given target.

Definition 1 (Quantity-targeting APs). *Under quantity-targeting APs, the central bank buys up to an announced quantity target b at market price Q according to*

$$b_{cb} = \begin{cases} b & \text{if } \tilde{S} \geq b \\ \tilde{S} & \text{if } \tilde{S} < b. \end{cases} \quad (5)$$

When $\tilde{S} \geq b$, some supply is absorbed by investors, and the equilibrium bond price Q is determined by the market. When instead $\tilde{S} < b$, we assume that the central bank buys all the supply at the price Q_{pas} and investors remain “passive.” We set $Q_{pas} = \theta$ to be the fully revealing price and show in Appendix A.5 that this is the market-clearing price in the limit when an infinitesimally small supply $\tilde{S} \in \text{Uniform}[0, \epsilon]$ is traded by investors, with $\epsilon > 0$.⁹

2.2 Market Clearing and Equilibrium

The bond market clearing condition is given by

$$\int_0^1 b_i di + b_{cb} + 1 - \tilde{S} = 1. \quad (6)$$

In a market equilibrium, the demand of investors equals the unit supplied by the government, net of the demand by the central bank (b_{cb}) and by noise traders ($1 - \tilde{S}$). We are now ready to give a formal definition of an equilibrium.

Definition 2 (Equilibrium). *Given an AP policy rule as per (5) and distributions (1)-(2), a Perfect Bayesian Equilibrium consists of demand schedules $b(\Omega_i)$, a price function $Q(\theta, \tilde{S}, b_{cb})$, and posterior beliefs $\mathbb{E}[\theta | \Omega_i]$ such that*

- (i) *the demand schedules solve investors’ problem (4) given their posterior beliefs;*

⁸This is equivalent to quantifying the AP objective in terms of face value, consistent with the accounting and communication practice of major central banks. As we show in Online Appendix I, this modeling choice is important to preserve the tractability of our risk-neutral model. In Section 3 we generalize the AP rule to a function of observables, such as the bond price.

⁹This is not an essential assumption: Q_{pas} could be simply treated as an exogenous parameter of the analysis without any loss of generality or qualitative change in our results.

- (ii) the price function $Q(\theta, \tilde{S}, b_{\text{cb}})$ clears the bond market, satisfying (6);
- (iii) posterior beliefs satisfy Bayes' law for all market clearing prices.

2.3 Investors' Strategies and Price Signals

We start by working out how the demand by investors and the central bank determines the equilibrium price and the information it contains.

Monotone threshold strategies. The posterior beliefs of investor i on the likelihood of a high payoff are given by $p_i := \text{Prob}(\theta = \theta_H | \Omega_i)$, and are weakly increasing in the private signal x_i in the sense of first-order stochastic dominance. In particular, given that x_i is the only piece of information that is heterogeneous across investors, it is true that

$$x_i \geq x_j \Leftrightarrow p_i \geq p_j \Leftrightarrow \mathbb{E}[\theta | \Omega_i] \geq \mathbb{E}[\theta | \Omega_j] \quad \forall i, j,$$

where $\mathbb{E}[\theta | \Omega_i] = (\theta_H - \theta_L)p_i + \theta_L$. This implies that each investor's demand follows a monotone threshold strategy of the form

$$b_i(x_m) = \begin{cases} 1 & \text{if } x_i \geq x_m \\ 0 & \text{if } x_i < x_m, \end{cases} \quad (7)$$

where $b_i(x_m)$ is the portfolio choice of agent i when the private signal threshold is x_m , which is endogenous to the equilibrium. By construction, x_m coincides with the private signal received by the marginal agent $i = m$, who is indifferent between buying bonds or not.¹⁰

Price signal mapping. We assume that a law of large numbers across investors applies as in Judd (1985): for a given value of the fundamental, the mass of investors buying bonds is given by the share of agents with a private signal larger than x_m , that is

$$\int_0^1 b_i \, di = \text{Prob}(x_i \geq x_m | \theta) = \Phi\left(\frac{\theta - x_m}{\sigma_x}\right), \quad (8)$$

where Φ denotes the standard normal cumulative distribution function. It follows that, conditional on x_m , aggregate bond demand is a function of θ , since investors' beliefs are

¹⁰Note that investors' optimal strategy (7) effectively amounts to the submission of a limit order, since—as we will show—there exists a one-to-one mapping between the private signal threshold x_m and the equilibrium price Q .

centered around it. We can thus rewrite the bond market clearing condition (6) as

$$\Phi\left(\frac{\theta - x_m}{\sigma_x}\right) = \tilde{S} - b_{cb}. \quad (9)$$

The left-hand side represents the mass of investors who buy bonds. The right-hand side represents the net supply of bonds available to investors, that is, government supply net of noise traders' demand and central bank purchases. For brevity, we define $S := \tilde{S} - b_{cb}$ and refer to it as *net supply*. For the market to clear for a given (θ, S) , the identity (or private signal) x_m of the marginal agent must adjust, which suggests that x_m and Q are related. Henceforth, we focus on equilibria where Q and x_m convey the same information.¹¹ Rearranging terms, we can express x_m as

$$x_m(\theta, S) = \theta + \sigma_x \Phi^{-1}(1 - S). \quad (10)$$

Equation (10) states that the marginal agent's private signal x_m must be equal, in equilibrium, to a function that is linear in the fundamental shock θ , and decreasing and nonlinear in net supply S . In other words, x_m is an endogenous public signal of θ that aggregates private information imperfectly, because of the noise introduced by unobservable net supply S . We thus refer to x_m as the price signal (or market signal).

The *crowding-out* and *revelation* effects of APs. Figure 1 illustrates the mapping from fundamental and gross supply shocks $(\theta, \tilde{S})$ and AP policy b to the equilibrium marginal investor (or price) signal x_m , as per equation (10). The figure plots the realization of x_m on the y-axis, as a function of the net supply shock S on the x-axis, for the two values of θ .¹²

The left panel illustrates the case with no APs: in this case, gross and net supply coincide ($\tilde{S} = S$). When $S \rightarrow 1$, the entire population of investors is needed to clear the market, so the most pessimistic investor, i.e., the one with the lowest private signal ($x_i \rightarrow -\infty$), is marginal. When instead $S \rightarrow 0$, there is an infinitesimal amount of supply, and only the most optimistic investor ($x_i \rightarrow +\infty$) buys bonds and is marginal. For any realization of net supply away from these limits, the mass of buyers required to clear the market and the identity of the marginal investor are interior. In all these cases, investors face an inference

¹¹This is equivalent to focusing on equilibria with continuous price functions. Pálvölgyi and Venter (2015) show that another class of equilibria with discontinuous and non-monotone price functions exists in noisy rational expectations economies.

¹²We use a common parametrization for all the figures in this section. We set $b = 0.1$ for the case with APs, which is purely chosen for illustrative purposes. We set the other parameters to $\theta_H = 2.87, \theta_L = -3.23, \gamma = 0.53, \sigma_x = 10.5$. These are chosen to match some features of the data, as we explain in detail in Online Appendix D.

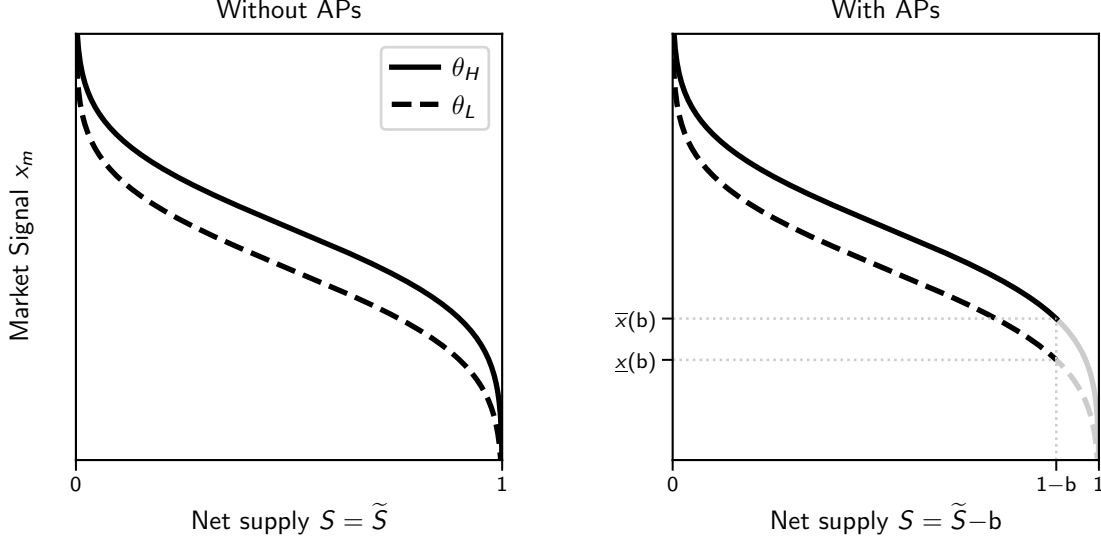


Figure 1: The price signal as a function of net supply S and fundamental θ .

problem, since any observed price signal can be the result of either high demand (due to a high payoff) and high supply, or low demand and low supply.

The right panel shows that positive APs ($b_{cb} = b > 0$) have two main effects. First, there is a *crowding-out* effect: for a given realization of the shocks $(\theta, \tilde{S})$, APs reduce net supply so that a relatively more optimistic investor is marginal, i.e., a smaller mass of buyers is needed to clear the market. In particular, some investors on the left tail of the private signal distribution will always be crowded out by APs and never be marginal, since the maximal net supply, $1 - b$, becomes smaller than one. It follows that the support of the price signal distribution is truncated from below and becomes $[\underline{x}(b), +\infty)$, where $\underline{x}(b) := x_m(\theta_L, 1 - b)$.

Second, there is a *revelation* effect specific to low payoff states: the truncation of the support of S induced by APs gives rise to an interval of price signals $[\underline{x}(b), \bar{x}(b))$, with $\bar{x}(b) := x_m(\theta_H, 1 - b)$, which only occur when $\theta = \theta_L$. Perfect revelation is an artifact of our simple assumption of a binary payoff, but the mechanism is more general. When the central bank is intervening, investors understand that a smaller mass of buyers is needed to clear the market, so the market signal will, in general, be higher. Observing a low market signal in this setting then implies a larger likelihood of low payoffs than in the absence of APs. This exact reasoning is also illustrated in Online Appendix G.2, where we consider the case where θ is a continuous random variable on the real line and revelation is never perfect.

Price signal distribution. We denote by $\mathcal{M}_b$ the distribution of the price signal x_m conditional on $\tilde{S} \geq b$, i.e., when the equilibrium bond price is determined by the market.

The p.d.f. is given by $\frac{1}{1-b}f_{\mathcal{M}_b}(x_m)$, where

$$f_{\mathcal{M}_b}(x_m) = \begin{cases} \frac{1}{\sigma_x} \left[(1-\gamma) \phi\left(\frac{x_m-\theta_L}{\sigma_x}\right) + \gamma \phi\left(\frac{x_m-\theta_H}{\sigma_x}\right) \right] & \text{for } x_m \in [\bar{x}(b), +\infty) \\ \frac{1}{\sigma_x} (1-\gamma) \phi\left(\frac{x_m-\theta_L}{\sigma_x}\right) & \text{for } x_m \in [\underline{x}(b), \bar{x}(b)), \end{cases} \quad (11)$$

and ϕ denotes the standard normal probability density function. Note that the market signal density is normalized by the probability $P(\tilde{S} \geq b) = 1 - b$ that the market is “active”. The remaining set of states has probability $P(\tilde{S} < b) = b$, and is such that the central bank buys all the supply and investors do not trade bonds nor learn from the price.

2.4 From Price Signal to Investors’ Beliefs

Having characterized how investors learn from prices, we now derive their posterior beliefs and how the equilibrium price is determined. Before digging into it, it is useful to consider, as an intermediate step, the posterior beliefs of an agent who only observes public information, i.e., the price signal, and receives no private information. We will refer to these as the “public” posterior beliefs.

Public posterior beliefs. The posterior probability of $\theta = \theta_H$ conditional only on the public price signal x_m is given by standard Bayesian updating

$$\hat{p}(x_m) := P(\theta_H | x_m \sim \mathcal{M}_b) = \begin{cases} \frac{\gamma \phi\left(\frac{\theta_H - x_m}{\sigma_x}\right)}{\gamma \phi\left(\frac{\theta_H - x_m}{\sigma_x}\right) + (1-\gamma) \phi\left(\frac{\theta_L - x_m}{\sigma_x}\right)} & \text{if } x_m \in [\bar{x}(b), +\infty), \\ 0 & \text{if } x_m \in [\underline{x}(b), \bar{x}(b)), \end{cases} \quad (12)$$

and reflects the fact that values of x_m in the fully revealing region are only consistent with θ_L .¹³ It is worth noting that APs do not directly affect the posterior probability conditional on x_m , but they do determine the fully- and non-revealing partitions of the support of the price signal, as well as its density function.

Investors’ posterior beliefs. We can now characterize the posterior beliefs of investor i conditional on their information set $\Omega_i = \{x_i \sim \mathcal{N}, x_m \sim \mathcal{M}_b\}$, that is, on both their private

¹³To lighten notation, we will often omit b as an argument when defining objects that are functions of the price signal x_m , since dependence on b is implicit in its effect on the distribution of x_m .

signal and the public price signal.¹⁴ Standard Bayesian updating yields

$$p(x_i, x_m) := P(\theta_H | x_i \sim \mathcal{N}, x_m \sim \mathcal{M}_b) =$$

$$= \begin{cases} \frac{\gamma \phi\left(\frac{\theta_H - \frac{x_i + x_m}{2}}{\sigma_x / \sqrt{2}}\right)}{\gamma \phi\left(\frac{\theta_H - \frac{x_i + x_m}{2}}{\sigma_x / \sqrt{2}}\right) + (1-\gamma) \phi\left(\frac{\theta_L - \frac{x_i + x_m}{2}}{\sigma_x / \sqrt{2}}\right)} & \text{if } x_m \in [\bar{x}(b), +\infty), \\ 0 & \text{if } x_m \in [\underline{x}(b), \bar{x}(b)). \end{cases} \quad (13)$$

Note that the investor posterior p is identical to the public posterior $\hat{p}$ —and equal to zero—in the fully revealing region, and is different otherwise. The difference derives from the fact that investors’ beliefs use two sources of information, while the public’s only use one, so the scale parameter of the former is smaller than that of the latter by a factor of $1/\sqrt{2}$.¹⁵

2.5 Market Prices and Fundamental Valuations

To characterize the equilibrium bond price, we consider the marginal investor whose private signal is at the threshold ($x_i = x_m$), and must thus be indifferent with respect to her bond position. We define the posterior probability of a high payoff for the marginal agent as

$$p(x_m) := p(x_i, x_m)|_{x_i=x_m}. \quad (14)$$

It is easy to see from (12) and (13) that $p(x_m) \neq \hat{p}(x_m)$, even though both depend solely on x_m . From the marginal investor’s indifference condition

$$Q = \mathbb{E}[\theta | x_i \sim \mathcal{N}, x_m \sim \mathcal{M}_b]|_{x_i=x_m} \quad (15)$$

we derive the equilibrium bond price

$$Q(p(x_m)) = p(x_m) \theta_H + (1 - p(x_m)) \theta_L, \quad (16)$$

which is a simple linear function of the marginal investor’s posterior beliefs. Combining (16) and (13), we can write down the inverse mapping between the equilibrium price and its

¹⁴Henceforth, we always characterize conditioning sets by indicating from which distribution— $\mathcal{N}$ or $\mathcal{M}_b$ —each realization is drawn. This is useful to avoid confusion and remind ourselves through which channels APs affect posterior beliefs and learning from prices.

¹⁵The price signal has the same conditional precision of the private signal, and is thus given the same Bayesian weight. This is due to the assumption that supply is uniformly distributed, which is made to preserve the normality of posterior beliefs in the presence of APs.

associated endogenous public signal

$$x_m(Q) = x^\dagger - \frac{\sigma_x^2}{2(\theta_H - \theta_L)} \log \left(\frac{\gamma}{1 - \gamma} \frac{\theta_H - Q}{Q - \theta_L} \right). \quad (17)$$

This concludes our derivation of how the state $(\theta, \tilde{S})$ maps into the marginal investor's signal via market clearing (10), and how the latter maps into the equilibrium bond price via the marginal investor's indifference condition (16).

Using (16), we construct an alternative object, which we define as *fundamental valuation*, and is given by the expected bond payoff once we plug in the public posterior beliefs we defined in (12):

$$Q(\hat{p}(x_m)) := \mathbb{E}[\theta | x_m \sim \mathcal{M}_b] = \hat{p}(x_m) \theta_H + (1 - \hat{p}(x_m)) \theta_L. \quad (18)$$

That is, $Q(\hat{p}(x_m))$ is the expected value of the fundamental θ , conditional on x_m being drawn from $\mathcal{M}_b$. The following proposition characterizes the difference between equilibrium bond price and fundamental valuation.

Proposition 1 (Single-crossing). *For any $x_m \in [\bar{x}(b), \infty)$, we have that*

$$Q(p(x_m)) > Q(\hat{p}(x_m)) \quad \text{if and only if} \quad x_m > x^\dagger := \frac{\theta_H + \theta_L}{2}.$$

For any $x_m \in [\underline{x}(b), \bar{x}(b))$, we have $Q(p(x_m)) = Q(\hat{p}(x_m)) = \theta_L$.

Proof. Postponed to Appendix A.1.

The proposition establishes that there exists a single crossing between market price and fundamental valuation occurring at $x^\dagger$, the unique value of the price signal that equalizes posterior and prior probabilities, $p(x^\dagger) = \hat{p}(x^\dagger) = \gamma$.

In Figure 2, we plot the market price $Q(p(x_m))$ in solid lines and the fundamental valuation $Q(\hat{p}(x_m))$ in dotted lines, as functions of the price signal x_m , for the same two scenarios as Figure 1. The figure shows that, when $x_m > x^\dagger$, the market prices bonds above its fundamental valuation, and below otherwise.

The effect of quantity-targeting APs can be seen by comparing the baseline of no intervention ($b = 0$) in the left panel to the case with intervention ($b > 0$) in the right panel, keeping in mind the mapping (10). Two channels operate at the level of the support of x_m and affect market prices $Q(p(x_m))$ and fundamental valuations $Q(\hat{p}(x_m))$ in the same way. By the *crowding-out* effect, APs truncate the left tail of market prices ($\underline{x}(b)$ rises from $-\infty$ to a finite number), excluding from the market the most pessimistic investors—those

with $x_i \in (-\infty, \underline{x}(b))$. By the *revelation* effect, APs make the price signals in the interval $[\underline{x}(b), \bar{x}(b))$ fully revealing of the low payoff state θ_L .

The two channels imply that the effect of APs on the equilibrium price is, in principle, ambiguous. Fix a state $(\theta, \tilde{S})$ and consider an increase in APs from zero to $b \in (0, \tilde{S})$. If the resulting price signal $x_m(\theta, \tilde{S} - b)$ falls in the non-revealing region, the crowding-out effect is at work: net supply is lower, a more optimistic investor is marginal, and x_m and the price have moved up. If instead x_m lies in the revealing region $[\underline{x}(b), \bar{x}(b))$, APs lower the equilibrium price down to θ_L since the observed signal reveals the low payoff state. Figure 2 also shows that the *wedge* between equilibrium price and fundamental valuation, namely $Q(p(x_m)) - Q(\hat{p}(x_m))$, is unaffected by APs for x_m belonging to the non-revealing region. The wedge is instead muted in the revealing region, as both price and valuation equal the same revealed value.

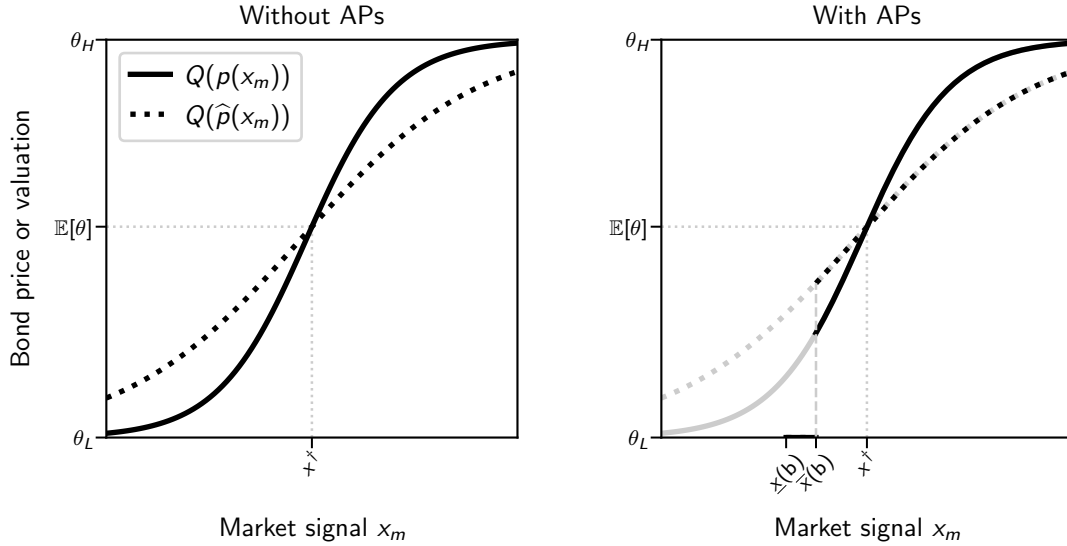


Figure 2: The equilibrium bond price $Q(p(x_m))$ and fundamental valuation $Q(\hat{p}(x_m))$ as a function of the price signal x_m . The x-axis is $x^\dagger \pm 3\sigma_x$.

Why a wedge? The inefficiencies generated by rational learning from prices have been studied by Palfrey (1985), Vives (1988), Amador and Weill (2010), Gaballo (2018) and Pavan et al. (2026), among others. They emerge because atomistic agents take as given the stochastic properties of the equilibrium price from which they learn, while these properties actually depend on aggregate behavior. More recently, Albagli et al. (2023, 2024) provided a useful characterization of the effects of learning from prices in the form of a wedge between the fundamental valuation and the market price for a general class of noisy rational expectations models, including economies where agents are risk-averse or have no position bounds.

Intuitively, in our framework the wedge is due to the fact that the equilibrium price is the expectation of the marginal trader, who is the one observing a private signal equal to the price signal. Thus, even though the price only depends on the realization of x_m , this is weighted twice in the marginal investor’s expectation: once because it is the realization of the price signal, and once because it is the realization of her own private signal. This is in contrast with the fundamental valuation, which is only conditional on one source of information—the price signal—and thus places less weight on x_m than the equilibrium price.

To gain a better understanding, let us highlight the dual role of x_m : informational, since it provides the public price signal (10), and allocational, since it determines the mass of informed buyers (8). Consider an increase in x_m due to a shock to θ or $\tilde{S}$. First, let us focus on the informational role of x_m , abstracting from market clearing forces. Since the price signal is public and investors receive information with the same precision, all investors’ posteriors react to x_m by moving up with the same elasticity, which depends on the weight given to the *public* source of information. The price must therefore increase, via the marginal investor’s indifference condition (15), because of the higher price signal. Second, consider the allocational role of x_m . Investors’ demand schedules (7) and market clearing jointly imply that the investor who is marginal must be the one whose private signal equals the price signal, so this investor must now be one with a higher private signal than before the shock.¹⁶ The price must thus increase further, again via the marginal investor’s indifference condition, by an amount determined by the elasticity of posterior beliefs to the *private* source of information. By the properties of Bayesian updating, the sum of the two elasticities we mentioned exceeds the elasticity of the fundamental valuation, the simple posterior conditional on the price signal alone.

2.6 Informativeness, Sensitivity, and Average Price

In this section, we characterize the properties of the equilibrium price by focusing on how APs impact three moments of interest.

The first moment is price *informativeness*, which is useful to understand how APs affect the information content of the price, independently of the use of such information by investors. We consider the reduction in Shannon entropy on the fundamental θ that is achieved by observing the price signal. This is also referred to as mutual information, and

¹⁶This will be the same investor with a higher x_i if the shock was to θ , or a different, more optimistic investor if the shock was to $\tilde{S}$.

is defined as

$$\begin{aligned} I(Q; \Theta)(b) &= H(\Theta) - H(\Theta | Q) \\ &= H_b(\gamma) - \int_{\bar{x}(b)}^{\infty} H_b(\hat{p}(x_m)) \frac{f_{\mathcal{M}_b}(x_m)}{1-b} dx_m, \end{aligned} \quad (19)$$

where $H(\Theta)$ denotes the prior entropy of the fundamental, $H(\Theta | Q)$ denotes the conditional entropy of the fundamental conditional on the equilibrium price, and $H_b(\gamma) := -\gamma \log \gamma - (1-\gamma) \log(1-\gamma)$ is the binary entropy function.

Our second moment of interest is the *sensitivity* of the price to a change in the fundamental. To get direct empirical relevance, we will consider the linear projection (OLS) coefficient, measuring the covariance between price and fundamental divided by the variance of the fundamental. Given the simple binary structure of the lottery (1), we get

$$\begin{aligned} \mathcal{S}(b) &:= \frac{\text{Cov}(Q, \theta)}{\text{Var}(\theta)} = \frac{\mathbb{E}[Q | \theta_H] - \mathbb{E}[Q | \theta_L]}{\theta_H - \theta_L} \\ &= \int_{\bar{x}(b)}^{\infty} p(x_m) \frac{1}{\sigma_x} \frac{1}{(1-b)} \left[\phi\left(\frac{x_m - \theta_H}{\sigma_x}\right) - \phi\left(\frac{x_m - \theta_L}{\sigma_x}\right) \right] dx_m, \end{aligned} \quad (20)$$

that is, sensitivity in our model boils down to the (normalized) difference between the price mean conditional on θ_H and θ_L , respectively.¹⁷

The third moment that we consider is the average value of the price, $\mathbb{E}[Q(p(x_m))]$, and of the fundamental valuation, $\mathbb{E}[Q(\hat{p}(x_m))]$, which we rewrite, respectively, as

$$\mathcal{Q}_{\text{qt}}(b) := \int_{\bar{x}(b)}^{\infty} Q(p(x_m)) \frac{f_{\mathcal{M}_b}(x_m)}{1-b} dx_m = \mathbb{E}[\theta] + W(b), \quad (21)$$

$$\hat{\mathcal{Q}}_{\text{qt}}(b) := \int_{\bar{x}(b)}^{\infty} Q(\hat{p}(x_m)) \frac{f_{\mathcal{M}_b}(x_m)}{1-b} dx_m = \mathbb{E}[\theta], \quad (22)$$

to emphasize their relation to the prior mean of the asset payoff. The average fundamental valuation $\hat{\mathcal{Q}}_{\text{qt}}(b)$ is equal to the prior mean, irrespective of the AP size b , because it is only conditional on public information, and the law of iterated expectations directly applies. However, this equivalence does not generally hold for the unconditional market price $\mathcal{Q}_{\text{qt}}(b)$, i.e., the aggregate wedge $W(b)$ generally differs from zero and changes with b .

The following proposition characterizes these three moments.

Proposition 2 (The Average effects of APs). *We make the following statements for all*

¹⁷Using $\mathbb{E}[Q] = \gamma \mathbb{E}[Q | \theta_H] + (1-\gamma) \mathbb{E}[Q | \theta_L]$, we get $\text{Cov}(Q, \theta) = \gamma(1-\gamma)(\theta_H - \theta_L) (\mathbb{E}[Q | \theta_H] - \mathbb{E}[Q | \theta_L])$, which we then simplify with $\text{Var}(\theta) = \gamma(1-\gamma)(\theta_H - \theta_L)^2$.

uncontingent AP programs $b \in [0, 1)$.

(i) Price informativeness is strictly increasing in APs

$$\frac{\partial I(Q; \Theta)(b)}{\partial b} = \frac{1}{1-b} \left[\frac{\gamma}{\hat{p}(\bar{x}(b))} H_b(\hat{p}(\bar{x}(b))) - H(\Theta | Q) \right] > 0.$$

(ii) Price sensitivity to the fundamental is strictly increasing in APs

$$\frac{\partial \mathcal{S}(b)}{\partial b} = \frac{1}{1-b} \left[\mathcal{S}(b) - p(\bar{x}(b)) \left(1 - \frac{\phi_L(\bar{x}(b))}{\phi_H(\bar{x}(b))} \right) \right] > 0.$$

(iii) The average wedge

$$W(b) = (\theta_H - \theta_L) \int_{\bar{x}(b)}^{+\infty} (p(x_m) - \hat{p}(x_m)) \frac{f_{\mathcal{M}_b}(x_m)}{1-b} dx_m \quad (23)$$

has the following properties: it is a single-peaked function of the AP size b , it is positive for b large enough, $W(0) \geq 0$ if and only if $\gamma \leq 1/2$, and $\lim_{b \rightarrow 1} W(b) = 0$.

Proof. See Appendix A.2.

The proposition states that uncontingent APs increase both price informativeness and price sensitivity, while they have an ambiguous effect on the average price and corresponding wedge. Figure 3 provides a graphical illustration. Solid lines represent price informativeness, price sensitivity, and the average wedge in the left, center, and right panels, respectively. In the latter two panels, shaded lines represent the corresponding objects for the fundamental valuation.

The effects of APs on these moments can be traced back to the importance of the revelation region relative to the overall support, as illustrated in Figure 2. The larger the AP size b , the larger the probability that a market price is fully revealing. This occurs since the size of the revealing region is of fixed width, whereas the size of the non-revealing region shrinks with APs. On the other hand, the likelihood of θ_H relative to θ_L at a given non-revealing state x_m is unaffected by b .

The implication for informativeness is immediate. The revealing region resolves all uncertainty about θ , while the posterior in the non-revealing region is unchanged. APs thus enlarge the share of states in which the price fully informs investors about θ .¹⁸

¹⁸Another way to see this is to directly consider the variance of x_m conditional on θ , which is decreasing in APs because the support of the *net* supply shock shrinks.

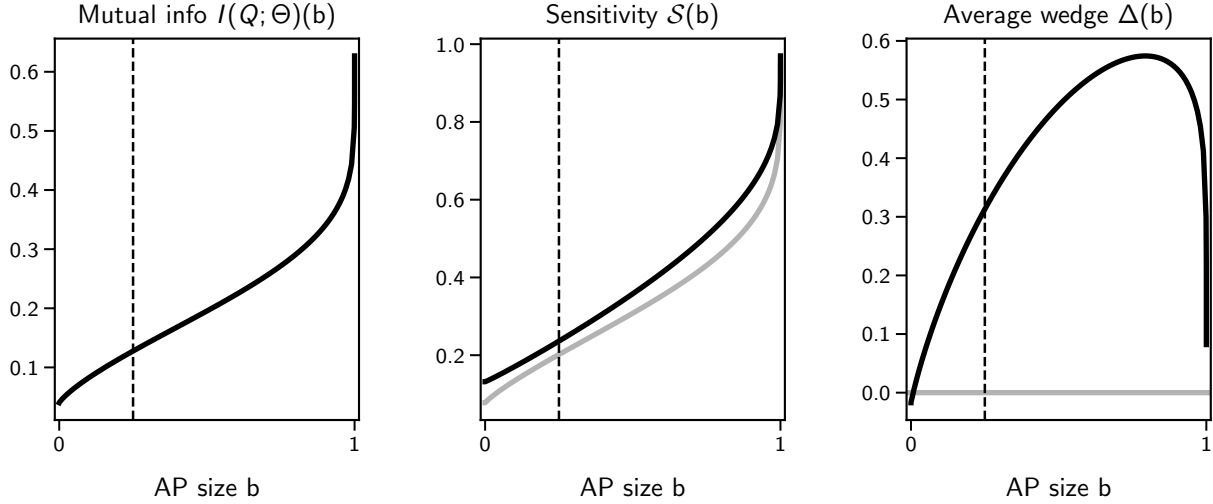


Figure 3: Average informativeness, sensitivity, and information wedge for the equilibrium price. Where present, shaded lines represent the fundamental valuation. The parameter values are: $\theta_H = 2.87, \theta_L = -3.23, \gamma = 0.53, \sigma_x = 10.5$. We choose model parameters to match the Q1-2009 distribution of 10-year inflation forecasts from the U.S. Survey of Professional Forecasters. The dashed vertical line represents the announced size of the first U.S. Federal Reserve Large-Scale Asset Purchase program that included Treasuries (March 18, 2009), as a fraction of outstanding Treasury bonds. The details of the calibration exercise can be found in Online Appendix D.

The intuition for sensitivity is analogous. In the non-revealing region, the equilibrium price pools high- and low-payoff states; in the revealing region, it separates them perfectly. By raising the probability of the revealing region, APs raise the average response of prices to θ . In addition, by truncating low signals, APs remove states that contribute little (or negatively) to sensitivity, raising its average within the non-revealing region as well. The graph shows that the effect of APs on the sensitivity of the equilibrium price and the fundamental valuation is qualitatively the same. This highlights the fact that sensitivity mainly depends on the extent to which APs affect price informativeness, whereas it does not hinge on the presence of the information wedge.

Finally, the impact of APs on the average price and valuation is best understood as trading off the crowding-out and revelation effects. As the figure shows, these effects exactly offset each other in the case of the fundamental valuation, which presents no information wedge, is independent of APs, and equals the prior mean of θ because it satisfies the law of iterated expectations. The average bond price departs from this benchmark through the average wedge $W(b)$, which is single-peaked in b . To see why, recall that the conditional wedge $Q(p(x_m)) - Q(\hat{p}(x_m))$ is positive for $x_m > x^\dagger$, and negative otherwise. Small interventions truncate the lower part of the support of x_m , removing states in which the market prices bonds below their fundamental value, and pushing $W(b)$ upward. Larger interventions rule

out less pessimistic signals, eventually starting to remove also states where the bonds are priced above their fundamental value, which drags $W(b)$ back down. The size of APs that maximizes the average price is therefore intermediate.

Some remarks are in order here. First, as mentioned earlier, the intuition behind our results is not specific to the current framework and the presence of full information revelation; Online Appendix G.2 shows that the results are qualitatively the same when the payoff is continuously distributed on the real line and revelation is never perfect. Second, we deliberately constructed the tractable version of the model to highlight the possibility that APs increase the informativeness and sensitivity of prices to changes in fundamentals. The monotonicity results in Proposition 2 are sharp because position bounds span a unit interval and APs are specified as quantity targets. However, in the more general version of the model with wider bounds $([-\underline{b}, \bar{b}] \supseteq [0, 1])$; see Online Appendix G.1) and in the CARA framework of Section 3, both moments are, in general, ambiguous functions of b . Put differently, the mechanism we emphasize can generate either an increase or a decrease in informativeness and sensitivity, depending on the details of the environment. We focus on the case in which APs increase sensitivity because it is the empirically relevant one, as we document later in Section 4. It is also the most policy-relevant case, as it contrasts with the conventional view that APs should generally calm down markets.

Similarly, the inverse-U shape of the aggregate wedge $W(b)$ relies on the unit position bounds and quantity-targeting policy of this section. We show that such a wedge is instead strictly increasing with APs in the price-targeting version of the current model (Online Appendix H), and in a CARA-binary model with unbounded supply under the same quantity-targeting rule (Section 3), which is consistent with the typical observation that market prices move up following an unanticipated *announcement* of an AP program (Krishnamurthy and Vissing-Jorgensen (2011)).

2.7 Asset Purchases and the Distribution of Gains

The existence of a wedge between prices and fundamentals due to learning from prices has implications for how APs affect the distribution of gains and losses for the government, noise traders, the central bank, and private investors.

The expected gains of an investor conditional on price signal x_m are given by

$$\mathbb{E}[\pi_{\text{inv}} | x_m] = \sum_{\theta \in \{\theta_H, \theta_L\}} \Phi\left(\frac{\theta - x_m}{\sigma_x}\right) (\theta - Q(p(x_m))) P(\theta | x_m), \quad (24)$$

where $\theta - Q(p(x_m))$ represents the unitary gain or loss, and $\Phi((\theta - x_m)/\sigma_x)$ represents the probability of being a buyer (i.e., of receiving a private signal above the price signal).¹⁹ Note that we consider a representative investor because all investors are identical *ex ante*, i.e., before receiving private information. Central bank expected gains conditional on x_m are

$$\mathbb{E}[\pi_{cb} | x_m] = b \sum_{\theta \in \{\theta_H, \theta_L\}} (\theta - Q(p(x_m))) P(\theta | x_m) = b [Q(\hat{p}(x_m)) - Q(p(x_m))], \quad (25)$$

where the second equality follows from the definition of the fundamental valuation in (18). The expected gains of the government are equal to the wedge, i.e., the revenue from the unit of issued debt in excess of its fundamental value, and are given by $\mathbb{E}[\pi_{gov} | x_m] = Q(p(x_m)) - Q(\hat{p}(x_m))$. The gains of the noise traders $\mathbb{E}[\pi_{nt} | x_m]$ obtain residually because of the zero-sum nature of the financial market: $\mathbb{E}[\pi_{nt} | x_m] + \mathbb{E}[\pi_{gov} | x_m] + \mathbb{E}[\pi_{inv} | x_m] + \mathbb{E}[\pi_{cb} | x_m] = 0$ for any state $x_m \in [\underline{x}(b), +\infty)$.²⁰ The following proposition outlines some properties of these conditional gains.

Proposition 3 (Conditional gains and losses). *For a given APs size $b \in [0, 1)$ and price signal $x_m \in [\underline{x}(b), +\infty)$ we have:*

- (i) *investors' expected gains are strictly positive and noise traders expected gains are strictly negative: $\mathbb{E}[\pi_{inv} | x_m] > 0$ and $\mathbb{E}[\pi_{nt} | x_m] < 0$;*
- (ii) *central bank and government expected gains depend on the conditional wedge as follows: $\mathbb{E}[\pi_{cb} | x_m] < 0$ and $\mathbb{E}[\pi_{gov} | x_m] > 0$ if and only if $x_m > x^\dagger$.*

Proof. Postponed to Appendix A.3.

The three panels in Figure 4 plot these conditional gains as a function of the market signal x_m for our baseline calibration. Shaded gray and solid black lines respectively denote the case without and with APs. Thin dashed lines refer to the no-wedge case, where transactions occur at the fundamental valuation rather than the market price, which is a useful benchmark to illustrate how results depend on the learning-from-prices externality.

Consider first the case without APs. The left panel shows that investors' expected gains are positive in all x_m states, regardless of the presence of a wedge between prices and fundamentals. Investors buy bonds only when they expect a payoff higher than their outside option of staying put (which delivers zero gains with certainty). This is effectively a call option whose payoff is positively correlated with the fundamental: conditional on x_m , each

¹⁹By the law of large numbers, $\Phi((\theta - x_m)/\sigma_x)$ also represents the mass of investors buying the bond conditional on θ and x_m .

²⁰This interval of price signals corresponds to states where $\tilde{S} \geq b$. When instead $\tilde{S} < b$ and the market is not active, we have zero gains for all agents since we assume $Q_{pas} = \theta$.

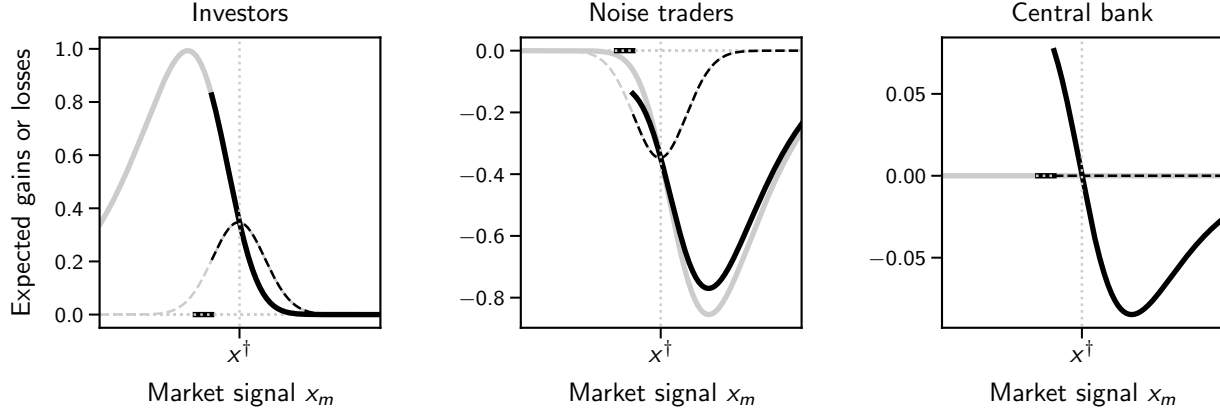


Figure 4: Expected gains conditional on x_m . Gray and black solid lines respectively denote the case without APs ($b = 0$) and with APs ($b = 0.1$). Thick and thin lines respectively denote the case with and without the information wedge.

investor is more likely to buy when fundamentals are good, and supply is in turn high. These two factors together imply that investors' gains are strictly positive even in the absence of the wedge due to learning, as shown by the dashed line in the left panel. In particular, positive gains are the result of limits to individual arbitrage that are due to the presence of position bounds and dispersed beliefs. The wedge induced by learning-from-prices introduces an additional force that generates skewness in the gains distribution: bonds are priced below their fundamental value when $x_m < x_m^\dagger$, which implies that investors' gains are larger than in the no-wedge case, and vice versa when $x_m > x_m^\dagger$.

Consider now the case with APs, denoted by black solid lines. In contrast to investors, the central bank buys unconditionally, making losses when $x_m > x_m^\dagger$ and gains otherwise, as the right panel shows. Central bank losses result in net gains for the government. In this sense, APs have the potential to engender implicit monetary financing, i.e., a transfer of resources from the central bank to the government. This result emerges because of the existence of the wedge. The effect of central bank APs on investors' and central bank gains is negative: the crowding-out effect truncates the left tail of the gains distribution, and the revelation effect makes gains drop to zero in the adjacent, fully revealing region. In the counterfactual where the wedge is muted—which is plotted as a dashed line in all three panels—APs are balance-sheet neutral, generating neither gains nor losses for the central bank or the government. We state the effect of APs on *average*—or unconditional—gains in the following proposition.

Proposition 4 (Average gains and losses). *Average investors' gains are positive and decreasing in APs. Average noise trader gains are negative for any AP size. Average central bank losses and government gains are proportional to the average wedge of Proposition 2,*

and are thus positive for any AP size if $\gamma \leq 1/2$, and for b large enough if $\gamma > 1/2$. The average gains and losses of all players converge to zero as $b \rightarrow 1$.

Proof. Postponed to Appendix A.4.

The proposition is illustrated in Figure 5: we plot the average gains of investors (dash-dotted line), central bank (dotted line), and noise traders (dashed line), as a function of the quantity target of the AP policy. The left panel represents the model, while the right panel shows what gains and losses would be in the absence of the wedge due to learning from prices.

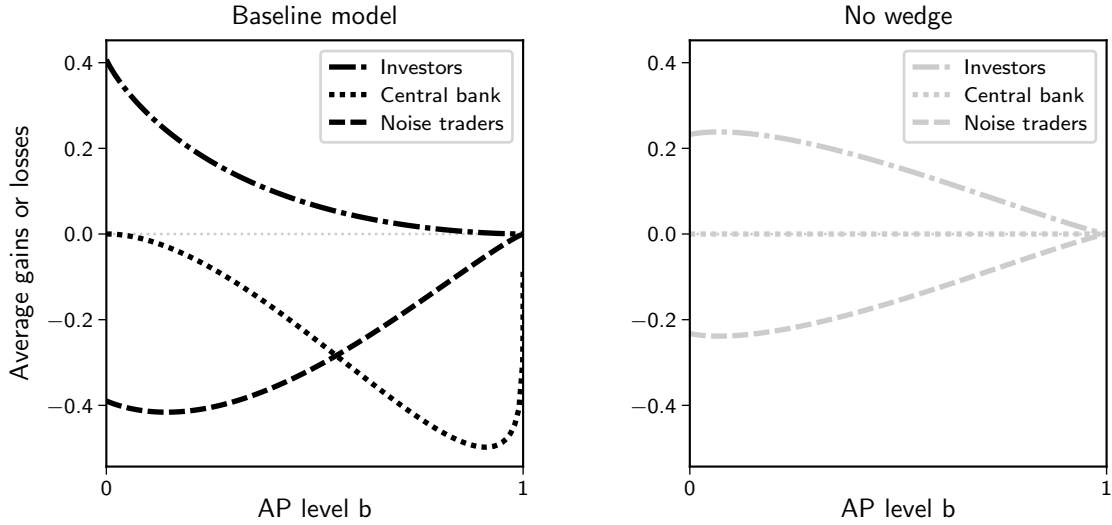


Figure 5: Average gains by investors, the central bank, and noise traders, as a function of the size of the AP program b . The left panel represents the model; the right panel represents what gains would be in the absence of the information wedge between bond prices and valuations.

As APs increase in size, investors' gains and noise traders' losses fall, and central bank losses rise, all to the benefit of the government (whose gains are given by the right panel of Figure 3). Thus, sufficiently large APs imply central bank monetary financing of the government. Net gains for the government emerge through two transfers of resources: directly from the central bank, which buys government debt at an average loss, and indirectly from investors, whose scope for buying assets priced below their fundamental value is reduced. As b approaches one, all gains and losses converge to zero, because in the limit the central bank is buying at the perfect information price, and there are no net transfers to the government.

It is important to note that these results are tied to the existence of the wedge, while they are independent of the degree of payoff skewness γ . In the no-wedge case, denoted by thin shaded lines in the right panel of Figure 5, the intervention is balance-sheet neutral for the central bank, while investors' gains are still decreasing in APs.

3 A CARA Model with Bounds

The tractable model of Section 2 relies on risk neutrality and a bounded bond supply to deliver closed-form characterizations. In the online appendix, we analyze various extensions of the risk-neutral model.²¹ In this section, we consider a richer asset pricing model with three ingredients: risk-averse informed investors who are possibly subject to position bounds, uninformed investors with a price-elastic asset demand, and an unbounded asset supply. We use this framework to show that the information channel of APs applies to a wider class of asset pricing models, and we characterize some of its general properties.

The analysis proceeds as follows. First, we show that, when agents are unconstrained, aggregate asset demand is linear in the fundamental, and APs have no impact on beliefs and only affect the price through the average premium for risk. Then, we consider the case where position bounds are finite and a varying mass of informed investors is constrained. In this case, aggregate demand is nonlinear, and APs affect the entire distribution of the price signal, its informativeness, and informed investors' beliefs. Finally, we relate this model to the risk-neutral framework of the previous section, and discuss the generality of our results.

Under the parameter configurations emphasized below, the properties of this CARA model align with those of the risk-neutral model illustrated in Section 2.6: larger APs make asset prices more informative and more sensitive to fundamentals, while increasing their unconditional mean.

3.1 Setup

We maintain the assumptions of the risk-neutral model of Section 2 regarding (i) the binary lottery (1) governing the stochastic bond payoff θ ; (ii) the structure of private signals (3); and (iii) the presence of position bounds $[b, \bar{b}]$ on investors' bond holdings. We instead depart from the risk-neutral model along three dimensions: (iv) informed investors have CARA utility with risk aversion coefficient $\rho > 0$, so that their demand is often interior and varies continuously with beliefs, rather than taking the bang-bang form of (7); (v) the supply shock is unbounded and normally distributed, $\tilde{S} \sim N(\mu_s, \sigma_s^2)$, rather than being uniformly distributed on $[0, 1]$; (vi) a second class of uninformed investors has aggregate demand $N(Q)$, a decreasing function of the bond price. One can think of this last assumption as capturing, in reduced form, the demand of investors who trade for reasons unrelated

²¹We consider wider position bounds in Online Appendix G.1; the case of a continuously-distributed fundamental in Online Appendix G.2; a price-targeting AP policy in Online Appendix H; and an expenditure-targeting AP policy in Online Appendix I.

to fundamental information—such as liquidity needs, regulatory requirements, or preferred-habitat motives—and whose aggregate behavior generates a downward-sloping residual demand curve that informed investors trade against.²² We assume that the central bank follows an uncontingent quantity-targeting policy, where APs are equal to b regardless of the state $(\theta, \tilde{S})$.

Informed investors' demand. The problem of informed investor i , with information set $\Omega_i = \{x_i, Q, b\}$, is

$$\max_{b_i \in [\underline{b}, \bar{b}]} \mathbb{E} \left[-e^{-\rho b_i(\theta - Q)} \mid \Omega_i \right], \quad (26)$$

which is a standard CARA portfolio problem subject to position bounds. The solution takes a censored form: the investor holds the unconstrained optimum whenever it falls within the bounds, and is pinned to the relevant bound otherwise:

$$b_i = \begin{cases} \bar{b} & \text{if } b^u(x_i, Q) > \bar{b}, \\ b^u(x_i, Q) & \text{if } b^u(x_i, Q) \in [\underline{b}, \bar{b}], \\ \underline{b} & \text{if } b^u(x_i, Q) < \underline{b}, \end{cases} \quad (27)$$

where the unconstrained demand

$$b^u(x_i, Q) := \frac{1}{\rho(\theta_H - \theta_L)} \left[\log \left(\frac{\gamma}{1 - \gamma} \frac{f(Q \mid \theta_H)}{f(Q \mid \theta_L)} \right) + (\theta_H - \theta_L) \frac{x_i - x^\dagger}{\sigma_x^2} - \log \frac{Q - \theta_L}{\theta_H - Q} \right] \quad (28)$$

depends on: (i) the pecuniary cost of the asset, measured by the logit transformation $\log \frac{Q - \theta_L}{\theta_H - Q}$ of the price, (ii) the public component $\log \left(\frac{\gamma}{1 - \gamma} \frac{f(Q \mid \theta_H)}{f(Q \mid \theta_L)} \right)$ of the posterior log-odds ratio, which combines the prior and the price signal log-likelihoods, and (iii) the private component of the posterior log-odds ratio, given by the distance between the private signal x_i and the uninformative signal $x^\dagger$, scaled by the private signal variance σ_x^2 .

The position bounds bind for investors with private signals that are sufficiently large in absolute value. Investor i is constrained at the upper bound whenever $x_i > x_H(Q)$ and at the lower bound whenever $x_i < x_L(Q)$, where $x_L(Q), x_H(Q)$ respectively solve $b^u(x_L, Q) = \underline{b}$ and $b^u(x_H, Q) = \bar{b}$.

²²This is a modeling device and should not be taken literally as a specific theory of uninformed demand; what matters for our results is only that $N(Q)$ is decreasing in Q . One can also think of $\tilde{S} - N(Q)$ as the price-elastic supply available to informed traders, as in Hellwig et al. (2006), or of $b + N(Q)$ as a price-contingent central bank AP rule. More generally, the results hold true for any extension in which N is a function of observables, like b , or commonly known statistics, like $x^\dagger$, provided N is weakly decreasing in the market price Q (otherwise, multiple equilibria obtain).

Market clearing. Finally, market clearing requires that aggregate demand equals net supply:

$$D^{\text{inf}}(\theta, Q) + N(Q) = \tilde{S} - b, \quad (29)$$

where $D^{\text{inf}}(\theta, Q) := \int_0^1 b_i di$ is the aggregate demand of informed investors. The market clearing equation (29) defines the key mapping between Q and $(\theta, \tilde{S}, b)$.

3.2 Unconstrained Case: Informational Neutrality

We work out here the case without bounds, illustrating how the information channel of APs is muted when aggregate asset demand is linear. In the absence of position bounds ($\underline{b} \rightarrow -\infty, \bar{b} \rightarrow +\infty$), all informed investors are on their unconstrained demand schedule $b^u(x_i, Q)$, so that $D^{\text{inf}}(\theta, Q) = b^u(\theta, Q)$ where $b^u(\theta, Q)$ is the average unconstrained demand across all informed investors. It follows from (28) that aggregate demand becomes a *linear* function of θ . This is the key property that makes an uncontingent, publicly observed AP policy have no impact on price informativeness. To see this, we can rearrange the market clearing condition to get

$$\underbrace{x^\dagger + \rho \sigma_x^2 (\mu_s - b - N(Q) - b^u(x^\dagger, Q))}_{x_m(Q)} = \theta - \rho \sigma_x^2 (\tilde{S} - \mu_s). \quad (30)$$

The left-hand side is a transformation of the price Q and other observables—including the AP quantity—which we denote with $x_m(Q)$. In analogy to (10), equation (30) characterizes $x_m(Q)$ as a public signal of θ that is Gaussian, linear, and unbiased. It also makes clear that APs (i) affect the mapping $x_m(Q)$ from the equilibrium price to the signal it provides, because they change average net supply ($\mu_s - b$) and in turn the average risk premium; but (ii) do not affect the information content of the price signal. An aggregate wedge due to a learning-from-price externality still exists, but uncontingent APs do not affect it.

3.3 Constrained Case

When bounds are finite, aggregate informed demand decomposes as

$$D^{\text{inf}}(\theta, Q) = P_L(Q, \theta) \underline{b} + P_H(Q, \theta) \bar{b} + \left(1 - P_H(Q, \theta) - P_L(Q, \theta)\right) \left(b^u(\theta, Q) - T(\theta, Q)\right), \quad (31)$$

where $P_L(Q, \theta) := \Phi\left(\frac{x_L(Q) - \theta}{\sigma_x}\right)$ and $P_H(Q, \theta) := 1 - \Phi\left(\frac{x_H(Q) - \theta}{\sigma_x}\right)$ denote the share of investors constrained at each bound, and $T(\theta, Q)$ is a correction that accounts for the two-sided

truncation of demand at the bounds.²³ The three summands in (31) have a transparent interpretation: the first two capture the demand of investors pinned at each bound, weighted by their respective population share, while the third captures average interior demand. When position bounds are finite, (31) shows that aggregate demand is a nonlinear, non-separable function of θ and Q . This does not allow a closed-form representation of the price signal analogous to (30), and implies that the conditional bias and variance of the price signal generally depend on APs.

We thus proceed by solving the model numerically, iterating on the aggregate demand function and the price signal log-likelihood ratio until convergence. To keep the exposition concise, we focus here on selected features of the model and the effect of APs on asset prices, their sensitivity, and their information content. Further details on the solution method, a further characterization of the model, and additional figures can be found in Online Appendix F.2.

Numerical results. Figure 6 illustrates some properties of the model; solid and dashed lines represent different levels of APs.²⁴ The three panels of the figure highlight in different ways the key mechanism of the model: how the fraction of informed agents that are constrained depends on the states. A sufficiently large supply shock pushes the equilibrium price close to its lower bound θ_L , and vice versa for a small supply shock. In these states, informed agents' demand is insufficient to clear the bond market, all informed agents are constrained (right panel), and uninformed demand is what adjusts. As a consequence, the equilibrium price is at its extremes; it is completely uninformative about the realization of θ , because it is determined by supply and not by the fundamental θ . It follows that posterior and prior beliefs coincide (central panel), because observing the price carries no information about the fundamental driving informed demand.

Since APs determine the net supply of the asset, they have a direct effect on when and how much informed traders are constrained by position bounds. The right panel shows that, conditional on the price, a larger AP policy makes traders less constrained by the long position bound, and more by the short position bound.

²³The analytical expression is $T(\theta, Q) := \frac{1}{\rho \sigma_x} \frac{\phi((x_H(Q) - \theta)/\sigma_x) - \phi((x_L(Q) - \theta)/\sigma_x)}{1 - P_H(Q, \theta) - P_L(Q, \theta)}$.

²⁴We use the following parametrization, chosen for illustrative purposes: $(\theta_L, \theta_H) = (0.5, 1), \gamma = 0.53, (d_L, d_H) = (-3.5, 2), \sigma_x = 1.5, \mu_s = 0, \sigma_s = 1.5, N(Q) = \log((\theta_H - Q)/(Q - \theta_L))$.

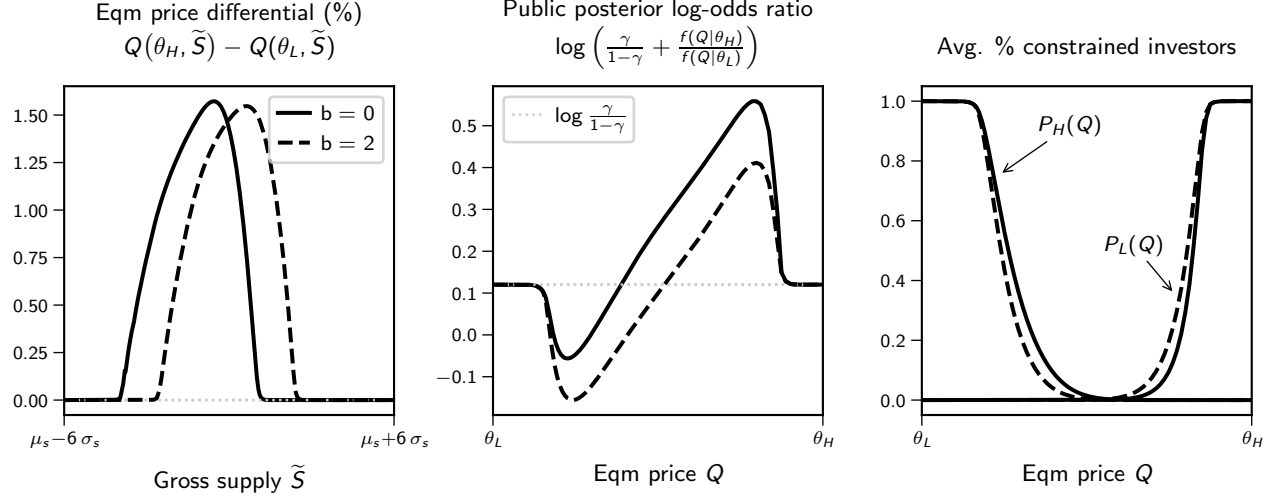


Figure 6: Selected model features for two different values of uncontingent APs, illustrated with dashed and solid lines. The left panel plots the differential between the equilibrium price conditional on θ_H and θ_L , as a function of gross supply. The central panel plots the public posterior log-odds ratio. The right panel plots the share of informed investors that are constrained, averaged across θ , as a function of the equilibrium price.

Demand constraints and the effect of APs. We now consider some key unconditional moments of the model, as a function of the uncontingent AP policy b and for both the case with and without position bounds. We study the same moments of Section 2.6: the average price, price informativeness, and price sensitivity. Since the current model features risk aversion, we also analyze risk-neutral prices to decompose the effect of APs on risk premia on one hand, and on the informational component of the price on the other hand.²⁵

Figure 7 provides an illustration. Shaded lines represent the *unconstrained* model, while solid lines represent the *constrained* model with finite bounds; the bottom-left panel only applies to the constrained model. The bottom-right panel shows that APs have an effect on the average equilibrium price (inclusive of the risk premium) that is similar in both models. The top panels show that the real difference comes from the behavior of beliefs, which in turn are the sole driver of risk-neutral prices. These are only affected by APs in the model with bounds, as is shown by the behavior of price informativeness, the sensitivity of the risk-neutral price to fundamentals, and its average value. The figure highlights how the informativeness of the price and the sensitivity of its risk-neutral version are the mirror image of the total share of informed agents that are constrained—illustrated in the bottom-left panel.

²⁵Risk-neutral prices are our main variable of interest also in our empirical analysis. Online Appendix F.2 explains in detail how price informativeness, price sensitivity, and risk-neutral prices are computed within the model at hand.

For this specific parametrization, positive APs reduce the average share of constrained investors and increase the average risk-neutral price, its sensitivity, and the informativeness of the price signal. The intuition is the following. APs reduce net supply and, in turn, the net demand required to clear the market. This means that informed investors become more likely to hit their short-selling constraint, and less likely to hit their demand upper bound. If the former are looser than the latter, then APs increase the average share of *unconstrained* agents. This in turn makes the price more informative, and the risk-neutral price more sensitive to fundamentals.

The effect is, however, non-monotonic: it flips once APs are large enough that the increase in the mass of investors at the short position bound dominates the decrease at the upper bound. In either case, the figure highlights our main result, that APs alter the state-contingent response of asset prices to shocks through an information channel that is absent in traditional models based on risk premia.

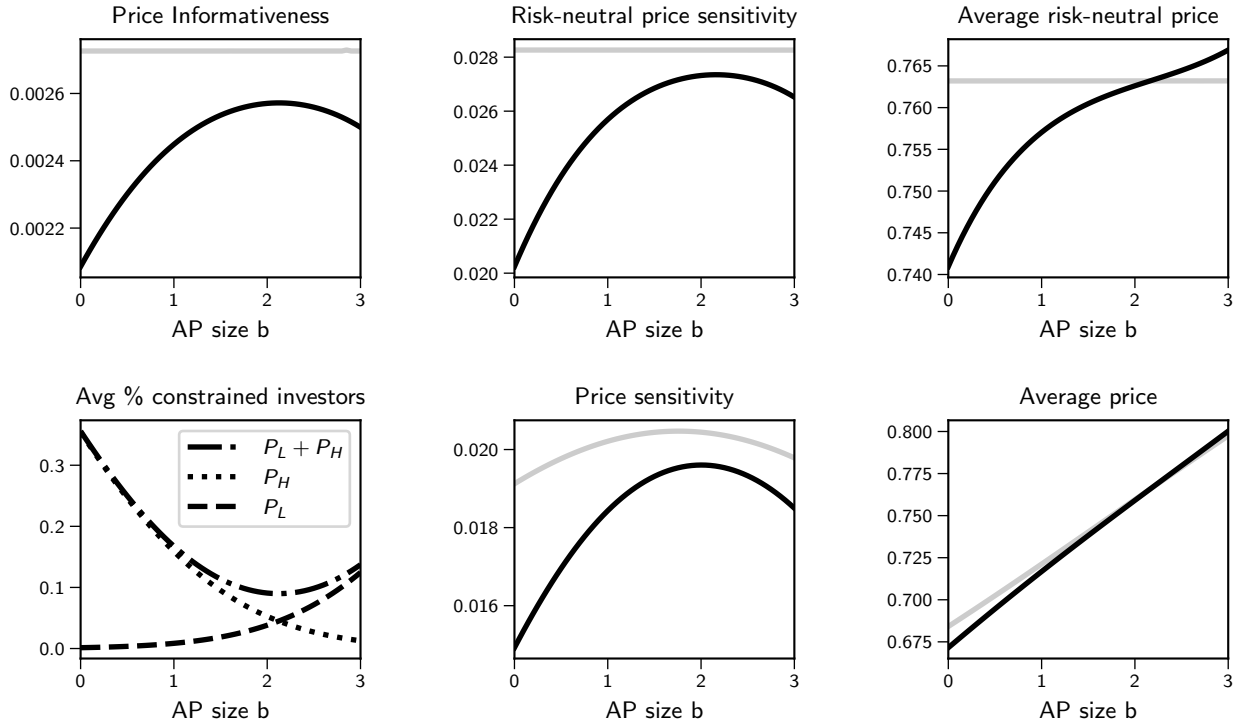


Figure 7: Average informativeness, share of constrained investors, sensitivity, and information wedge for the equilibrium price and the risk-neutral price. Black and gray lines represent the model with and without bounds, respectively.

3.4 A Unifying View

To see the connection between the model of this section and that of Section 2, we consider a parsimonious framework that nests both and allows us to illustrate the generality of our results.

Consider an asset market characterized by the market clearing condition

$$D(\theta, Q) = \tilde{S} - b \quad (32)$$

where θ is the fundamental shock that follows the binary distribution in (1), b are central bank uncontingent APs, $\tilde{S}$ is the random supply shock that follows an exogenous distribution with density f_S and support that is independent of θ and b , and D is an aggregate demand function that is increasing in θ ($D_\theta > 0$), decreasing in Q ($D_Q < 0$), and such that there exists an equilibrium price that clears the market for any $(\theta, \tilde{S}, b)$. To study the effect of APs on our outcomes of interest, it is useful to consider the price signal distribution and, specifically, its log-likelihood ratio $\ell(q) := \log \frac{f(q|\theta_H)}{f(q|\theta_L)}$, since it is the main driver of posterior beliefs, the informativeness of the price signal, and the risk-neutral price sensitivity.²⁶

Informational neutrality. In the CARA model without position bounds, we have seen that aggregate demand is a linear combination of the fundamental and the price, and APs have no effect on the information content of the price signal, and in turn on beliefs. This result can be generalized to a broader class of models, which also includes the well-known CARA framework with a linear normal payoff, and the risk-neutral setting with position bounds of Hellwig et al. (2006), as we show in Online Appendix F.3.

Proposition 5 (Information neutrality). *Under the assumptions of this section and given market clearing condition (32), if aggregate demand $D(\theta, Q)$ is additively separable in θ and Q , uncontingent APs have no effect on the price signal log-likelihood ratio, and so on investors' posterior beliefs about θ , at any equilibrium price.*

Proof. Let $D(\theta, Q) \equiv a(\theta) - g(Q)$ where $a(\cdot)$ and $g(\cdot)$ are suitable monotone functions. Market clearing (32) implies that there exists a known transformation $z(Q, b) := g(Q) - b$ that gives the price signal $z = a(\theta) - \tilde{S}$, whose likelihood given θ is $f_{Z|\theta}(z|\theta) = f_S(a(\theta) - z)$ whenever $a(\theta) - z$ belongs to the support of $\tilde{S}$, and zero otherwise. Since z and Q are one-to-one given b , conditioning on Q is equivalent to conditioning on z . The likelihood of the price signal z is independent of APs, and so are posterior beliefs on θ conditional on z . ■

²⁶Note that $\ell(q)$ is invariant to considering price signals that are a one-to-one transformation of the price, since such mapping cancels out when taking the ratio of the signal likelihoods.

The price signal defined in the proof corresponds to equation (30) in our unconstrained CARA example, where the log-likelihood ratio is given by

$$\ell(q) = \frac{\theta_H - \theta_L}{\sigma_q^2} (x_m(q) - x^\dagger) \quad (33)$$

where $\sigma_q^2 := \sigma_s^2(\rho\sigma_x^2)^2$, and APs thus affect the mapping $x_m(q)$ from the equilibrium price to the price signal, but not the information content of the latter.

Informational non-neutrality. When instead aggregate demand is not additively separable, APs affect the entire shape of the equilibrium price likelihood. This is the case in the CARA model when investors are constrained by position bounds, as we showed earlier: the log-likelihood ratio is given by $\ell(q) = \log \frac{D_Q(\theta_H, q)}{D_Q(\theta_L, q)} + \log \frac{f_S(D(\theta_H, q) + b)}{f_S(D(\theta_L, q) + b)}$, and the aggregate demand slopes depend on the fundamental and the equilibrium price, so when APs affect the latter, they alter the whole distribution of the price likelihood ratio. The risk-neutral model of Section 2 also features a non-separable aggregate demand, in particular $D(\theta, Q) = \Phi\left(\frac{\theta - x_m(Q)}{\sigma_x}\right)$ where the mapping $x_m(Q)$ is given by (17). The model, however, remains analytically tractable because the assumed functional forms allow us to retain much of the Gaussian structure of Bayesian updating typical of standard unconstrained CARA models. The log-likelihood ratio is

$$\ell(q) = \begin{cases} -\infty, & x_m(q) \in [\underline{x}(b), \bar{x}(b)), \\ \frac{\theta_H - \theta_L}{\sigma_x^2} (x_m(q) - x^\dagger), & x_m(q) \in [\bar{x}(b), +\infty). \end{cases} \quad (34)$$

Here APs do not affect the log-likelihood ratio of the price when it belongs to the set of non-revealing prices—which in fact coincides with (33)—but they do change such set itself, thus effectively changing $\ell(q)$ from a finite value to $-\infty$ for some states. This feature of the risk-neutral model makes it a particularly tractable application of our mechanism.

4 Empirical Insights

Our theory shows that APs affect asset prices to the extent that they alter their distribution conditional on fundamentals. Our informational channel—unlike in traditional mechanisms based on risk-premia microfoundations—may induce a state-contingent response of risk-neutral prices to APs.

To provide empirical support to our conclusions, we examine the effects of U.S. Federal

Reserve AP flows on Treasury yields from August 25th, 2005, to December 31st, 2019, conditional on the arrival of macroeconomic news, which we use as a proxy for changes in the fundamental state. Focusing on the U.S. Treasury market provides a particularly informative test, as it is arguably the most stable and resilient bond market in the world. We find robust evidence that Fed purchases of Treasury bonds amplify the sensitivity of yields to labor market news at daily frequencies. That is, a price change in either direction induced by a labor market surprise tends to be larger, the greater the AP purchase.

These results are consistent with the main prediction of our models, and suggest that APs may increase the economy’s sensitivity to changes in fundamentals rather than calm down markets. More broadly, they point to the potential for APs to accelerate fundamentals-driven crises rather than buffer against them.

Further details about the data sources and additional regression specifications are available in Appendix [B](#).

4.1 Yields, APs, and News

Treasury yields. To abstract from confounding movements in risk premia, we use as dependent variable the daily change in the risk-neutral component of U.S. Treasury yields, $\Delta y_t^{\text{rn}} \equiv y_t^{\text{rn}} - y_{t-1}^{\text{rn}}$. The risk-neutral yield y_t^{rn} is obtained from the Federal Reserve Bank of New York and constructed following Adrian et al. (2013).²⁷ Focusing on the risk-neutral component allows us to interpret yield changes Δy_t^{rn} —consistent with the mechanism highlighted by our theory—as revisions in expectations about the future path of short-term interest rates, abstracting from fluctuations in term premia.

Asset purchase operations. Daily data on Federal Reserve outright Treasury purchases at the security level are recorded by the New York Fed in the Federal Reserve System Open Market Account (SOMA). Publicly available data records start on August 25th, 2005. We construct daily AP flows, denoted by AP_t , by summing the face value of all securities traded by the Fed across all maturities in a given day. We exclude inflation protected bonds (TIPS) since the risk-neutral yield measures are computed for nominal bonds only. Our sample consists of 998 daily observations with non-zero flows. These operations are pre-announced: the New York Fed’s Open Market Trading Desk publishes, roughly monthly, a schedule with the date, the eligible maturity sector, and an approximate size for each operation (for more

²⁷These yields are available at a daily frequency for maturities from 2 to 10 years. We use the 3-year maturity in our baseline specification, and show results for the 10-year maturity in the appendix.

details see Song and Zhu (2018)). At daily frequency, these operations represent the natural empirical counterpart of b , measuring the day’s absorption by the central bank.

Macroeconomic news. U.S. announcement surprise data are taken from Bloomberg. For each economic release $e \in \mathcal{E}$, Bloomberg provides the consensus median forecast $F_{e,t}$, the cross-sectional standard deviation $\text{Std}_{e,t}$, and the actual released value $A_{e,t}$. We consider Bloomberg surprises, which are defined as $U_{e,t} = (A_{e,t} - F_{e,t})/\text{Std}_{e,t}$. Among all the news events reported in the dataset, we decide to focus on the single Nonfarm Payrolls (NFP henceforth) announcement, and on the first principal component of the surprises in the 14 indicators of the Employment Situation report (NFP is among them) simultaneously released on the first Friday of each month at 8:30 a.m. (ET).²⁸ We now describe these two news measures.

Our baseline news variable is the monthly surprise in Nonfarm Payrolls ($e = \text{NFP}$). This choice rests on three considerations. First, the Nonfarm Payrolls statistic is widely considered the single most market-moving U.S. data release, providing the statistical power needed to identify the effects of interest. Second, the distribution of the surprises is stable over time, so changes in the estimated yield sensitivity to news can only reflect changes in the transmission mechanism and not in the distribution of surprises.²⁹ Third, the news releases occur on a known fixed calendar, which provides enough regularly-spaced observations for testing different subsamples.

We also construct a broader labor market news index by extracting the first principal component (PCL) from the surprises in the 13 indicators of the Employment Situation report that are released simultaneously on each announcement day.³⁰ Because all 13 components are released simultaneously, the PCL index is exactly zero on every non-announcement day by construction.

²⁸We effectively use 13 of the 14 indicators, since for one of them—the two-month payroll net revision—Bloomberg does not collect forecasts and thus does not compute the surprise.

²⁹Appendix B.2 compares the distribution of surprises in and out of the zero lower bound period (2009-2015) and finds they are not statistically different.

³⁰These are: nonfarm payrolls, manufacturing payrolls, private payrolls, household employment, unemployment rate, underemployment rate (U_6), labor force participation rate, average hourly earnings (MoM and YoY), average weekly hours (all employees), average hourly earnings of production workers (MoM and YoY), average weekly hours of production workers. We exclude the two-month payroll net revision, since Bloomberg does not collect forecasts for this event.

4.2 Yield Regressions

Our baseline test estimates, for the 3-year maturity, the following simple interaction model:

$$\Delta y_t^{\text{rn}} = \alpha_{q,t} + \beta U_t + \kappa \text{AP}_t + \psi(U_t \times \text{AP}_t) + \phi^\top C_t + \varepsilon_t, \quad (35)$$

where U_t is the standardized surprise in either the NFP or the PCL news, $\alpha_{q,t}$ denotes year-quarter fixed effects, and C_t is a vector of controls. All standard errors are Newey-West heteroskedasticity and autocorrelation consistent, with automatic lag selection according to the data-adaptive Bartlett rule for each regression.

The key object of interest is the coefficient ψ , which captures how the total *sensitivity* ($\beta + \psi \text{AP}_t$) of yields to news varies with the volume of Fed purchases. This sensitivity is the empirical counterpart of the price sensitivity to the fundamental we define with (20) in our models. A significant ψ implies that Fed purchases alter the pass-through of macroeconomic fundamentals to the risk-neutral component of Treasury yields.

The regression results of our baseline test are reported in Table 1. We estimate six specifications, varying three design choices simultaneously. Columns (1)–(3) use the Nonfarm Payrolls surprise; columns (4)–(6) replace it with the PCL surprise. Columns (1) and (4) use all the daily observations in the full sample period; columns (2) and (5) restrict to the zero lower bound (ZLB) period; columns (3) and (6) use the full sample period but restrict estimation to the 172 announcement days. Finally, the full-sample and ZLB columns include year-quarter fixed effects and the actual released values (not surprises) of *all* labor market indicators at each date (which we denote with $\bar{A}$) as controls. The announcement-day columns instead include only the actual release value of the focal indicator (A)—either the Nonfarm Payrolls or the PCL—and no year-quarter dummies, given the reduced sample size.

The first row of Table 1 shows that a one-unit standardized surprise in Nonfarm Payrolls or in the first principal component of the Employment Situation indicators significantly increases the risk-neutral component of Treasury yields by 1-1.5 basis points across all specifications. The direct level coefficients on AP (κ) are negative, and only significant in the full-sample specifications (1) and (4). This finding supports the idea that AP flows unconditionally increase asset prices (and thus lower yields), *even in their risk-neutral component*, consistent with our theory. However, their insignificance in the restricted-sample specifications (2), (3), (5), and (6) suggests that this effect may primarily manifest at lower frequencies, particularly during the QE regime in the ZLB, implying a limited scope at daily frequencies.

Table 1: Baseline specification.

	Nonfarm Payroll			PC Labor News		
	(1)	(2)	(3)	(4)	(5)	(6)
Surprise	0.011*** (0.003)	0.014*** (0.005)	0.010*** (0.003)	0.012*** (0.004)	0.017*** (0.006)	0.013*** (0.004)
AP	-0.006** (0.003)	-0.004 (0.003)	-0.001 (0.010)	-0.007** (0.003)	-0.004 (0.003)	-0.003 (0.011)
Surprise $\times$ AP	0.013*** (0.005)	0.015*** (0.005)	0.013*** (0.005)	0.022*** (0.006)	0.029*** (0.006)	0.024*** (0.006)
Obs	3590	1752	172	3590	1752	172
R^2	0.093	0.162	0.179	0.088	0.150	0.173
Years	05-19	09-15	05-19	05-19	09-15	05-19
# News	172	84	172	172	84	172
Event days only	No	No	Yes	No	No	Yes
Controls	$\bar{A}$	$\bar{A}$	A	$\bar{A}$	$\bar{A}$	A
Time FE	Yr-Qtr	Yr-Qtr	No	Yr-Qtr	Yr-Qtr	No

Notes: Dependent variable: daily change in risk-neutral 3-year yields. Robust standard errors in parentheses. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Our central finding is a positive and significant interaction coefficient, showing a state-contingent effect of APs on the risk-neutral component of yields. The size of the estimated coefficient is large and robust across all specifications. In terms of magnitudes, an increase in Fed Treasury purchases of one standard deviation (between \$2 and \$3 billion depending on the specification) increases the risk-neutral yield impact of Nonfarm Payrolls by 0.3-0.4 basis points, or 25-35% of the direct surprise effect, and of PCL by 0.5-0.8 basis points, or 40-50% of the direct surprise effect. This finding makes a strong case for the possibility that AP flows can heighten, rather than dampen, the sensitivity of yields to fundamental news.

Some features of the design suggest a causal reading of the interaction coefficient ψ . First, AP flows are predetermined with respect to news surprises. Consistent with the model's assumption that b is observable, the New York Fed releases operation calendars, specifying operation dates, maturity sectors, and planned purchase amounts. The operations themselves are conducted in late-morning auction windows—after the 8:30 a.m. release of the Employment Situation report; reverse regressions show that flows do not respond to surprises on the announcement day (Table B.2). In Appendix B we show that our results hold across a variety of different specifications. In Table B.3 we augment controls to include prior statistical releases; in Table B.4 we run our baseline test on 10-year risk-neutral bond yields; in Table B.5 we include lags of daily yield changes and AP flows; in Table B.6 we consider lagged or cumulated AP flows; in Table B.7 we use the change in raw nominal

yields (rather than just their risk-neutral component) as the dependent variable; in Table B.8 we control for realized yield volatility or the level of the VIX index; and in Table B.9 we include program-specific surprise interactions and announcement-date dummies. Finally, we perform a leave-one-out analysis to show that our results are not driven by any single outlier observation.

5 Welfare Rationale for Asset Purchases

In this section, we extend our framework to address the natural question of what frictions justify central bank intervention through large-scale asset purchases. These frictions are often modeled using nominal rigidities, a zero lower bound on interest rates, or a financially constrained banking sector—none of which are essential for our mechanism. We demonstrate that the same frictions underpinning our approach—namely, position bounds and dispersed information—are sufficient, on their own, to generate inefficiencies in a simple consumption-saving model where the investors are financial intermediaries. Specifically, investors make inefficiently high average gains in the financial market. This inefficiency stems from dispersed information and limited arbitrage, which is distinct from the learning-from-prices externality and is present even without any informational wedge in prices, as we showed in Section 2.7. AP policies can effectively mitigate this inefficiency by lowering investor gains.

Environment. Time is discrete and there are two periods, $t = 0, 1$. We consider an economy populated by a government, a central bank, and two continua of measure one of households and investors. For simplicity, we abstract from noise traders and assume no time discounting. In the first period, the government issues bonds to finance spending. Households receive an endowment, which they either consume or deposit in financial institutions—such as banks—that we refer to as investors. These investors observe noisy signals and allocate funds between government bonds and central bank reserves. The central bank, in turn, issues reserves to investors for the purpose of purchasing bonds or saving through storage. In the second period, the government levies lump-sum taxes to finance spending and debt service and to make transfers to the central bank; investors remunerate deposits and transfer their trading gains to households; and households liquidate deposits, consume, and pay taxes. We now consider the problem of each class of agents in detail.

Households. Households are homogeneous and enjoy utility from consumption in two periods. In the first period, household j solves the following consumption-saving problem:

$$\begin{aligned} & \max_{c_{j,0}, c_{j,1}, \{s_{j,i}\}_{i \in [0,1]}} u(c_{j,0}) + u(c_{j,1}) \\ \text{s.t. } & c_{j,0} = y - \int_0^1 s_{j,i} \, di \quad \text{and} \quad c_{j,1} = \int_0^1 \mathcal{R}_i s_{j,i} \, di + D - \tau. \end{aligned} \tag{36}$$

where y is the initial endowment of the consumption good;³¹ $s_{j,i}$ is the amount of savings that household j deposits with investor i and which earns the fixed rate $\mathcal{R}_i$; D denotes dividends paid out by investors; and τ is the amount of lump-sum taxes paid by the household to the government. Deposit contracts are signed before any shock is realized and allow households to get any quantity of savings remunerated at a fixed rate.³² Since investors are ex-ante identical and compete perfectly for savings, they all set the same interest rate $\mathcal{R}$ in equilibrium, such that they make zero profits in expectation. It follows that $s_{j,i} = s$, since households are identical and indifferent with respect to which investor they save with. Finally, we assume that households hold an equal ownership share of each investor, so the dividends they receive are independent of their saving choice, and are thus taken as given. Note that dividends are generally non-zero because investors as a whole always make gains or losses ex post.

Government and central bank. The government issues an amount of debt whose face value is stochastic and given by $\tilde{S}$, according to equation (2).³³ The total amount of government spending in the two periods must sum up to a fixed quantity G of units of the consumption good, so the random fraction consumed in $t = 0$ depends on debt issuance revenues according to the budget constraint $g_0 = \tilde{S} Q$, and the remaining fraction $g_1 = G - g_0$ relies on tax revenues net of central bank transfers and debt service, according to $g_1 = \tau - \tilde{S} \theta - \tau_{cb}$.

In the first period, the central bank issues reserves a_{cb} to purchase bonds b_{cb} at price Q , or to save in a safe storage technology k_{cb} , so its budget constraint is given by $a_{cb} = Q b_{cb} + k_{cb}$. Reserves and storage are assumed to offer a zero net real interest rate. In the second period, the central bank uses the proceeds from APs, storage, and government transfers to reimburse reserves, and its budget constraint is given by $\theta b_{cb} + k_{cb} + \tau_{cb} = a_{cb}$.

The second-period budget constraints of the government and the central bank can be

³¹We assume that the endowment is sufficiently large to allow liquidity to the financial sector in any state of the world.

³²This assumption is an implicit restriction on contracts that, beyond improving tractability, also captures realistic features of the banking system.

³³Abstracting from noise traders and interpreting $\tilde{S}$ directly as government debt issuance keeps the model parsimonious and avoids specifying the noise traders' problem. Our ultimate object of interest, investors' profits, is unaffected by this assumption, as shown in Section 2.7.

respectively written as

$$\tau = \tau_{\text{cb}} + G + \pi_{\text{gov}}, \quad (37)$$

and

$$-\tau_{\text{cb}} = \pi_{\text{cb}}, \quad (38)$$

where $\pi_{\text{gov}} := \tilde{S}(\theta - Q)$ and $\pi_{\text{cb}} := b_{\text{cb}}(\theta - Q)$, matching the payoff functions we use in Section 2.7. Joining these two conditions, we get the second-period budget constraint of the consolidated public sector

$$\tau - G = (\tilde{S} - b_{\text{cb}})(\theta - Q), \quad (39)$$

which states that the public sector surplus must equal the net losses from bond issuance. In the absence of information frictions, $Q = \theta$, the government runs a balanced budget, and debt is only needed because of a mismatch between the timing of taxes and government spending.

Investors. Before observing any information, investors receive funds $s_i = \int s_{j,i} dj$ from households. After learning their private signal, each investor i allocates their funds between central bank reserves a_i and government debt b_i , according to $s_i = a_i + Q b_i$, taking the bond price Q and the return on savings $\mathcal{R}$ as given. Their objective is to maximize expected dividends conditional on their information set

$$\begin{aligned} \max_{b_i \in [0,1], d_i} \quad & \mathbb{E}[d_i \mid \Omega_i] \\ \text{s.t.} \quad & d_i = b_i(\theta - Q) - s_i(\mathcal{R} - 1) \end{aligned} \quad (40)$$

where bond position bounds and the information set Ω_i are as in Section 2.1, so that investors' bond demand follows (7).

Market clearing. The bond market clears as in (6). Central bank reserves and household savings clear according to $\int_0^1 a_i di = a_{\text{cb}}$ and $\int_0^1 s_{j,i} dj = s_i$, respectively. Goods market clearing in periods 0 and 1 requires $c_0 + g_0 + k_{\text{cb}} = y$ and $c_1 + g_1 = k_{\text{cb}}$, respectively. The equilibrium remuneration of household savings is given by the investors' *ex-ante* zero-profit condition

$$\mathcal{R} = 1 + \frac{1}{s} \mathbb{E}[\pi_{\text{inv}}], \quad (41)$$

where $\mathbb{E}[\pi_{\text{inv}}]$ denotes investors' average gains in the bond market, and is the same object we characterized in Proposition 4.³⁴ Finally, dividends $D = \int d_i di$ are such that ex-post gains are rebated to—and losses recapitalized by—households, ensuring that all investors can remunerate their deposit obligations. Note that, because of (39), the exogenous component of households' second-period consumption (that is, $D - \tau$) is deterministic: both dividends and taxes depend on the gains and losses realized by the investors and the public sector in the bond market, which offset each other.

Competitive equilibrium. A competitive equilibrium in this extended model consists of two blocks. First, consumption and loan allocations for households and investors, government tax and spending policies, central bank transfer policies, and an interest rate $\mathcal{R}$ such that: the allocations solve the household problem (36) and the ex-ante investor problem; government and central bank policies satisfy the budget constraints (37) and (38); the interest rate satisfies equation (41); and the markets for goods, loans, and reserves clear. Second, a generic AP policy, a bond price function Q , and investors' portfolio allocations and posterior beliefs that form a Perfect Bayesian Equilibrium of the financial market according to Definition 2.

Efficiency. The Euler equation for the representative household is given by

$$u'(c_0) = \mathcal{R} u'(c_1), \quad (42)$$

taking dividends D and lump-sum taxes τ as given. Plugging in market clearing conditions and budget constraints, equation (42) becomes

$$u'(y - s) = \mathcal{R} u'(s - G). \quad (43)$$

In contrast, the planner problem is given by $\max_s u(y - s) + u(s - G)$, which yields the first best condition

$$u'(y - s) = u'(s - G). \quad (44)$$

The key observation here is that the planner ignores the return on savings offered by investors because—in contrast to households—it internalizes the effect of aggregate savings on taxes and investors' gains. The following proposition states our main result on the welfare effects of APs.

³⁴That is, average investor profits obtain by integrating (24) with respect to the market signal distribution $\mathcal{M}_b$. The analytical derivation can be found in Appendix A.4.

Proposition 6. *Household welfare is increasing in the central bank quantity-target b insofar as*

$$\mathcal{R} > 1 \quad \Leftrightarrow \quad \mathbb{E}[\pi_{inv}] > 0. \quad (45)$$

Proof. The equivalence in condition (45) follows from (41). Comparing equation (44) with (43), one can see that the competitive equilibrium is not efficient when average investor gains are positive. Proposition 4 then proves that such gains are decreasing in the AP quantity target. ■

The key takeaway from the proposition is that the very same assumptions—limited arbitrage and dispersed information—that make APs non-neutral in our model also provide a microfoundation for the rationale behind central bank interventions, without resorting to additional frictions such as price rigidities or inequality motives. In particular, policy has a role because households fail to internalize that investor gains in the financial market lead to higher taxes, thus saving too much relative to the social optimum. Proposition 6 states that APs reduce these expected returns in the financial market and thus increase welfare.

In this setting, the optimal asset purchase policy would be the one inducing the largest possible reduction in returns on savings, since the central bank bears no cost from conducting APs or transferring resources directly to the treasury. Introducing inflation and limits to the transfer policy between central bank and treasury would be one way of thinking about the potential inflationary costs of central bank APs and balance sheet expansion. We leave this as an interesting direction for future research.

6 Conclusion

This paper characterizes the information channel of APs: central bank intervention moves bond markets because it affects the amount of information contained by asset prices.

This channel is orthogonal to the effect of APs on term premia. It operates through investors' beliefs when they learn from prices, and it vanishes only when aggregate asset demand is additively separable in the price and fundamental—a property of the standard CARA model that position bounds naturally break. The same mechanism implies that non-contingent APs can raise average prices above fundamentals, generating expected losses for the central bank and implicit transfers to the government.

A theoretical implication of our channel is that APs can amplify, rather than dampen, the response of prices to fundamentals. We empirically document this possibility: Federal Reserve purchases strengthened the reaction of Treasury yields to labor market news.

We further show that the frictions responsible for this non-neutrality also rationalize central bank intervention: limited arbitrage and dispersed information alone make investors' expected returns inefficiently high, and APs increase welfare by reducing them.

Taken together, our results caution against treating the transmission of APs—and the demand elasticities used to evaluate it—as invariant to the informational environment in which purchases occur.

References

- Adrian, Tobias, Richard K. Crump, and Emanuel Moench**, “Pricing the term structure with linear regressions,” *Journal of Financial Economics*, 2013, *110* (1), 110–138.
- Albagli, Elias, Christian Hellwig, and Aleh Tsyvinski**, “Imperfect Financial Markets and Investment Inefficiencies,” *American Economic Review*, September 2023, *113* (9), 2323–54.
- , —, —, and —, “Information Aggregation with Asymmetric Asset Payoffs,” *The Journal of Finance*, 2024, *79* (4), 2715–2758.
- Amador, Manuel and Pierre-Olivier Weill**, “Learning from Prices: Public Communication and Welfare,” *Journal of Political Economy*, 2010, *118* (5), 866–907.
- Bassetto, Marco and Carlo Galli**, “Is Inflation Default? The Role of Information in Debt Crises,” *American Economic Review*, 2019, *109* (10), 3556–3584.
- Bhattarai, Saroj and Christopher J. Neely**, “An Analysis of the Literature on International Unconventional Monetary Policy,” *Journal of Economic Literature*, June 2022, *60* (2), 527–597.
- , **Gauti B Eggertsson, and Bulat Gafarov**, “Time Consistency and Duration of Government Debt: A Model of Quantitative Easing,” *The Review of Economic Studies*, 09 2022, *90* (4), 1759–1799.
- Candian, Giacomo, Pierre De Leo, and Luca Gemmi**, “Managing Expectations with Exchange Rate Policy,” *manuscript, HEC Montreal and University of Maryland and University of Bologna*, 2023.
- Christensen, Jens and Glenn Rudebusch**, “The Response of Interest Rates to US and UK Quantitative Easing,” *Economic Journal*, 2012, *122* (564), F385–F414.
- Cochrane, John H.**, *Asset Pricing: Revised Edition*, Princeton, NJ: Princeton University Press, 2005.
- Cole, Harold, Daniel Neuhann, and Guillermo Ordoñez**, “Asymmetric Information and Sovereign Debt: Theory Meets Mexican Data,” *Journal of Political Economy*, 2022, *130* (8), 2055–2109.
- Cole, Harold L, Daniel Neuhann, and Guillermo Ordoñez**, “Information Spillovers and Sovereign Debt: Theory Meets the Eurozone Crisis,” *The Review of Economic Studies*, 05 2024, p. rdae017.
- Cui, Wei and Vincent Sterk**, “Quantitative easing with heterogeneous agents,” *Journal of Monetary Economics*, 2021, *123* (C), 68–90.

- Cúrdia, Vasco and Michael Woodford**, “The central-bank balance sheet as an instrument of monetary policy,” *Journal of Monetary Economics*, January 2011, 58 (1), 54–79.
- Diamond, Douglas W. and Robert E. Verrecchia**, “Constraints on Short-Selling and Asset Price Adjustment to Private Information,” *Journal of Financial Economics*, 1987, 18 (2), 277–311.
- Fontanier, Paul**, “Sovereign Bond Purchases and Rollover Crises,” *manuscript, Yale School of Management*, 2023.
- Gabaix, Xavier and Ralph S.J. Koijen**, “In Search of the Origins of Financial Fluctuations: The Inelastic Markets Hypothesis,” *NBER Working Paper*, 2021, (28967).
- Gaballo, Gaetano**, “Price Dispersion, Private Uncertainty, and Endogenous Nominal Rigidities,” *The Review of Economic Studies*, April 2018, 85 (2), 1070–1110.
- Gagnon, Joseph E., Matthew Raskin, Julie Remache, and Brian P. Sack**, “Large-scale asset purchases by the Federal Reserve: did they work?,” *Economic Policy Review*, 2011, 17 (May), 41–59.
- Geanakoplos, John**, “Liquidity, Default, and Crashes: Endogenous Contracts in General Equilibrium,” in Mathias Dewatripont, Lars Peter Hansen, and Stephen J. Turnovsky, eds., *Advances in Economics and Econometrics: Theory and Applications, Eighth World Congress*, Vol. 2, Cambridge University Press, 2003, pp. 170–205.
- , “The Leverage Cycle,” *NBER Macroeconomics Annual*, 2010, 24 (1), 1–65.
- Gertler, Mark and Peter Karadi**, “Monetary Policy Surprises, Credit Costs, and Economic Activity,” *American Economic Journal: Macroeconomics*, 2015, 7 (1), 44–76.
- Glosten, Lawrence R. and Paul R. Milgrom**, “Bid, ask and transaction prices in a specialist market with heterogeneously informed traders,” *Journal of Financial Economics*, 1985, 14 (1), 71–100.
- Goldstein, Itay and Alexander Guembel**, “Manipulation and the Allocational Role of Prices,” *Review of Economic Studies*, 2008, 75 (1), 133–164.
- Greenwood, Robin and Dimitri Vayanos**, “Bond Supply and Excess Bond Returns,” *Review of Financial Studies*, 2014, 27 (3), 663–713.
- Gromb, Denis and Dimitri Vayanos**, “Equilibrium and Welfare in Markets with Constrained Arbitrageurs,” *Journal of Financial Economics*, 2002, 66 (2–3), 361–407.
- and —, “Limits of Arbitrage: The State of the Theory,” *Annual Review of Financial Economics*, 2010, 2, 251–275.
- Grossman, Sanford J.**, “On the Efficiency of Competitive Stock Markets Where Trades Have Diverse Information,” *Journal of Finance*, 1976, 31 (2), 573–585.
- and **Joseph E. Stiglitz**, “On the Impossibility of Informationally Efficient Markets,” *American Economic Review*, 1980, 70 (3), 393–408.
- Gürkaynak, Refet S., Brian Sack, and Eric Swanson**, “Do Actions Speak Louder Than Words? The Response of Asset Prices to Monetary Policy Actions and Statements,” *International Journal of Central Banking*, 2005, 1 (1), 55–93.
- Harrison, J. Michael and David M. Kreps**, “Speculative Investor Behavior in a Stock Market with Heterogeneous Expectations,” *Quarterly Journal of Economics*, 1978, 92 (2), 323–336.

- Hellwig, Christian, Arijit Mukherji, and Aleh Tsyvinski**, “Self-Fulfilling Currency Crises: The Role of Interest Rates,” *The American Economic Review*, 2006, *96* (5), 1769–1787.
- Hellwig, Martin F.**, “On the Aggregation of Information in Competitive Markets,” *Journal of Economic Theory*, 1980, *22* (3), 477–498.
- Hong, Harrison, José A. Scheinkman, and Wei Xiong**, “Asset Float and Speculative Bubbles,” *Journal of Finance*, 2006, *61* (3), 1073–1117.
- Iovino, Luigi and Dmitriy Sergeyev**, “Central Bank Balance Sheet Policies Without Rational Expectations,” *The Review of Economic Studies*, 02 2023, *90* (6), 3119–3152.
- Jeanne, Olivier and Lars E. O. Svensson**, “Credible Commitment to Optimal Escape from a Liquidity Trap: The Role of the Balance Sheet of an Independent Central Bank,” *American Economic Review*, March 2007, *97* (1), 474–490.
- Jiang, Zhengyang, Robert J. Richmond, and Tony Zhang**, “Understanding the Strength of the Dollar,” *Journal of Financial Economics*, 2024. Accepted.
- Judd, Kenneth L.**, “The Law of Large Numbers With a Continuum of IID Random Variables,” *Journal of Economic Theory*, 1985, *35*, 19–25.
- Koijen, Ralph S.J. and Motohiro Yogo**, “A Demand System Approach to Asset Pricing,” *Journal of Political Economy*, 2019, *127* (4), 1475–1515.
- , **François Koulischer, Benoît Nguyen, and Motohiro Yogo**, “Inspecting the mechanism of quantitative easing in the euro area,” *Journal of Financial Economics*, 2021, *140* (1), 1–20.
- Krishnamurthy, Arvind and Annette Vissing-Jorgensen**, “The Effects of Quantitative Easing on Interest Rates: Channels and Implications for Policy,” *Brookings Papers on Economic Activity*, 2011, *42* (2 (Fall)), 215–287.
- Kyle, Albert S.**, “Continuous Auctions and Insider Trading,” *Econometrica*, 1985, *53* (6), 1315–1335.
- Lim, Bryan Y.**, “Short-Sale Constraints and Price Bubbles,” *Journal of Banking and Finance*, 2011, *35* (9), 2443–2453.
- Marin, Jose M. and Jacques P. Olivier**, “The Dog That Did Not Bark: Insider Trading and Crashes,” *Journal of Finance*, 2008, *63* (5), 2429–2476.
- Miller, Edward M.**, “Risk, Uncertainty, and Divergence of Opinion,” *Journal of Finance*, 1977, *32* (4), 1151–1168.
- Mussa, Michael**, *The Role of Official Intervention*, Group of Thirty New York, 1981.
- Nakamura, Emi and Jón Steinsson**, “High-Frequency Identification of Monetary Non-Neutrality: The Information Effect,” *The Quarterly Journal of Economics*, 2018, *133* (3), 1283–1330.
- Ottaviani, Marco and Peter Norman Sørensen**, “Price Reaction to Information with Heterogeneous Beliefs and Wealth Effects: Underreaction, Momentum, and Reversal,” 2015, *105* (1), 1–34.
- Palfrey, Thomas R.**, “Uncertainty Resolution, Private Information Aggregation and the Cournot Competitive Limit,” *The Review of Economic Studies*, January 1985, *52* (1), 69–83.
- Pálvölgyi, Dömötör and Gyuri Venter**, “Multiple Equilibria in Noisy Rational Expectations

Economies,” <http://dx.doi.org/10.2139/ssrn.2524105> 2015.

Pavan, Alessandro, Savitar Sundaresan, and Xavier Vives, “(In)efficiency in Information Acquisition and Aggregation through Prices,” Working Paper, IESE Business School 2026. Revision and resubmission at The Review of Economic Studies.

Ray, Walker, Michael Droste, and Yuriy Gorodnichenko, “Unbundling Quantitative Easing: Taking a Cue from Treasury Auctions,” *Journal of Political Economy*, 2024, 132 (9), 3115–3172.

Shleifer, Andrei and Robert W. Vishny, “The Limits of Arbitrage,” *Journal of Finance*, 1997, 52 (1), 35–55.

Song, Zhaogang and Haoxiang Zhu, “QE auctions of Treasury bonds,” *Journal of Financial Economics*, 2018, 128 (1), 103–124.

Vayanos, Dimitri and Jean-Luc Vila, “A Preferred-Habitat Model of the Term Structure of Interest Rates,” *Econometrica*, January 2021, 89 (1), 77–112.

Vives, Xavier, “Aggregation of Information in Large Cournot Markets,” *Econometrica*, July 1988, 56 (4), 851–876.

—, *Information and Learning in Markets: The Impact of Market Microstructure*, Princeton, NJ: Princeton University Press, 2010.

Wallace, Neil, “A Hybrid Fiat-Commodity Monetary System,” *Journal of Economic Theory*, 1981, 25 (3), 421–430.

Young, Alwyn, “Channeling Fisher: Randomization Tests and the Statistical Insignificance of Seemingly Significant Experimental Results,” *The Quarterly Journal of Economics*, 2019, 134 (2), 557–598.

Appendix

A Proofs and Derivations for Section 2

Before starting, let us introduce a few objects that will be useful. Let $\ell(x_m) := \log \frac{f_{X_m|\Theta}(x_m|\theta_H)}{f_{X_m|\Theta}(x_m|\theta_L)}$ denote the price signal log-likelihood ratio, and $\ell_0 := \log \frac{\gamma}{1-\gamma}$ denote the prior log-odds ratio. In the non-revealing region $x_m \in [\bar{x}(b), \infty)$ we have

$$\ell(x_m) = \frac{\theta_H - \theta_L}{\sigma_x^2} (x_m - x^\dagger). \quad (\text{A.1})$$

A.1 Proof of Proposition 1 (Single crossing)

Using equation (12) for $\hat{p}(x_m)$ and (13) and (14) for $p(x_m)$, we can write the public and market posterior odds ratio as

$$\frac{\hat{p}(x_m)}{1 - \hat{p}(x_m)} = \exp\{\ell_0 + \ell(x_m)\}, \quad \frac{p(x_m)}{1 - p(x_m)} = \exp\{\ell_0 + 2\ell(x_m)\}.$$

Therefore,

$$\frac{p(x_m)/(1 - p(x_m))}{\hat{p}(x_m)/(1 - \hat{p}(x_m))} = \exp\{\ell(x_m)\}.$$

Since posterior probabilities are increasing in posterior odds,

$$p(x_m) > \hat{p}(x_m) \iff \ell(x_m) > 0 \iff x_m > x^\dagger.$$

Finally,

$$Q(p(x_m)) - Q(\hat{p}(x_m)) = (\theta_H - \theta_L)[p(x_m) - \hat{p}(x_m)],$$

so the same crossing applies to the market price and the fundamental valuation. For $x_m \in [\underline{x}(b), \bar{x}(b))$, both posteriors equal zero, hence both prices equal θ_L .

A.2 Proof of Proposition 2 (Average effects of APs)

A.2.1 Part (i): Price Informativeness

Since $H_b(\gamma)$ does not depend on b , we have $\frac{\partial I}{\partial b} = -\frac{\partial}{\partial b}H(\Theta | Q)$. Differentiate $H(\Theta | Q)$ with respect to b

$$\frac{\partial H(\Theta | Q)}{\partial b} = \frac{1}{1 - b}H(\Theta | Q) - H_b(\hat{p}(\bar{x}(b))) \frac{f_{\mathcal{M}_b}(\bar{x}(b))}{1 - b} \frac{\partial \bar{x}(b)}{\partial b}. \quad (\text{A.2})$$

The moving boundary is $\bar{x}(b) = \theta_H - \sigma_x \Phi^{-1}(1 - b)$, and $\partial \bar{x}(b)/\partial b = \sigma_x / \phi\left(\frac{\bar{x}(b) - \theta_H}{\sigma_x}\right) > 0$. Using the market signal density in the non-revealing region we have

$$f_{\mathcal{M}_b}(\bar{x}(b)) \frac{\partial \bar{x}(b)}{\partial b} = \frac{\gamma}{\hat{p}(\bar{x}(b))}, \quad (\text{A.3})$$

so we can rewrite (A.2) as

$$\frac{\partial H(\Theta | Q)}{\partial b} = \frac{1}{1 - b} (H(\Theta | Q) - \eta(\hat{p}(\bar{x}(b)))) \quad (\text{A.4})$$

where $\eta(p) = \frac{\gamma}{p} H_b(p)$. Since η is strictly decreasing ($\eta'(p) = \frac{\gamma \log(1-p)}{p^2} < 0$), we have

$$\begin{aligned} H(\Theta | Q) &= \frac{1}{1-b} \int_{\bar{x}(b)}^{\infty} H_b(\hat{p}(x_m)) f_{\mathcal{M}_b}(x_m) dx_m \\ &= \frac{1}{1-b} \int_{\bar{x}(b)}^{\infty} \eta(\hat{p}(x_m)) \frac{1}{\sigma_x} \phi\left(\frac{x_m - \theta_H}{\sigma_x}\right) dx_m \\ &< \eta(\hat{p}(\bar{x}(b))). \end{aligned}$$

It follows that $\partial H(\Theta | Q)/\partial b < 0$ and therefore $\partial I(Q; \Theta)(b)/\partial b > 0$.

A.2.2 Part (ii): Price Sensitivity

Write (20) as

$$\mathcal{S}(b) = \frac{1}{1-b} \left[\int_{\bar{x}(b)}^{\infty} p(x_m) \phi_H(x_m) dx_m - \int_{\bar{x}(b)}^{\infty} p(x_m) \phi_L(x_m) dx_m \right].$$

Apply the change of variable $x_m = \theta + \sigma_x z$ for $\theta \in \{\theta_H, \theta_L\}$

$$\mathcal{S}(b) = \frac{1}{1-b} \left[\int_{\Phi^{-1}(b)}^{\infty} p(\theta_H + \sigma_x z) \phi(z) dz - \int_{\Phi^{-1}(b) + (\theta_H - \theta_L)/\sigma_x}^{\infty} p(\theta_L + \sigma_x z) \phi(z) dz \right]$$

which is strictly positive because p is strictly increasing.

Differentiating with respect to b gives

$$\frac{\partial \mathcal{S}}{\partial b} = \frac{1}{1-b} \left[\mathcal{S}(b) - p(\bar{x}(b)) \left(1 - \frac{\phi_L(\bar{x}(b))}{\phi_H(\bar{x}(b))} \right) \right] \quad (\text{A.5})$$

where note that $\frac{\phi(\Phi^{-1}(b) + (\theta_H - \theta_L)/\sigma_x)}{\phi(\Phi^{-1}(b))} = \frac{\phi_L(\bar{x}(b))}{\phi_H(\bar{x}(b))}$.

If $\bar{x}(b) \leq x^\dagger$, then $\phi_L(\bar{x}(b)) \geq \phi_H(\bar{x}(b))$, so the whole square bracket in (A.5) is strictly positive.

If instead $\bar{x}(b) \geq x^\dagger$, note we can rewrite $\mathcal{S}(b)$ as

$$\mathcal{S}(b) = \frac{1}{1-b} \int_{\bar{x}(b)}^{\infty} p(x_m) \left(1 - \frac{\phi_L(x_m)}{\phi_H(x_m)} \right) \phi_H(x_m) dx_m,$$

and $p(x_m) \left(1 - \frac{\phi_L(x_m)}{\phi_H(x_m)} \right) \geq p(\bar{x}(b)) \left(1 - \frac{\phi_L(\bar{x}(b))}{\phi_H(\bar{x}(b))} \right)$ for all $x_m \geq \bar{x}(b)$. Therefore

$$\mathcal{S}(b) > p(\bar{x}(b)) \left(1 - \frac{\phi_L(\bar{x}(b))}{\phi_H(\bar{x}(b))} \right),$$

and (A.5) again gives $\partial \mathcal{S} / \partial b > 0$.

A.2.3 Part (iii): Average Price and Valuation

Derivation of equations (22)-(23). First, we derive (22). Since $\hat{p}(x_m) = 0$ on the revealing region $[\underline{x}(b), \bar{x}(b))$, and since on $[\bar{x}(b), \infty)$

$$\hat{p}(x_m) f_{\mathcal{M}_b}(x_m) = \gamma \frac{1}{\sigma_x} \phi\left(\frac{x_m - \theta_H}{\sigma_x}\right),$$

the average fundamental valuation is

$$\begin{aligned} \mathbb{E}[Q(\hat{p}(x_m))] &= \theta_L + \frac{\theta_H - \theta_L}{1 - b} \int_{\bar{x}(b)}^{+\infty} \hat{p}(x_m) f_{\mathcal{M}_b}(x_m) dx_m \\ &= \theta_L + \gamma \frac{\theta_H - \theta_L}{1 - b} \int_{\bar{x}(b)}^{+\infty} \frac{1}{\sigma_x} \phi\left(\frac{x_m - \theta_H}{\sigma_x}\right) dx_m \\ &= \theta_L + \gamma(\theta_H - \theta_L) = \mathbb{E}[\theta], \end{aligned}$$

where the last equality uses $\bar{x}(b) = \theta_H + \sigma_x \Phi^{-1}(b)$.

Second, since $p(x_m) = \hat{p}(x_m) = 0$ on $[\underline{x}(b), \bar{x}(b))$, we have

$$\begin{aligned} \mathbb{E}[Q(p(x_m))] &= \mathbb{E}[Q(\hat{p}(x_m))] + \frac{\theta_H - \theta_L}{1 - b} \int_{\bar{x}(b)}^{+\infty} (p(x_m) - \hat{p}(x_m)) f_{\mathcal{M}_b}(x_m) dx_m \\ &= \mathbb{E}[\theta] + W(b), \end{aligned}$$

which gives (21) and (23).

Average wedge without APs. We now prove that $W(0) \geq 0$ if and only if $\gamma \leq 1/2$, with equality at $\gamma = 1/2$. Since $\theta_H - \theta_L > 0$, it is enough to sign

$$\int_{-\infty}^{+\infty} (p(x_m) - \hat{p}(x_m)) f_{\mathcal{M}_0}(x_m) dx_m.$$

Fix $h > 0$, let $r_h := \exp\{\ell(x^\dagger + h)\} > 1$, and $f_i(x) := \phi((x - \theta_i)/\sigma_x)$ for $i \in \{L, H\}$. By symmetry of the Gaussian likelihoods around $x^\dagger$, there is a constant $c_h > 0$ such that

$$f_H(x^\dagger + h) = r_h c_h, \quad f_L(x^\dagger + h) = c_h, \quad f_H(x^\dagger - h) = c_h, \quad f_L(x^\dagger - h) = r_h c_h.$$

Hence

$$\widehat{p}(x^\dagger + h) = \frac{\gamma r_h}{\gamma r_h + 1 - \gamma}, \quad p(x^\dagger + h) = \frac{\gamma r_h^2}{\gamma r_h^2 + 1 - \gamma},$$

and

$$\widehat{p}(x^\dagger - h) = \frac{\gamma}{\gamma + (1 - \gamma)r_h}, \quad p(x^\dagger - h) = \frac{\gamma}{\gamma + (1 - \gamma)r_h^2}.$$

The contribution of the symmetric pair $x^\dagger + h$ and $x^\dagger - h$ to the integral is therefore

$$\begin{aligned} & \left[p(x^\dagger + h) - \widehat{p}(x^\dagger + h) \right] f_{\mathcal{M}_0}(x^\dagger + h) + \left[p(x^\dagger - h) - \widehat{p}(x^\dagger - h) \right] f_{\mathcal{M}_0}(x^\dagger - h) \\ &= \frac{c_h \gamma (1 - \gamma) r_h (r_h - 1) (r_h^2 - 1) (1 - 2\gamma)}{(\gamma r_h^2 + 1 - \gamma) (\gamma + (1 - \gamma)r_h^2)}. \end{aligned}$$

All factors except $1 - 2\gamma$ are strictly positive. Integrating over $h > 0$ gives $W(0) \geq 0$ if and only if $\gamma \leq 1/2$, with equality at $\gamma = 1/2$.

Properties of the average wedge. To rewrite the average wedge in a more convenient way, we perform the following change of variable

$$x_m = \theta_H + \sigma_x \Phi^{-1}(y),$$

which we previously defined as $\bar{x}(y)$. This implies that $dx_m = \sigma_x \frac{1}{\phi\left(\frac{x_m - \theta_H}{\sigma_x}\right)} dy$, so the expression for the average price becomes

$$\mathcal{Q}_{\text{qt}}(b) = \mathbb{E}[\theta] + \gamma(\theta_H - \theta_L) \left(\frac{1}{1 - b} \int_b^1 \frac{p(\bar{x}(y))}{\widehat{p}(\bar{x}(y))} dy - 1 \right)$$

and we define the following convenient linear transformation of the average wedge

$$\widetilde{W}(b) := \frac{W(b)}{\gamma(\theta_H - \theta_L)} + 1 = \frac{1}{1 - b} \int_b^1 w(\bar{x}(y)) dy,$$

where $w(x) := \frac{p(x)}{\widehat{p}(x)}$ denotes the *conditional wedge ratio* function. We now derive some properties of the function $w(\cdot)$, which will be useful to prove the properties of the average wedge $W(\cdot)$ that are stated in the Proposition.

Lemma A.1. *The conditional wedge ratio $w(x_m) := \frac{p(x_m)}{\widehat{p}(x_m)}$*

(i) is single-peaked, has maximizer $x^ := x^\dagger + \frac{\sigma_x^2}{\theta_H - \theta_L} \log\left(\frac{1 + \sqrt{\gamma}}{\sqrt{\gamma}}\right)$, and maximum $w(x^*) = \frac{1 + \sqrt{\gamma}}{2\sqrt{\gamma}}$.*

(ii) is increasing (decreasing) in x_m to the left (right) of x^* , and is such that $\lim_{x \rightarrow -\infty} w(x) = 0$ and $\lim_{x \rightarrow +\infty} w(x) = 1$.

Proof. On the non-revealing region, public and marginal-investor posterior odds are

$$\frac{\hat{p}(x)}{1 - \hat{p}(x)} = \frac{\gamma}{1 - \gamma} e^{\ell(x)}, \quad \frac{p(x)}{1 - p(x)} = \frac{\gamma}{1 - \gamma} e^{2\ell(x)}.$$

Therefore, setting $r = e^{\ell(x)}$,

$$w(x) = \frac{p(x)}{\hat{p}(x)} = \frac{r(\gamma r + 1 - \gamma)}{\gamma r^2 + 1 - \gamma}.$$

Since $\ell'(x) = (\theta_H - \theta_L)/\sigma_x^2 > 0$, the sign of $w'(x)$ is the sign of

$$\frac{\partial w}{\partial \ell} = \frac{(1 - \gamma)r[(1 - \gamma) + 2\gamma r - \gamma r^2]}{(\gamma r^2 + 1 - \gamma)^2}.$$

Thus $w'(x) \geq 0$ if and only if

$$r \leq \frac{1 + \sqrt{\gamma}}{\sqrt{\gamma}},$$

which gives the stated maximizer x^* . Evaluating w at this value gives

$$w(x^*) = \frac{1 + \sqrt{\gamma}}{2\sqrt{\gamma}}.$$

The limits follow directly from the displayed formula for w : as $x \rightarrow -\infty$, $r \rightarrow 0$, so $w(x) \rightarrow 0$; as $x \rightarrow +\infty$, $r \rightarrow +\infty$, so $w(x) \rightarrow 1$. ■

We now return to the proof of the proposition, where we will use the results of the lemma.

Limit of $W(\mathbf{b})$ for $\mathbf{b} \rightarrow 1$. Using L'Hôpital's rule,

$$\lim_{\mathbf{b} \rightarrow 1} \widetilde{W}(\mathbf{b}) = \lim_{\mathbf{b} \rightarrow 1} w(\bar{x}(\mathbf{b})) = 1,$$

where the last equality follows from Lemma A.1. Hence $\lim_{\mathbf{b} \rightarrow 1} W(\mathbf{b}) = 0$.

Single-peakedness of $W(\mathbf{b})$. The derivative of the (transformed) average wedge with respect to APs is given by

$$\widetilde{W}'(\mathbf{b}) = \frac{1}{1 - \mathbf{b}} (\widetilde{W}(\mathbf{b}) - v(\mathbf{b})) \tag{A.6}$$

where $v(b) = w(\bar{x}(b))$.

Since $\bar{x}(y)$ is increasing, Lemma A.1 implies that v is single-peaked, with maximizer $b(x^*) = \Phi((x^* - \theta_H)/\sigma_x)$.

If $b \leq b(x^\dagger)$, then $v(b)$ is the minimum value of v on $[b, 1]$, so $\widetilde{W}(b) > v(b)$ and $\widetilde{W}'(b) > 0$. If $b \geq b(x^*)$, then $v(b)$ is the maximum value of v on $[b, 1]$, so $\widetilde{W}(b) < v(b)$ and $\widetilde{W}'(b) < 0$.

It remains to consider $b \in (b(x^\dagger), b(x^*))$. At any stationary point, $\widetilde{W}'(b) = 0$, so $\widetilde{W}(b) = v(b)$. Differentiating once more gives

$$\widetilde{W}''(b) = \frac{1}{1-b} [2\widetilde{W}'(b) - v'(b)].$$

At a stationary point in this interval, $v'(b) > 0$, hence $\widetilde{W}''(b) < 0$. Therefore every stationary point is a strict local maximum. Since $\widetilde{W}' > 0$ at $b(x^\dagger)$ and $\widetilde{W}' < 0$ at $b(x^*)$, there is exactly one stationary point, and it is the unique maximizer.

Finally, the result for the sign of $W(b)$ follows from the sign of $W(0)$ proved above, continuity, $\widetilde{W}(b(x^\dagger)) > 1$, and the fact that $\widetilde{W}(b) > 1$ for all $b \in [b(x^*), 1)$. Thus $W(b)$ is positive for all $b \geq \tilde{b}(\gamma)$, with $\tilde{b}(\gamma) = 0$ when $\gamma \leq 1/2$ and $\tilde{b}(\gamma) \in (0, b(x^\dagger))$ otherwise.

A.3 Proof of Proposition 3 (Conditional gains and losses)

Note that the proposition focuses on the non-revealing region $x_m \in [\bar{x}(b), +\infty)$, because otherwise the fundamental is fully revealed, $Q(p(x_m)) = \theta$, and gains and losses are zero.

Investors. Since $\mathbb{E}[\theta - Q(p(x_m)) | x_i, x_m]$ is increasing in x_i , equals zero for the investor with private signal $x_i = x_m$, and is strictly positive when $x_i > x_m$. Since investors buy exactly on this set, $\mathbb{E}[\pi_{\text{inv}} | x_m] > 0$ for every x_m in the non-revealing region.

Central bank and government. From (25) it is straightforward to see that the sign of central bank gains and government losses conditional on x_m solely depends on the sign of the conditional wedge which we characterize in Proposition 1. It follows that the central bank realizes a gain, and the government a loss, if and only if $x_m < x^\dagger$ and bonds are priced below their fundamental valuation by the market. The converse is true if $x_m > x^\dagger$.

Noise traders. Using the market clearing condition (6) we can write noise traders' gains as

$$\begin{aligned}\mathbb{E}[\pi_{\text{nt}} | x_m] &= \sum_{\theta \in \{\theta_H, \theta_L\}} (1 - b) \left(\theta - Q(p(x_m)) \right) P(\theta | x_m \sim \mathcal{M}_b) \\ &\quad - \sum_{\theta \in \{\theta_H, \theta_L\}} \left[\Phi \left(\frac{\theta - x_m}{\sigma_x} \right) \right] \left(\theta - Q(p(x_m)) \right) P(\theta | x_m \sim \mathcal{M}_b).\end{aligned}$$

The purpose of this representation is the following. The first line represents gains or losses on the bonds supplied by the government net of central bank APs, and is strictly positive whenever $x_m < x^\dagger$ and bonds are priced below their fundamental value, as was the case for central bank gains. The second line represents—as a loss—the missed gains on the bonds that are purchased by investors, and is strictly negative for all (x_m, b) because it is the flip side of investors' gains.

We now characterize when noise traders make *net* gains or losses. The expression above can be rewritten as

$$\begin{aligned}\frac{\mathbb{E}[\pi_{\text{nt}} | x_m]}{\theta_H - \theta_L} &= \hat{p}(x_m)(1 - p(x_m)) \left[1 - b - \Phi \left(\frac{\theta_H - x_m}{\sigma_x} \right) \right] \\ &\quad - p(x_m)(1 - \hat{p}(x_m)) \left[1 - b - \Phi \left(\frac{\theta_L - x_m}{\sigma_x} \right) \right].\end{aligned}$$

If $1 - b - \Phi((\theta_H - x_m)/\sigma_x) = 0$, the right-hand side of the equation above is strictly negative. If instead $1 - b - \Phi((\theta_H - x_m)/\sigma_x) > 0$, noise trader gains are negative if and only if

$$\frac{1 - b - \Phi \left(\frac{\theta_L - x_m}{\sigma_x} \right)}{1 - b - \Phi \left(\frac{\theta_H - x_m}{\sigma_x} \right)} > \frac{\hat{p}(x_m)(1 - p(x_m))}{p(x_m)(1 - \hat{p}(x_m))}. \quad (\text{A.7})$$

Since $x_m \geq \bar{x}(b)$, we have $1 - b \geq \Phi((\theta_H - x_m)/\sigma_x)$, so

$$\frac{1 - b - \Phi \left(\frac{\theta_L - x_m}{\sigma_x} \right)}{1 - b - \Phi \left(\frac{\theta_H - x_m}{\sigma_x} \right)} \geq \frac{1 - \Phi \left(\frac{\theta_L - x_m}{\sigma_x} \right)}{1 - \Phi \left(\frac{\theta_H - x_m}{\sigma_x} \right)}.$$

The last ratio is strictly larger than the right-hand side of (A.7). To see this, use the posterior odds-ratio to note that

$$\frac{\hat{p}(x_m)(1 - p(x_m))}{p(x_m)(1 - \hat{p}(x_m))} = e^{-\ell(x_m)}.$$

Next, we can write

$$1 - \Phi\left(\frac{\theta_L - x_m}{\sigma_x}\right) = \int_{(\theta_H - x_m)/\sigma_x}^{\infty} \phi\left(z - \frac{\theta_H - \theta_L}{\sigma_x}\right) dz > e^{-\ell(x_m)} \int_{(\theta_H - x_m)/\sigma_x}^{\infty} \phi(z) dz,$$

because, for every $z > (\theta_H - x_m)/\sigma_x$,

$$\frac{\phi\left(z - \frac{\theta_H - \theta_L}{\sigma_x}\right)}{\phi(z)} > e^{-\ell(x_m)}.$$

Hence $\mathbb{E}[\pi_{\text{nt}} | x_m] < 0$ for every x_m in the non-revealing region.

A.4 Proof of Proposition 4 (Average gains and losses)

Conditional on $\tilde{S} \geq b$ and the market being active, average gains for $j \in \{\text{inv}, \text{cb}, \text{nt}, \text{gov}\}$ are

$$\mathbb{E}[\pi_j] = \int_{\bar{x}(b)}^{\infty} \mathbb{E}[\pi_j | x_m] \frac{f_{\mathcal{M}_b}(x_m)}{1 - b} dx_m, \quad (\text{A.8})$$

where the integration interval is the non-revealing region $[\bar{x}(b), +\infty)$, because gains and losses in the fully revealing region $[\underline{x}(b), \bar{x}(b))$ are zero for all players.

For the central bank and the government, averaging the conditional payoff definitions over the active-market distribution gives

$$\mathbb{E}[\pi_{\text{gov}}] = W(b), \quad \mathbb{E}[\pi_{\text{cb}}] = -b W(b),$$

and the sign statements thus follow from the same properties of $W(b)$ as in Proposition 2.

With respect to noise traders, since Proposition 3 gives $\mathbb{E}[\pi_{\text{nt}} | x_m] < 0$ for every non-revealing signal and zero gains in the fully revealing region, average noise-trader gains conditional on an active market are strictly negative.

For investors, Proposition 3 shows that $\mathbb{E}[\pi_{\text{inv}} | x_m] > 0$ for all $x_m \in [\bar{x}(b), +\infty)$, which implies that investors' average gains are strictly positive. To show they are decreasing in b , let $y = \Phi((\theta_H - x_m)/\sigma_x)$, so that $x_m = \theta_H - \sigma_x \Phi^{-1}(y)$ and $y \in [0, 1 - b]$ in the non-revealing region of x_m . Using the posterior definitions to simplify the low-state term, investor gains can be written as

$$\begin{aligned} \mathbb{E}[\pi_{\text{inv}}] &= \frac{\gamma(\theta_H - \theta_L)}{1 - b} \int_0^{1-b} \left(1 - p(\theta_H - \sigma_x \Phi^{-1}(y))\right) \times \\ &\quad \times \left[y - \Phi\left(\Phi^{-1}(y) - \frac{\theta_H - \theta_L}{\sigma_x}\right) \frac{\phi(\Phi^{-1}(y))}{\phi\left(\Phi^{-1}(y) - \frac{\theta_H - \theta_L}{\sigma_x}\right)} \right] dy. \end{aligned} \quad (\text{A.9})$$

The integrand is positive and strictly increasing in y : the first factor is increasing because $p(x_m)$ is increasing in x_m , while $x_m = \theta_H - \sigma_x \Phi^{-1}(y)$ is decreasing in y ; and the term in square brackets has limit zero as $y \downarrow 0$ and is increasing because, writing $u = \Phi^{-1}(y)$, its derivative with respect to u is

$$\frac{\theta_H - \theta_L}{\sigma_x} \Phi\left(u - \frac{\theta_H - \theta_L}{\sigma_x}\right) \frac{\phi(u)}{\phi\left(u - \frac{\theta_H - \theta_L}{\sigma_x}\right)} > 0.$$

Denote the integrand above by $g(y)$. Differentiating the normalized integral gives

$$\frac{\partial}{\partial b} \mathbb{E}[\pi_{\text{inv}}] = \frac{\gamma(\theta_H - \theta_L)}{(1 - b)^2} \left[\int_0^{1-b} g(y) dy - (1 - b)g(1 - b) \right] < 0,$$

because g is strictly increasing, so its average on $[0, 1 - b]$ is strictly below its right-endpoint value.

To see that average gains converge to zero as $b \rightarrow 1$, note first that the government and central-bank formulas below imply convergence for those two players because $W(b) \rightarrow 0$ by Proposition 2. For investors, from equation (A.9) we can see that the integrand $g(y) \rightarrow 0$ as $y \downarrow 0$, so the whole integral converges to zero as $b \rightarrow 1$. Noise trader gains then converge to zero by the zero-sum identity.

A.5 Microfoundation of Q_{pas} when $\tilde{S} < b$

Here we provide explicit microfoundations for the assumption we make in Section 2.1 that $Q_{\text{pas}} = \theta$ when the market is passive.

We assume there exists a small disturbance in the bond market, such that there is always an infinitesimal residual quantity of bonds traded by investors, even when the central bank quantity target exceeds supply. Formally, net supply is

$$S = \epsilon \quad \text{if} \quad \tilde{S} < b,$$

where ϵ is a random variable uniformly distributed in $[0, \bar{\epsilon}]$, with $\bar{\epsilon}$ arbitrarily small.

First, notice that investors correctly recognize states where $\tilde{S} < b$, because they observe central bank purchases below the quantity target ($b_{\text{cb}} < b$). In these states, x_m has the same distribution that it would have when the central bank has a target $b = 1 - \bar{\epsilon}$ in the main text, as discussed in the paragraph around equation (11).

Proposition A.1 (Q_{pas} microfoundation). *In the residual market where net supply is $S \sim$*

$U[0, \bar{\epsilon}]$, the equilibrium price $Q = Q(p(x_m))$ satisfies, for each $\theta \in \{\theta_L, \theta_H\}$,

$$\mathbb{E}[Q \mid \theta] \longrightarrow \theta \quad \text{as } \bar{\epsilon} \rightarrow 0.$$

Proof. The price signal $x_m(\theta, S)$ and its bounds $\underline{x}, \bar{x}$ are given by (10) in the main text and the surrounding discussion, with the difference that here $S \sim U[0, \bar{\epsilon}]$. Let $\delta := (\theta_H - \theta_L)/\sigma_x > 0$ and $a := \Phi^{-1}(1 - \bar{\epsilon})$ for notational convenience, so $a \rightarrow \infty$ as $\bar{\epsilon} \rightarrow 0$. Let $\mathcal{R} := [\underline{x}(\bar{\epsilon}), \bar{x}(\bar{\epsilon})] = [\theta_L + \sigma_x a, \theta_H + \sigma_x a]$ denote the revealing region.

We start by proving the proposition conditional on θ_L . First, we show that the probability mass of the revealing region converges to one.

The conditional probability of the non-revealing region is

$$P(x_m \geq \bar{x}(\bar{\epsilon}) \mid \theta_L) = \frac{1 - \Phi(a + \delta)}{1 - \Phi(a)}.$$

Using the Gaussian tail bounds $\frac{a}{1+a^2}\phi(a) < 1 - \Phi(a) < \frac{1}{a}\phi(a)$ and noting that $\frac{\phi(a+\delta)}{\phi(a)} = e^{-\ell(\bar{x}(\bar{\epsilon}))}$ we get

$$\frac{1 - \Phi(a + \delta)}{1 - \Phi(a)} < \frac{1 + a^2}{a(a + \delta)} e^{-\ell(\bar{x}(\bar{\epsilon}))} \rightarrow 0,$$

because $\ell(\bar{x}(\bar{\epsilon})) = a\delta + \delta^2/2 \rightarrow \infty$. It follows that $P(x_m \in \mathcal{R} \mid \theta_L) \rightarrow 1$, and the conditional expectation of the price converges to θ_L :

$$|\mathbb{E}[Q \mid \theta_L] - \theta_L| \leq (\theta_H - \theta_L)P(x_m \notin \mathcal{R} \mid \theta_L) \rightarrow 0.$$

We now prove the proposition conditional on θ_H . In this case, the support of the price signal is $[\bar{x}(\bar{\epsilon}), \infty)$, and $x_m \geq \bar{x}(\bar{\epsilon}) \rightarrow \infty$ which implies $\ell(x_m) \geq \ell(\bar{x}(\bar{\epsilon})) \rightarrow \infty$. We have

$$1 - p(x_m) = \frac{1}{1 + \exp\{\ell_0 + 2\ell(x_m)\}} \leq \exp\{-\ell_0 - 2\ell(x_m)\} \leq \exp\{-\ell_0 - 2\ell(\bar{x}(\bar{\epsilon}))\}.$$

Since $\theta_H - Q = (\theta_H - \theta_L)(1 - p(x_m))$,

$$0 \leq \theta_H - \mathbb{E}[Q \mid \theta_H] \leq (\theta_H - \theta_L) \exp\{-\ell_0 - 2\ell(\bar{x}(\bar{\epsilon}))\} \rightarrow 0.$$

We have thus shown that $\mathbb{E}[Q \mid \theta] \rightarrow \theta$ for all θ , which justifies setting $Q_{\text{pas}} = \theta$. ■

B Empirics: Robustness

This appendix collects additional details on the data and several robustness exercises related to the main findings of Table 1.

B.1 Data

All code used in our empirical analysis can be found in our replication package. The original data sources are the following: the data on yields and their risk-neutral decomposition are available at https://www.newyorkfed.org/research/data_indicators/term-premia-tabs; the data on AP flows is available at <https://www.newyorkfed.org/markets/desk-operations/treasury-securities>; the data on the news surprises from Bloomberg is proprietary and not publicly available.

B.2 Stability of the News Surprise Distributions

We test whether the distribution of standardized NFP surprises is stable over time by comparing the ZLB period, which is when most of the Fed APs took place, against the rest of our sample. Figure B.1 plots the empirical density of NFP surprises inside the ZLB period (Jan 2009-Dec 2015, 84 observations) and outside of it (25 Aug 2005-31 Dec 2008 and 01 Jan 2016-31 Dec 2019, 88 observations). We find that the two distributions are statistically indistinguishable: a Welch t -test fails to reject equal means ($p = 0.70$), a Brown-Forsythe Levene test fails to reject equal variances ($p = 0.42$), the standardized mean difference is small (Cohen's $d = 0.058$), and both the standard and permutation versions of the Kolmogorov-Smirnov test fail to reject equal distribution shapes ($p = 0.47$ and $p = 0.46$, respectively). We only perform this test for NFP surprises because the principal component for the Employment Situation indicators is estimated for each sample specification. However, comparable stability results obtain for the PC of Labor News in all estimations.

B.3 Exogeneity of AP Flows

Table B.1 shows the results of removing the Surprise $\times$ AP interaction term from our baseline specification. The coefficient on the news surprise remains large, stable, and highly significant, while the R^2 is only marginally lower than in the baseline model. This confirms that the interaction term does not capture spurious correlations with the standalone Surprise and AP variables.

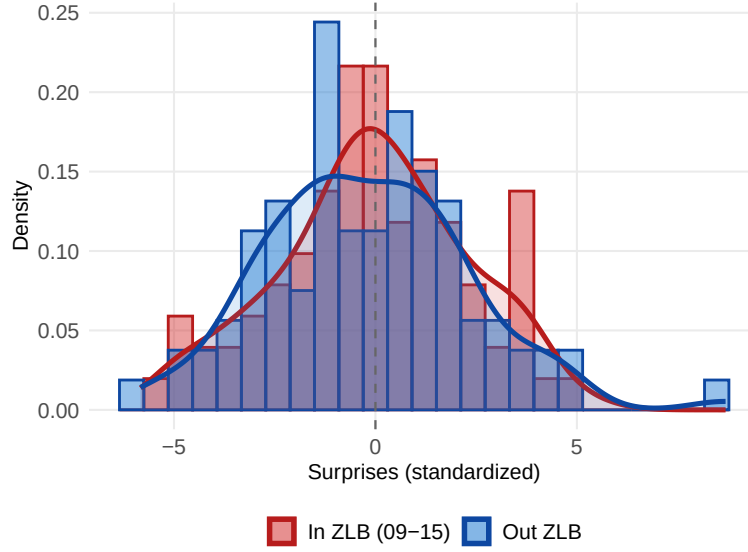


Figure B.1: Distribution of Nonfarm Payrolls surprises inside (red) and outside (blue) the ZLB period (2009–2015).

Table B.1: Robustness: no Surprise $\times$ AP interaction term.

	Nonfarm Payroll			PC Labor News		
	(1)	(2)	(3)	(4)	(5)	(6)
Surprise	0.011*** (0.003)	0.014*** (0.005)	0.010*** (0.003)	0.011*** (0.004)	0.013** (0.007)	0.012*** (0.004)
AP	-0.006** (0.003)	-0.004 (0.003)	-0.003 (0.015)	-0.007** (0.003)	-0.004 (0.003)	-0.008 (0.015)
Obs	3590	1752	172	3590	1752	172
R^2	0.091	0.155	0.158	0.086	0.140	0.143
Years	05-19	09-15	05-19	05-19	09-15	05-19
# News	172	84	172	172	84	172
Event days only	No	No	Yes	No	No	Yes
Controls	$\bar{A}$	$\bar{A}$	A	$\bar{A}$	$\bar{A}$	A
Time FE	Yr-Qtr	Yr-Qtr	No	Yr-Qtr	Yr-Qtr	No

Notes: Dependent variable: daily change in risk-neutral 3-year yields. Robust standard errors in parentheses. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table B.2 directly tests the orthogonality of AP flows to the news surprises by regressing AP flows on surprises, under the same sample splits and fixed-effects structures of the baseline regression in Table 1. Across all 6 coefficient estimates, none is statistically significant.

These results confirm that the Federal Reserve’s daily purchase decisions were orthogonal to concurrent labor market news surprises.

Table B.2: AP flows on news surprises.

	(1)	(2)	(3)	(4)	(5)	(6)
Surprise	-0.012 (0.008)	-0.016 (0.017)	-0.010 (0.008)	0.002 (0.015)	0.020 (0.033)	-0.003 (0.018)
R^2	0.317	0.304	0.009	0.316	0.304	0.170
Obs	3590	1752	172	3590	1752	172
Years	05-19	09-15	05-19	05-19	09-15	05-19
Time FE	Yr-Qtr	Yr-Qtr	No	Yr-Qtr	Yr-Qtr	No

Notes: Dependent variable: AP flows (AP_t). Robust standard errors in parentheses. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

B.4 Robustness of the Baseline Specification

Augmenting controls with prior release values. In Table B.3 we add the statistical release prior to the current one as an additional control in every specification. We control for the prior release of *all* labor market indicators (which we denote with $\bar{P}$) in the full sample columns (1), (2), (4), and (5), and we only include the prior release value of the focal indicator (P) in the event-day-only columns (3) and (6), as in the baseline specification. Including prior releases raises the R^2 by about 50%, reflecting the predictive power of past releases for future yield changes, but does not otherwise affect our results.

Looking at the 10-year maturity. Table B.4 replicates our baseline specification for changes in the 10-year risk-neutral yield, the longest maturity available in the risk-neutral decomposition data by Adrian et al. (2013). The results closely mirror the 3-year findings reported in the main text. The surprise coefficient is of similar magnitude, while the interaction coefficient is slightly smaller but remains statistically significant. The unconditional effect of AP flows is stronger and remains significant even in the restricted ZLB sample. Overall, the main findings extend to longer maturities, with somewhat stronger unconditional effects.

Lagged yield changes and AP flows. Table B.5 augments the baseline specification with two lags of the daily yield change ($\Delta y_{t-1}^{\text{rn}}$, $\Delta y_{t-2}^{\text{rn}}$), and one lag of the daily AP flow (AP_{t-1}). Despite the substantial increase in the number of regressors, the coefficient of the (contemporaneous) Surprise $\times$ AP interaction term is nearly unchanged. Among the added

Table B.3: Robustness: controls include the prior news release.

	Nonfarm Payroll			PC Labor News		
	(1)	(2)	(3)	(4)	(5)	(6)
Surprise	0.010*** (0.003)	0.017*** (0.005)	0.010*** (0.002)	0.010** (0.004)	0.015** (0.008)	0.010*** (0.004)
AP	-0.007** (0.003)	-0.004 (0.003)	-0.001 (0.009)	-0.007** (0.003)	-0.004 (0.003)	-0.006 (0.011)
Surprise $\times$ AP	0.014*** (0.005)	0.014** (0.006)	0.013** (0.005)	0.023*** (0.007)	0.026*** (0.007)	0.022*** (0.006)
Obs	3590	1752	172	3590	1752	172
R^2	0.138	0.243	0.179	0.135	0.230	0.267
Years	05-19	09-15	05-19	05-19	09-15	05-19
# News	172	84	172	172	84	172
Event days only	No	No	Yes	No	No	Yes
Controls	$\bar{A}, \bar{P}$	$\bar{A}, \bar{P}$	A, P	$\bar{A}, \bar{P}$	$\bar{A}, \bar{P}$	A, P
Time FE	Yr-Qtr	Yr-Qtr	No	Yr-Qtr	Yr-Qtr	No

Notes: Dependent variable: daily change in risk-neutral 3-year yields. Robust standard errors in parentheses. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table B.4: Robustness: risk-neutral yields for the 10-year maturity.

	Nonfarm Payroll			PC Labor News		
	(1)	(2)	(3)	(4)	(5)	(6)
Surprise	0.009*** (0.003)	0.013*** (0.004)	0.009*** (0.002)	0.010*** (0.003)	0.015*** (0.005)	0.011*** (0.004)
AP	-0.006** (0.003)	-0.004* (0.002)	-0.002 (0.009)	-0.007*** (0.003)	-0.005** (0.002)	-0.006 (0.009)
Surprise $\times$ AP	0.011** (0.004)	0.012*** (0.004)	0.010** (0.004)	0.018*** (0.005)	0.024*** (0.005)	0.019*** (0.005)
Obs	3590	1752	172	3590	1752	172
R^2	0.099	0.169	0.204	0.094	0.156	0.201
Years	05-19	09-15	05-19	05-19	09-15	05-19
# News	172	84	172	172	84	172
Event days only	No	No	Yes	No	No	Yes
Controls	$\bar{A}$	$\bar{A}$	A	$\bar{A}$	$\bar{A}$	A
Time FE	Yr-Qtr	Yr-Qtr	No	Yr-Qtr	Yr-Qtr	No

Notes: Dependent variable: daily change in risk-neutral 10-year yields. Robust standard errors in parentheses. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

lagged regressors, some are significant, suggesting that allowing for an autoregressive structure seems useful to explain persistence in the data; however, controlling for this persistence leaves the contemporaneous interaction broadly unaffected.

Table B.5: Robustness: lagged yield changes and AP flows as extra regressors.

	Nonfarm Payroll			PC Labor News		
	(1)	(2)	(3)	(4)	(5)	(6)
Surprise	0.011*** (0.003)	0.014*** (0.005)	0.010*** (0.003)	0.012*** (0.004)	0.017*** (0.006)	0.013*** (0.004)
AP	-0.006* (0.003)	-0.002 (0.003)	0.004 (0.009)	-0.006* (0.003)	-0.003 (0.003)	-0.003 (0.011)
Surprise $\times$ AP	0.013** (0.005)	0.015*** (0.005)	0.013** (0.006)	0.021*** (0.006)	0.029*** (0.006)	0.024*** (0.005)
$\Delta y_{t-1}^{\text{rn}}$	-0.048 (0.035)	0.000 (0.037)	-0.146*** (0.048)	-0.048 (0.035)	0.004 (0.037)	-0.163** (0.071)
$\Delta y_{t-2}^{\text{rn}}$	-0.064* (0.034)	-0.059** (0.029)	0.071 (0.085)	-0.062* (0.034)	-0.062** (0.030)	0.074 (0.075)
AP_{t-1}	0.001 (0.003)	0.005** (0.003)	-0.008 (0.012)	0.001 (0.003)	0.006** (0.003)	-0.001 (0.011)
Obs	3590	1752	172	3590	1752	172
R^2	0.099	0.166	0.208	0.094	0.155	0.206
Years	05-19	09-15	05-19	05-19	09-15	05-19
# News	172	84	172	172	84	172
Event days only	No	No	Yes	No	No	Yes
Controls	$\bar{A}$	$\bar{A}$	A	$\bar{A}$	$\bar{A}$	A
Time FE	Yr-Qtr	Yr-Qtr	No	Yr-Qtr	Yr-Qtr	No

Notes: Dependent variable: daily change in risk-neutral 3-year yields. Robust standard errors in parentheses. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Alternative AP timing windows. Our baseline specification uses AP flows on the same day of the news surprise as the measure of the Fed’s balance-sheet policy. Table B.6 compares two alternative timing definitions side by side: the flows on working day $t - 1$ only (columns (1)–(5)) and a four-day backward average from working day $t - 3$ to working day t (columns (1′)–(5′)). The sample split and fixed-effects structure of each column is the same as that of the corresponding column number in the baseline specification.

Using the AP flows of the prior day, the interaction term becomes smaller, remains positive, and loses significance in all four specifications, and the direct effect coefficient flips sign and loses significance. Using the four-day backward average, the interaction term becomes much smaller and is only mildly significant in the ZLB sample for PCL surprises, and the direct effect coefficient becomes smaller and loses significance across specifications. Overall, these results indicate that the direct and interacted effects of APs that we measured in our main results are tied to the simultaneous nature of AP flows.

Table B.6: Robustness: different time window for AP flows.

	Nonfarm Payroll		PC Labor News		Nonfarm Payroll		PC Labor News	
	(1)	(2)	(4)	(5)	(1')	(2')	(4')	(5')
Surprise	0.009*** (0.003)	0.010*** (0.003)	0.010*** (0.004)	0.010* (0.006)	0.009*** (0.003)	0.010*** (0.004)	0.010** (0.004)	0.010* (0.005)
AP	0.002 (0.003)	0.005* (0.003)	0.002 (0.003)	0.005* (0.003)	-0.004* (0.003)	-0.002 (0.002)	-0.004* (0.002)	-0.002 (0.002)
Surprise $\times$ AP	0.016 (0.018)	0.017 (0.015)	0.016 (0.023)	0.019 (0.022)	0.006 (0.005)	0.008* (0.004)	0.011* (0.007)	0.015** (0.006)
Obs	3590	1752	3590	1752	3590	1752	3590	1752
R^2	0.092	0.161	0.086	0.142	0.094	0.163	0.089	0.149
Years	05-19	09-15	05-19	09-15	05-19	09-15	05-19	09-15
# News	172	84	172	84	172	84	172	84
Event days only	No	No	No	No	No	No	No	No
Controls	$\bar{A}$	$\bar{A}$	$\bar{A}$	$\bar{A}$	$\bar{A}$	$\bar{A}$	$\bar{A}$	$\bar{A}$
Time FE	Yr-Qtr	Yr-Qtr	Yr-Qtr	Yr-Qtr	Yr-Qtr	Yr-Qtr	Yr-Qtr	Yr-Qtr

Notes: Dependent variable: daily change in risk-neutral 3-year yields. In columns (1)–(5), $AP = AP_{t-1}$ (APs on previous trading day); in columns (1')–(5'): $AP = \frac{1}{4} \sum_{j=0}^3 AP_{t-j}$ (APs average over 4 prior trading days, $\{t, t-1, t-2, t-3\}$). Non-operation days contribute zero in both cases. Event-day-only specifications excluded. Robust standard errors in parentheses. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Raw nominal yields. Table B.7 repeats the 3-year baseline specification using the raw nominal yield (ACMY in the labeling of Adrian et al. (2013)) in place of the risk-neutral component (ACMRNY). The AP interaction term coefficient retains its positive sign in all columns, but it shrinks considerably and loses significance in all specifications, except for PCL surprises in the ZLB sample, and especially so for the NFP measure. On the other hand, we observe a more negative and more significant unconditional effect of APs, except for the event-day-only specifications. Relative to the risk-neutral baseline specification, the reinforcement of the unconditional effect and the attenuation of the interaction mechanism of APs are consistent with the presence of a risk (or term) premium channel offsetting the informational one. Investigating this mechanism more in depth would require a structural

model that accounts for changes in the structure of macroeconomic uncertainty during the ZLB period and links these to first-order movements in asset prices. We leave this interesting question for future research.

Table B.7: Robustness: raw nominal yields.

	Nonfarm Payroll			PC Labor News		
	(1)	(2)	(3)	(4)	(5)	(6)
Surprise	0.016*** (0.003)	0.022*** (0.004)	0.016*** (0.002)	0.017*** (0.005)	0.022*** (0.007)	0.018*** (0.005)
AP	-0.010*** (0.004)	-0.009*** (0.003)	0.002 (0.007)	-0.011*** (0.004)	-0.010*** (0.004)	-0.013 (0.011)
Surprise $\times$ AP	0.003 (0.005)	0.004 (0.005)	0.001 (0.004)	0.010** (0.005)	0.018*** (0.006)	0.009** (0.005)
Obs	3590	1752	172	3590	1752	172
R^2	0.106	0.192	0.279	0.099	0.175	0.258
Years	05-19	09-15	05-19	05-19	09-15	05-19
# News	172	84	172	172	84	172
Event days only	No	No	Yes	No	No	Yes
Controls	$\bar{A}$	$\bar{A}$	A	$\bar{A}$	$\bar{A}$	A
Time FE	Yr-Qtr	Yr-Qtr	No	Yr-Qtr	Yr-Qtr	No

Notes: Dependent variable: daily change in raw nominal 3-year yields. Robust standard errors in parentheses. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Uncertainty. Table B.8 tests whether AP operations cluster in turbulent periods, when the sensitivity of yields to news may be higher for reasons unrelated to our channel. The right object is therefore a measure of how volatile the market is in the run-up to each announcement. We use two complementary measures, each predetermined and entered one at a time, as a standalone regressor and interacted with the surprise. The first measure is the standard deviation of daily, risk-neutral yield changes over the prior two weeks, lagged by one day (columns (2) and (5)), which is a function of realized volatility. The second measure is the lagged level of the VIX index (columns (3) and (6)), which is itself a forward-looking measure of expected volatility. The surprise $\times$ AP interaction coefficient ψ is unchanged across specifications (around 0.013 for NFP and 0.020–0.022 for the PC labor index), and neither surprise $\times$ volatility interaction is statistically significant. The amplification we document is therefore not an omitted volatility-regime effect. Results are similar using the contemporaneous level of the VIX index, and for one- and four-week yield volatility windows.

Table B.8: Robustness: volatility regime (realized yield volatility and VIX level) as controls and competing interaction.

	Nonfarm Payroll			PC Labor News		
	(1)	(2)	(3)	(4)	(5)	(6)
Surprise	0.011*** (0.003)	0.011*** (0.003)	0.011*** (0.003)	0.012*** (0.004)	0.013*** (0.005)	0.012*** (0.005)
AP	-0.006* (0.003)	-0.006* (0.003)	-0.006* (0.003)	-0.007** (0.003)	-0.007** (0.003)	-0.007** (0.003)
Surprise $\times$ AP	0.013*** (0.005)	0.013*** (0.005)	0.013*** (0.005)	0.022*** (0.006)	0.021*** (0.006)	0.022*** (0.006)
Yield vol_{t-1}		-0.000 (0.003)			-0.000 (0.003)	
Surprise $\times$ Yield vol_{t-1}		0.001 (0.002)			0.006 (0.007)	
VIX $_{t-1}$			0.004 (0.004)			0.004 (0.004)
Surprise $\times$ VIX $_{t-1}$			0.000 (0.004)			0.002 (0.006)
Obs	3590	3590	3590	3590	3590	3590
R^2	0.093	0.093	0.094	0.088	0.089	0.090

Notes: Dependent variable: daily change in risk-neutral 3-year yields, full sample (2005–2019), year-quarter fixed effects and $\bar{A}$ controls as in Table 1. *Yield vol_{t-1}* is the standard deviation of daily risk-neutral yield changes over the prior two weeks (10 trading days, lagged one day). *VIX $_{t-1}$* is the lagged level of the CBOE volatility index, itself a forward-looking measure of expected volatility. Columns (1) and (4) repeat the baseline; (2) and (5) add the yield-volatility level and its interaction; (3) and (6) do the same for the VIX level. Robust standard errors in parentheses. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

QE programs and announcement effects. Table B.9 isolates the Fed’s purchase programs. Columns (2) and (5) add interactions between news surprises and program-specific dummies, so the response of yields to news may differ arbitrarily across the different Fed Quantitative Easing programs: QE1 (first round), QE2 (second round), MEP (Maturity Extension Program, or Operation Twist), and QE3 (third round). The surprise-AP flows interaction ψ is then identified only from within-program variation in the timing and size of operations. Columns (3) and (6) further add a level dummy equal to one on each program’s

announcement date, following standard practice in the QE event-study literature.³⁵ Our interaction coefficient of interest ψ remains positive and significant across all columns (about 0.010 for NFP and 0.017–0.018 for the PC labor index), so the amplification is not an artifact of program-specific sensitivity or of the announcement-day price jumps that dominate the QE literature.

Two of the program-specific interactions are worth interpreting. The Surprise $\times$ QE1 coefficient is large and positive—for the PC index several times the baseline slope—so yields were far more sensitive to labor news during 2009–2010. Because no QE1 operation coincided with an Employment Situation release, this is a program-specific effect that the dummy absorbs and that ψ does not claim. It is, however, the type of correlation our channel predicts: yields should react more strongly to news when the central bank is purchasing more intensively, and QE1 was by far the most intensive program. Being a comparison across periods, it cannot be separated from other features of the 2009–2010 crisis; this is precisely why we identify the mechanism from the within-program flow interaction ψ rather than from the program dummies. The Surprise $\times$ MEP coefficient is instead negative. The Maturity Extension Program was balance-sheet neutral by design: it sold short-dated and bought long-dated Treasuries in equal amounts, leaving the overall size of the portfolio—and hence the net flow—essentially unchanged. Its effect on news sensitivity was however not neutral, because the amplification is stronger at shorter maturities (compare the 3- and 10-year estimates in Table B.4). The Fed was a net seller of 3-year bonds during the program, and since the short-maturity interaction dominates the long-maturity one, the MEP *lowered* the sensitivity of yields to news—hence the negative sign. Half of the operation announcement days fall within the MEP; reassuringly, ψ survives the inclusion of the program dummy and the two are separately identified because sale sizes vary within the program.

The announcement-dummy coefficient estimates are themselves sensible and validate the specification: risk-neutral 3-year yields fall on QE1, QE2, and QE3 announcement dates (by about 7, 5, and 2 basis points) and rise on MEP dates (by about 6 basis points), the latter as expected for a program that sold the short-to-medium sector of the yield curve.

A note on sample size. The interaction ψ is identified from the 28 announcement days on which the Fed was also operating; these cluster in QE2, the MEP, and QE3, with none in QE1. Such a small and grouped effective sample is intrinsic to high-frequency identification and is the norm in this literature, where the price effects of QE are routinely estimated from a comparable or smaller number of events (Gagnon et al. (2011), Krishnamurthy and Vissing-

³⁵We use the following announcement dates: QE1 on 25 Nov, 1 Dec, and 16 Dec 2008, and 28 Jan and 18 Mar 2009; QE2 on 10 Aug, 21 Sep, and 3 Nov 2010; and QE3 on 13 Sep and 12 Dec 2012; augmented with the Maturity Extension Program announcements of 21 Sep 2011 and its 20 Jun 2012 extension.

Jorgensen (2011), Gürkaynak et al. (2005), Nakamura and Steinsson (2018)). We accordingly stress-test ψ with the leave-one-out and randomization-inference diagnostics reported below.

Outliers and randomization inference. By construction, the interaction term is non-zero only on announcement days with non-zero operation flows: 28 days in our sample, including days with negative flows from the sales of short-dated Treasuries under the 2012 Maturity Extension Program. We verify that no single day drives the result by re-estimating the baseline specification dropping each of these 28 days in turn. For the PC labor index, the interaction coefficient remains within $[0.017, 0.025]$ and significant at the 1% level in every case. For the NFP measure, the coefficient remains within $[0.009, 0.017]$ and is significant at the 5% level in all but one case (dropping June 1, 2012, the day of a large negative payroll surprise during the MEP, yields a t -statistic of 1.85 and a p -value of 0.06). The breadth of the PC labor index, which aggregates thirteen indicators, makes it the more robust measure on this dimension.

Because asymptotic HAC inference can overstate precision when the effective number of interaction observations is small, we complement it with randomization inference in the spirit of Young (2019). The design-based logic of our test treats the news surprise as the as-good-as-random element: forecast errors are unforecastable by construction, and their distribution is stable across policy regimes (Appendix B.2). The operation calendar, by contrast, is a deliberate, regime-dependent policy choice, and is therefore held fixed. We re-estimate the baseline specification 9,999 times, randomly permuting the realized surprises across announcement days while keeping the AP path and all controls unchanged, and compare the studentized interaction coefficient with its permutation distribution. The two-sided randomization p -value is 0.02 for the NFP measure and below 0.01 for the PC labor index. The pairing of news realizations and operation flows observed in the data is thus unlikely to have arisen by chance.

Table B.9: Robustness: QE program interactions and announcement-date dummies.

	Nonfarm Payroll			PC Labor News		
	(1)	(2)	(3)	(4)	(5)	(6)
Surprise	0.011*** (0.003)	0.009*** (0.003)	0.009*** (0.003)	0.012*** (0.004)	0.009** (0.004)	0.009** (0.004)
AP	-0.006* (0.003)	-0.007* (0.004)	-0.007** (0.003)	-0.007** (0.003)	-0.007* (0.004)	-0.007** (0.003)
Surprise $\times$ AP	0.013*** (0.005)	0.010** (0.004)	0.010** (0.004)	0.022*** (0.006)	0.017*** (0.006)	0.018*** (0.007)
Surprise $\times$ QE1		0.034** (0.015)	0.034** (0.015)		0.138** (0.054)	0.138** (0.054)
Surprise $\times$ QE2		-0.004 (0.004)	-0.005 (0.004)		-0.003 (0.006)	-0.003 (0.006)
Surprise $\times$ MEP		-0.010*** (0.003)	-0.010*** (0.003)		-0.008 (0.005)	-0.008 (0.005)
Surprise $\times$ QE3		-0.002 (0.004)	-0.002 (0.004)		-0.005 (0.007)	-0.005 (0.007)
QE1 announcement			-0.068*** (0.020)			-0.068*** (0.020)
QE2 announcement			-0.053* (0.027)			-0.054** (0.027)
MEP announcement			0.060** (0.026)			0.060** (0.026)
QE3 announcement			-0.021*** (0.008)			-0.021*** (0.008)
Obs	3590	3590	3590	3590	3590	3590
R^2	0.093	0.105	0.111	0.088	0.102	0.107

Notes: Dependent variable: daily change in risk-neutral 3-year yields, full sample (2005–2019), year-quarter fixed effects and $\bar{A}$ controls as in Table 1. Columns (1) and (4) repeat the baseline; (2) and (5) add free Surprise $\times$ program interactions; (3) and (6) further add a level dummy equal to one on each program’s announcement dates (canonical LSAP announcement dates following Krishnamurthy and Vissing-Jorgensen, 2011, augmented with the MEP announcement dates). The reported dummy coefficients are daily yield changes in basis points on those dates. Program-specific levels are included with their interactions. Robust standard errors in parentheses. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Online Appendix

C Analytical details

C.1 Price Neutrality

We discuss here the conditions under which we obtain that central bank APs are neutral with respect to prices and allocations, as in Wallace (1981). To do so, we consider a simple household consumption-saving problem, assuming that the household is the investor, the utility function has curvature, and position bounds are absent.

The household-investor has a generic increasing (and possibly concave) utility function $u(\cdot)$, and an information set $\Omega_i = \{x_i, Q, b_{cb}\}$, which includes an exogenous private signal, the equilibrium price Q , and the AP quantity b_{cb} . Households have an initial endowment y , can buy a risk-free asset with a unitary gross rate of return, and a risky bond b that has a price of Q and a stochastic payoff of θ . The bond is issued by the government in random supply $\tilde{S}$, and the central bank purchases a publicly observed quantity b_{cb} . The consolidated public sector levies lump-sum taxes $\tau = (\tilde{S} - b)(\theta - Q) + G$ to pay for exogenous spending and net gains or losses from bond issuance.

The portfolio allocation of the investor is

$$\max_{c_i, b_i} \mathbb{E} [u(c_i) \mid \Omega_i],$$

subject to

$$c_i = b_i(\theta - Q) + y - \tau.$$

Taking the first-order condition with respect to b_i and plugging in the expression for taxes, we get the optimality condition

$$\mathbb{E} \left[u' \left((b_i + b_{cb} - \tilde{S})(\theta - Q) + y \right) (\theta - Q) \mid \Omega_i \right] = 0,$$

which pins down individual bond demand as a function of beliefs, expected net bond supply, and the equilibrium price.

It is easy to see that, everything else equal, the investor only cares about their known net risk exposure $b_i + b_{cb}$, so any changes in central bank APs will be met by a one-for-one change in bond demand. This is true for all investors in the market. Formally, we can use the implicit function theorem to show that $\frac{\partial b_i}{\partial b_{cb}} = -1$, that is, central bank APs perfectly

crowd out each investor's demand in exactly the same way.

Let $b_i^*(b_{cb}, Q)$ denote individual bond demand as a function of APs and the bond price. The reasoning above implies that

$$b_i^*(0, Q) = b_i^*(b_{cb}, Q) + b_{cb} \quad \text{for all } b_{cb}, Q.$$

It follows that the total quantity demanded by investors and the central bank is invariant to b_{cb} and is given by $\int b_i^*(b_{cb}, Q) di + b_{cb} = \int b_i^*(0, Q) di$. As a consequence, the price Q that clears the market (i.e., such that $\int b_i^*(0, Q) di = \tilde{S}$) is also invariant to APs. In this setting, APs are thus neutral because they crowd out all investors in the same way. That is, all investors participate in the bond market, and APs have a homogeneous effect on their demand (the intensive margin).

A direct implication of this reasoning is that any departure from the model such that aggregate demand does vary with APs delivers the result that central bank intervention moves the asset price, i.e. it is price non-neutral. Yet, being price non-neutral does not imply being information non-neutral. As we show in Section 3.4, the relevant condition for information non-neutrality is that aggregate demand is not additively separable. The assumption of heterogeneous beliefs and position bounds is one such departure, because it implies that the bounds bind for some investors, who therefore do not respond one-for-one to APs. This is precisely what happens in the CARA model of Section 3, where the fraction of constrained investors varies with the shocks and is never one. The risk-neutral model of Sections 2 and 5 instead features an extreme version of this mechanism, because risk neutrality implies that *all* investors are against their position constraint. As a result, APs crowd out a specific part of the investor distribution, namely the most pessimistic agents among those that would buy bonds absent APs. That is, APs affect the market participation of some investors (the extensive margin), while leaving the demand of all other investors unaffected.

C.2 Distribution of x_m conditional on x_i

The p.d.f. of the market signal x_m conditional on x_i is given by

$$\begin{aligned} f_{\mathcal{M}_b|\mathcal{N}}(x_m|x_i) &= \sum_{j \in \{H,L\}} f_{\mathcal{M}_b|\mathcal{N},\Theta}(x_m|x_i,\theta_j) f_{\Theta|\mathcal{N}}(\theta_j|x_i) \\ &= \sum_{j \in \{H,L\}} f_{\mathcal{M}_b|\Theta}(x_m|\theta_j) f_{\Theta|\mathcal{N}}(\theta_j|x_i) \\ &= \sum_{j \in \{H,L\}} f_{\mathcal{M}_b|\Theta}(x_m|\theta_j) \frac{f_{\mathcal{N}|\Theta}(x_i|\theta_j)\varpi_j}{f_{\mathcal{N}}(x_i)} \end{aligned}$$

where the second line follows from the conditional independence of x_m and x_i given θ , the third line from Bayes' law, and the conditional distributions $\mathcal{M}_b|\theta$, $\mathcal{N}|\theta$ can be backed out from their marginals.

D Calibration and Interpretation

Our result on the impact of APs on asset prices in Section 2.6 does not assess whether the implied price elasticity is quantitatively significant. To provide a first pass on this question, we set our stylized model to target key aspects of the U.S. Federal Reserve's Large-Scale Asset Purchases announcement on March 18, 2009 (the first including Treasuries, LSAP1 henceforth). With a simple calibration exercise we explain how we map elements of our risk-neutral model to observables, and show that the model-predicted magnitudes are empirically plausible, even though the framework is not designed for quantitative accuracy.

Data. To compute expected real bond returns at the individual forecaster level, we take the nominal yield to maturity of the 10-year constant-maturity US Treasury bond on March 17th, 2009, and subtract expected average inflation over the maturity of the bond (i.e., 10 years), which we take from the 2009 Q1 release of the Survey of Professional Forecasters (SPF) run by the Philadelphia Fed.³⁶ This captures a structurally slow-moving dimension of forecast disagreement: in Q1-2009, the distribution had a standard deviation of 0.6%, the same as in Q1-2010, both lying one standard deviation (0.15) above the historical mean for the 1992-2024 period (0.44).

³⁶An alternative approach to derive the distribution of real return forecasts would be to take bond yield forecasts and subtract inflation forecasts. We choose not to follow this approach because, as is well known, in the SPF the survey question about long-term forecasts of Treasury yields was ambiguously phrased—and the wording was indeed changed from 2014:Q1.

As a reference for the empirical effect of the LSAP1 announcements on real yields, we take the two-day change in the real yield of 10-year TIPS from Table 4 of Krishnamurthy and Vissing-Jorgensen (2011), which is -59 basis points. An alternative approach would be to take the two-day change in the yield of nominal 10-year bonds from Table 1, which is -41 basis points, and then subtract the 22 basis point increase in 10-year inflation swaps, which delivers a comparable total change of -63 basis points.

As a reference for the size of APs in the model, we take the size of the LSAP1 announcement relative to the face value of marketable, non-indexed U.S. Treasury bonds and notes with residual maturity larger than 5 years as of February 2009, amounting to \$585.82 billions in Treasury notes and \$609.35 billions in Treasury bonds, as stated in the monthly statement of the public debt of the United States on the 28 of February 2009. The Fed announcement of purchasing up to “300 billion of longer-term Treasury securities over the next six months” is thus equivalent to about 25% of the existing marketable stock of debt (or about 20% if we include TIPS).

Model interpretation. In terms of model counterparts, we interpret θ as the present value of the realized real gross payoff on 10-year U.S. Treasury bonds, and Q as their price. To see that, consider a textbook asset pricing model, such as that in Chapter 1 of Cochrane (2005). Take a nominal bond that, at the end of the period, delivers a stochastic real payoff denoted by $\tilde{\theta}$. This random payoff reflects various sources of risk, including inflation or default during the holding period, as well as resale price fluctuations in the case of a long-term bond. The equilibrium bond price conditional on a generic aggregate state x_m satisfies $Q(x_m) = \mathbb{E}[\zeta \tilde{\theta} | x_m]$, where ζ is a stochastic discount factor such that the real risk-free rate is given by $\mathbb{E}[\zeta | x_m] = 1/(1+r)$. Using the definition of covariance and letting $\theta := \tilde{\theta}/(1+r)$, we can rewrite the bond pricing equation as

$$\lambda(x_m) = \mathbb{E}[\theta | x_m] - Q(x_m) \tag{D.1}$$

where $\lambda(x_m) := -(1+r)\text{Cov}(\zeta, \theta | x_m)$ is a generic risk adjustment, which includes various premia such as those for term, liquidity, default, and inflation risk.

Consider now our risk-neutral model, where equation (21) implies $-W(b) = \mathbb{E}[\theta] - Q$. It follows that the average wedge is the natural (negative) counterpart of the average equilibrium bond premium in the simple generic bond pricing framework of (D.1), that is, $W(b) = -\mathbb{E}[\lambda]$. To account for the distribution of individual beliefs in our model, we define the model analog of the SPF individual forecasts as the individual expectation of the

conditional wedge according to

$$\mathbb{E}[\theta - Q(p(x_m)) \mid x_i] \Big|_{b=0} = (\theta_H - \theta_L) \int (p(x_i, x_m) - p(x_m)) f_{\mathcal{M}_0|\mathcal{N}}(x_m \mid x_i) dx_m, \quad (\text{D.2})$$

where we are averaging over x_m states, and $f_{\mathcal{M}_0|\mathcal{N}}$ is the distribution of the price signal conditional on the private signal, in the absence of APs.³⁷ Thus, by equation (D.1), $\mathbb{E}[\theta - Q(p(x_m)) \mid x_i]$ represents the individual conditional expectation of the equilibrium bond premium, $\mathbb{E}[\lambda \mid x_i]$, conditional on observing x_i and averaged across all possible x_m states.³⁸

Note that we adopt an ex-ante perspective and focus on average across x_m states—that is, before any shock realizes—to address the dynamic nature of the data. This approach also avoids taking a stand on what identifies x_m in the data. In this case, investors would agree the bond premium is given by $\mathbb{E}[\lambda] = \int \mathbb{E}[\lambda \mid x_i] f_{\mathcal{N}}(x_i) di$.

Quantification. We choose model parameters to approximate the cross-sectional distribution of individual forecasts for 10-year real Treasury yields, which we described above. Figure D.1 plots the binned empirical distribution of these forecasts with circle markers, and the model-implied forecast distribution density with a solid line.³⁹ The parameter values are: $\theta_H = 2.87, \theta_L = -3.23, \gamma = 0.53, \sigma_x = 10.5$. Importantly, the absolute values of θ do not affect the average wedge—only the 610 basis point $\theta_H - \theta_L$ range matters—highlighting that our model captures price elasticities rather than price levels. We choose the value of θ_H such that the prior mean is zero.

According to an ex-ante perspective, we think of the average wedge $W(b) = -(\mathbb{E}[\theta] - \mathcal{Q}_{\text{qt}}(b))$ as capturing the *realized* yield premium averaged over all possible aggregate states x_m , and look at its change before ($b = 0$) and after ($b > 0$) an AP intervention. In practice, $\mathcal{Q}_{\text{qt}}$ represents the price investors would agree upon ex ante—before aggregate and idiosyncratic shocks are realized. This is consistent with our assumption that APs convey no new information about fundamentals, or equivalently, that any such information is commonly known.

The left panel of Figure 3 shows how the average price $\mathcal{Q}_{\text{qt}}$ —specifically, its wedge component—responds to interventions of size b . The model predicts that intermediate-scale APs

³⁷See Online Appendix C.2 for the derivation.

³⁸Market clearing implies that this premium is expected to be zero for the marginal agent: $Q(p(x_m)) = \mathbb{E}[\theta \mid x_i, x_m] \Big|_{x_i=x_m}$.

³⁹Specifically, the line plots the density $f_{\mathcal{N}}(x_i(\varphi))$ where $x_i(\varphi)$ is the model-implied mapping from an individual forecast φ to the corresponding private signal x_i . Defining $z_j := \theta_j - \int Q(p(x_m)) f_{\mathcal{M}_0|\theta}(x_m \mid \theta_j) dx_m$, for $j \in \{L, H\}$, then $x_i(\varphi)$ is given by the private signal x_i that solves $\varphi = z_L + (z_H - z_L)P(\theta_H \mid x_i)$ if $\varphi \in (z_L, z_H)$ (which is always the case in our parametrization).

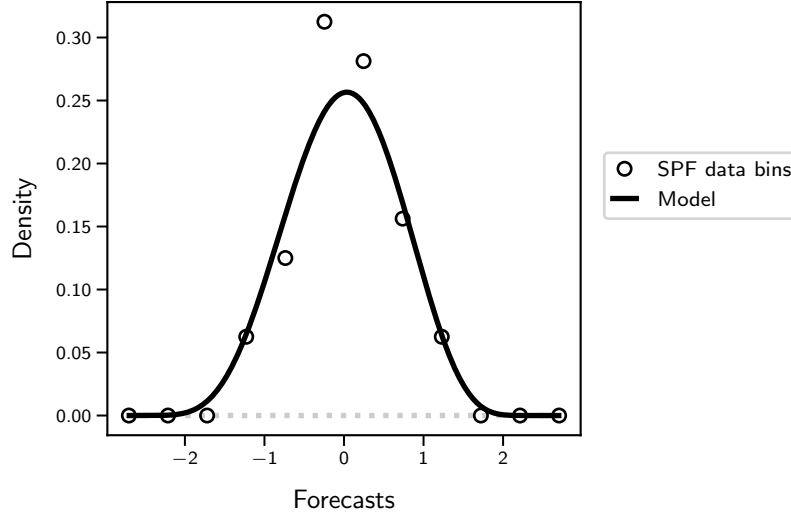


Figure D.1: Cross-section dispersion of individual real yield forecasts. Circle markers denote the empirical distribution constructed using SPF 10-year inflation forecasts; the solid line represents the corresponding model-implied distribution.

can lower yields by up to 47 basis points. The dashed vertical line at $b = 0.25$ marks the LSAP1 intervention size as a fraction of outstanding Treasury bonds. For this intervention size, the model predicts an increase in Q_{qt} , that is, a drop in yields of approximately 31 basis points ($W(0.25) - W(0) \approx 31$), about half of the 59 basis point decline in real yields observed around the LSAP1 announcement, as documented for example by Krishnamurthy and Vissing-Jorgensen (2011).

E Communication Policies

The exercises in this section endow the central bank with superior information about the state $(\theta, \tilde{S})$ and ask how much of the AP mechanism it can reproduce through communication alone. It is important to note that this constitutes a significant departure from our setting, where the central bank does not possess private information and, indeed, does not require it, as asset purchase (AP) policies are implemented via a limit order. We consider two instruments: a state-contingent announcement that withholds or reveals supply (Online Appendix E.1), and a public signal about the fundamental (Online Appendix E.2). The first replicates the informational effects of APs but reassigns the central bank's gains and losses to the noise traders; the second compresses, without eliminating, the wedge between prices and fundamentals.

E.1 Replicating Asset Purchases through Announcements

Can a central bank that observes $(\theta, \tilde{S})$ but cannot purchase bonds replicate the AP equilibrium through communication alone? Consider the following rule.

Definition E.1 (Announcement policy $\mathcal{A}(b)$). *The central bank publicly announces $\tilde{S}$ if and only if $\tilde{S} \in [1 - b, 1]$, and remains silent otherwise.*

The announced region $[1 - b, 1]$ has mass b , matching the mass of the passive region $[0, b]$ under APs. We show that $\mathcal{A}(b)$ replicates the size- b AP equilibrium in prices, beliefs, and gains, up to the central bank's balance-sheet position.

Announced region ($\tilde{S} \in [1 - b, 1]$). When $\tilde{S}$ is announced, investors observe both the announcement and the price signal $x_m = \theta + \sigma_x \Phi^{-1}(1 - \tilde{S})$, from which they recover θ exactly. The bond price equals the fundamental value θ and all trading gains are zero. This mirrors the passive region under APs, where $Q^{\text{pas}} = \theta$ and gains are likewise zero; both regions carry mass b .

Silent region ($\tilde{S} \in (0, 1 - b)$). Because the rule is public, silence is itself informative: conditional on it, $\tilde{S} \sim \text{Uniform}(0, 1 - b)$ and the price signal is $x_m^A = \theta + \sigma_x \Phi^{-1}(1 - \tilde{S})$. Under APs, conditional on an active market ($\tilde{S} \geq b$), net supply $S = \tilde{S} - b \sim \text{Uniform}(0, 1 - b)$ and $x_m = \theta + \sigma_x \Phi^{-1}(1 - S)$. The two policies are therefore identical in distribution. It follows that, at any realized x_m , the posterior beliefs $p(x_m)$ and $\hat{p}(x_m)$, the equilibrium price $Q(p(x_m))$, and the partition into revealing region $[\underline{x}(b), \bar{x}(b)]$ and non-revealing region $[\bar{x}(b), +\infty)$ all coincide with their AP counterparts.

Distribution of gains. Because the equilibrium is replicated state by state, the gains that depend on the price signal alone are unchanged. Investor gains (24) and government gains, $\mathbb{E}[\pi_{\text{gov}} | x_m] = Q(p(x_m)) - Q(\hat{p}(x_m))$, are functions of x_m only and therefore coincide with their AP counterparts: $\mathbb{E}[\pi_{\text{inv}}^A | x_m] = \mathbb{E}[\pi_{\text{inv}}^{AP} | x_m]$ and $\mathbb{E}[\pi_{\text{gov}}^A | x_m] = \mathbb{E}[\pi_{\text{gov}}^{AP} | x_m]$ for every x_m . The central bank, however, now holds no bonds, so $\pi_{\text{cb}}^A = 0$ in place of $\pi_{\text{cb}}^{AP} = b[Q(\hat{p}(x_m)) - Q(p(x_m))]$. By the zero-sum identity $\pi_{\text{nt}} + \pi_{\text{gov}} + \pi_{\text{inv}} + \pi_{\text{cb}} = 0$, this difference is absorbed entirely by the noise traders, who end up holding the position the central bank would have taken under APs:

$$\mathbb{E}[\pi_{\text{nt}}^A | x_m] = \mathbb{E}[\pi_{\text{nt}}^{AP} | x_m] + \mathbb{E}[\pi_{\text{cb}}^{AP} | x_m], \quad (\text{E.1})$$

which also holds unconditionally. Relative to size-b APs, the announcement policy thus leaves investor and government gains unchanged and sets central bank gains to zero, but it does not make the associated trading gains or losses disappear: it shifts them from the central bank's balance sheet onto the noise traders. The information channel of APs—the crowding-out and revelation effects that reshape the price-signal distribution—is fully replicable by communication at zero balance-sheet cost; what communication cannot reproduce is the central bank's risk-bearing role. Under actual purchases the central bank absorbs the losses that learning from prices generates on its holdings, whereas under pure communication those losses fall on the private non-fundamental traders who take its place.

E.2 A Public Signal

We now enrich the information structure of the risk-neutral model by modeling central bank communication about fundamentals with a *public* signal y , observed by all investors before trading, and given by

$$y = \theta + \sigma_y \varepsilon_y, \quad \varepsilon_y \sim N(0, 1), \quad (\text{E.2})$$

with $\sigma_y > 0$ and ε_y independent of all other shocks. Each investor's information set becomes $\Omega_i = \{x_i, b, Q, y\}$.

Price signal. Because y is observed by everyone, it shifts all investors' posteriors uniformly without altering their ranking by the private signal x_i . The monotone threshold strategy and the market-clearing condition (9) therefore carry over unchanged, and the price signal mapping is still (10).

Posterior beliefs. On the non-revealing region $x_m \in [\bar{x}(b), +\infty)$, the signals (x_m, x_i, y) are conditionally independent given θ with Gaussian likelihoods, and Bayesian updating yields the public posterior

$$\hat{p}(x_m, y) := \frac{\gamma \phi\left(\frac{x_m - \theta_H}{\sigma_x}\right) \phi\left(\frac{y - \theta_H}{\sigma_y}\right)}{\gamma \phi\left(\frac{x_m - \theta_H}{\sigma_x}\right) \phi\left(\frac{y - \theta_H}{\sigma_y}\right) + (1 - \gamma) \phi\left(\frac{x_m - \theta_L}{\sigma_x}\right) \phi\left(\frac{y - \theta_L}{\sigma_y}\right)}, \quad (\text{E.3})$$

and the marginal investor's posterior

$$p(x_m, y) := \frac{\gamma \phi\left(\frac{x_m - \theta_H}{\sigma_x}\right)^2 \phi\left(\frac{y - \theta_H}{\sigma_y}\right)}{\gamma \phi\left(\frac{x_m - \theta_H}{\sigma_x}\right)^2 \phi\left(\frac{y - \theta_H}{\sigma_y}\right) + (1 - \gamma) \phi\left(\frac{x_m - \theta_L}{\sigma_x}\right)^2 \phi\left(\frac{y - \theta_L}{\sigma_y}\right)}. \quad (\text{E.4})$$

The price signal likelihood enters *squared* in p because the marginal investor observes $x_i = x_m$ alongside the price signal x_m —two independent draws of precision σ_x —whereas the public signal y is only observed once. This squaring is the direct counterpart of the $\sigma_x/\sqrt{2}$ scale compression in (13). On the revealing region $x_m \in [\underline{x}(b), \bar{x}(b))$, the price signal already reveals θ_L , so $\hat{p}(x_m, y) = p(x_m, y) = 0$ irrespective of y . The bond price and fundamental valuation are $Q(p(x_m, y))$ and $Q(\hat{p}(x_m, y))$, similarly to the baseline model of Section 2.

The average price. Since y is random, the object of interest is the average equilibrium price at a given x_m , $\mathbb{E}[Q(p(x_m, y)) \mid x_m]$, taken over the conditional density of y ,

$$f(y \mid x_m) = \hat{p}(x_m) \frac{1}{\sigma_y} \phi\left(\frac{y - \theta_H}{\sigma_y}\right) + (1 - \hat{p}(x_m)) \frac{1}{\sigma_y} \phi\left(\frac{y - \theta_L}{\sigma_y}\right), \quad (\text{E.5})$$

a Gaussian mixture weighted by the public posterior $\hat{p}(x_m)$ of the baseline model. For the *public* posterior, averaging over y returns the fundamental valuation of the baseline model,

$$\mathbb{E}[Q(\hat{p}(x_m, y)) \mid x_m] = Q(\hat{p}(x_m)), \quad (\text{E.6})$$

by the law of iterated expectations: revealing y and then averaging it out leaves the fundamental valuation unchanged. The equilibrium price behaves differently. Because $p(x_m, y)$ places squared weight on the price-signal likelihood while $f(y \mid x_m)$ weighs it only linearly, the two do not cancel, and

$$\mathbb{E}[Q(p(x_m, y)) \mid x_m] \neq Q(p(x_m)). \quad (\text{E.7})$$

The gap is the extra wedge that learning from prices produces once a public signal is present; this integral has no closed form and is computed numerically. The following result sharpens (E.7): the information wedge is always smaller than in the baseline and shrinks as the public signal becomes more precise.

Proposition E.1 (Public signals compress the wedge). *For any $x_m \neq x^\dagger$ in the non-revealing region and any finite σ_y , the average market price $\bar{Q}(x_m) := \mathbb{E}[Q(p(x_m, y)) \mid x_m]$ lies strictly between the fundamental valuation $Q(\hat{p}(x_m))$ and the baseline market price $Q(p(x_m))$, and the average wedge $|\bar{Q}(x_m) - Q(\hat{p}(x_m))|$ is strictly increasing in σ_y , with $\bar{Q}(x_m) \rightarrow Q(\hat{p}(x_m))$ as $\sigma_y \rightarrow 0$ and $\bar{Q}(x_m) \rightarrow Q(p(x_m))$ as $\sigma_y \rightarrow \infty$.*

Proof. Let $\ell_x := \ell(x_m) = \frac{\theta_H - \theta_L}{\sigma_x^2} (x_m - x^\dagger)$ be the price-signal log-likelihood ratio of the main text and $\ell_y := \frac{\theta_H - \theta_L}{\sigma_y^2} (y - x^\dagger)$ its public-signal counterpart, so that the public and market

posteriors have log-odds $\ell_0 + \ell_x + \ell_y$ and $\ell_0 + 2\ell_x + \ell_y$, with $\ell_0 = \log \frac{\gamma}{1-\gamma}$. Put $u := \hat{p}(x_m, y)$; then $\mathbb{E}[u \mid x_m] = \hat{p}(x_m)$ by the law of iterated expectations (as in (E.6)), and

$$p(x_m, y) = \psi(u), \quad \psi(u) := \frac{u e^{\ell_x}}{u e^{\ell_x} + 1 - u},$$

since adding ℓ_x to the log-odds multiplies the odds by e^{ℓ_x} ; note that $\psi(\hat{p}(x_m)) = p(x_m)$. For $x_m > x^\dagger$ one has $\ell_x > 0$, so ψ is strictly increasing, strictly concave, and $\psi(u) > u$ on $(0, 1)$; hence

$$\hat{p}(x_m) = \mathbb{E}[u] < \mathbb{E}[\psi(u)] < \psi(\mathbb{E}[u]) = p(x_m),$$

the first inequality from $\psi(u) > u$ and the second from Jensen's inequality, both strict because u is non-degenerate. Since Q is affine and increasing and $\bar{Q}(x_m) = Q(\mathbb{E}[\psi(u)])$, the betweenness and wedge claims follow. Lowering σ_y makes y Blackwell-more-informative, so the posterior u spreads out in the mean-preserving sense, with mean fixed at $\hat{p}(x_m)$; concavity of ψ then lowers $\mathbb{E}[\psi(u)]$ toward $\hat{p}(x_m)$, giving the monotonicity and the two limits. The case $x_m < x^\dagger$ is symmetric, with ψ convex and $\psi(u) < u$. ■

Intuitively, the average market price is a concave transform—multiplication of the public posterior's odds by e^{ℓ_x} —of a belief whose mean is the fundamental valuation: Jensen's inequality pulls it below the baseline market price, while the marginal trader's double weight on the price keeps it above the fundamental valuation, and a more precise public signal disperses the posterior and shrinks this gap.

Quantitative illustration. Figure E.1 illustrates Proposition E.1 for the calibration of Figure 2, at two values of σ_y . The fundamental valuation $Q(\hat{p}(x_m))$ (dotted) is invariant to σ_y ; for each finite σ_y the average market price lies between it and the baseline market price $Q(p(x_m))$, and is closer to the fundamental valuation, the more precise the signal.

This attenuation is separate from the effect of APs. Comparing the two panels, APs truncate the left tail of the price-signal distribution and create a fully-revealing region (the crowding-out and revelation effects of Section 2.3), and both mechanisms operate independently of σ_y , since the mapping from (θ, S) to x_m is unaffected by the public signal. A more informative public signal therefore shifts the average market price toward the fundamental valuation in every state, without altering the non-monotone response of average prices to the size of APs characterized in Proposition 2.

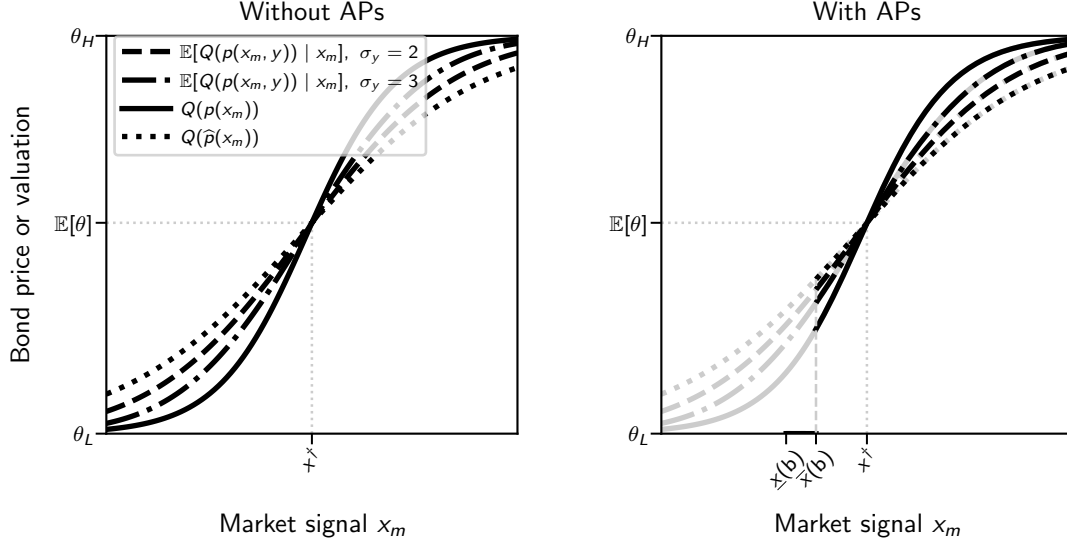


Figure E.1: Market price and fundamental valuation with a public signal y . The solid and dotted lines plot the market price and fundamental valuation of the model of Section 2, invariant to σ_y . The dashed and dash-dotted lines plot the average market price in the presence of a central bank public signal with different precision. The parametrization is as in Figure 2.

F Additional Derivations for Section 3

This online appendix is structured as follows: F.1 solves for the equilibrium price in the unconstrained version of the model; F.2 describes the numerical procedure used to solve the constrained version of the model, and provides a further characterization of the model; F.3 shows which models satisfy the hypotheses of Proposition 5.

Before starting, let us introduce a few objects that will be useful in the analysis that follows. Let $u(Q) := \log \frac{Q - \theta_L}{\theta_H - Q}$ denote the logit transformation of the price, and recall ℓ_0 and $\ell(Q)$ as defined at the beginning of Appendix A.⁴⁰ The posterior log-odds ratio of an informed trader with private signal x_i is given by

$$\log \frac{P(\theta_H | x_i, Q)}{P(\theta_L | x_i, Q)} = \ell_0 + \ell(Q) + \frac{(\theta_H - \theta_L)}{\sigma_x^2} (x_i - x^\dagger), \quad (\text{F.1})$$

which coincides with the first two terms of $b^u(x_i, Q)$ in (28).

⁴⁰Note that $\ell(\cdot)$ is invariant to any monotone transformation of the price, since the gradient cancels out in the ratio.

F.1 Unconstrained Case

Start from the observation that the price signal $x_m(Q)$ in (30) is a one-to-one transformation of the equilibrium price, and in equilibrium is an endogenous public signal of the fundamental θ , with noise $\rho \sigma_x^2(\tilde{S} - \mu_s)$. Let $1/\tau_q := \sigma_s^2(\rho \sigma_x^2)^2$ and $1/\tau_x = \sigma_x^2$ respectively denote the price and private signal variance. We can write the log-likelihood ratio of the price signal as

$$\ell(Q) = (\theta_H - \theta_L) \tau_q (x_m(Q) - x^\dagger).$$

Unconstrained demand in (27) becomes

$$b^u(x_i, Q) = \frac{1}{\rho(\theta_H - \theta_L)} \left[\ell_0 + \ell(Q) + (\theta_H - \theta_L) \tau_x (x_i - x^\dagger) - u(Q) \right]. \quad (\text{F.2})$$

Evaluating (F.2) at $x_i = x^\dagger$ and using the mapping defined by the left-hand side of (30), we can solve for $x_m(Q)$ explicitly:

$$x_m(Q) = x^\dagger + \frac{1}{\tau_x + \tau_q} \left[\rho(\mu_s - b - N(Q)) - \frac{\ell_0 - u(Q)}{(\theta_H - \theta_L)} \right].$$

Rearranging this equation we get that the equilibrium price satisfies⁴¹

$$u(Q) = \ell_0 + (\theta_H - \theta_L) \left[(\tau_x + \tau_q)(x_m - x^\dagger) - \rho(\mu_s - b - N(Q)) \right]$$

This highlights clearly how, in this context, uncontingent APs only affect the average risk premium $\rho(\mu_s - b)$, and have no state-contingent effect on beliefs or on the logistic transformation of the equilibrium price. The binary shape of the payoff, however, implies that the price is the following non-linear function of beliefs and of the risk premium

$$Q = \theta_L + \frac{\theta_H - \theta_L}{1 + \exp \left\{ -\ell_0 - (\theta_H - \theta_L) \left[(\tau_x + \tau_q)(x_m - x^\dagger) - \rho(\mu_s - b - N(Q)) \right] \right\}}. \quad (\text{F.3})$$

F.2 Constrained Case

F.2.1 Numerical Procedure

Our goal is to solve for the price signal log-likelihood ratio $\ell(Q)$, since that allows us to write down unconstrained demand (27) and aggregate demand (31) explicitly, and solve for the equilibrium price. We follow a procedure similar to Albagli et al. (2024), iterating on the

⁴¹We can solve for Q in closed form if we assume that $N(Q) = u(Q)$, as we do when we solve the constrained model numerically.

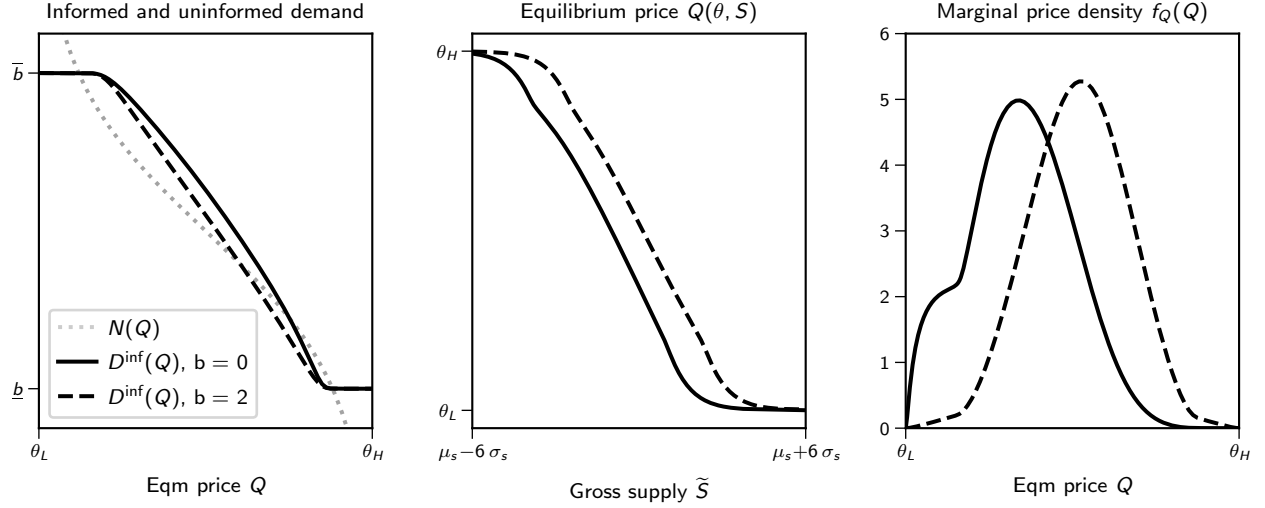


Figure F.1: Selected model features for two different values of uncontingent APs, illustrated with black solid and dashed lines. The left panel plots informed demand averaged over θ , as well as uninformed demand (dotted gray line), as a function of the equilibrium price. The central panel plots the equilibrium price as a function of gross supply. The right panel plots the marginal density of the equilibrium price.

price signal likelihood ratio until convergence. First, we create a grid for Q and start from a guess $D^0(\theta, Q)$ for aggregate demand. Second, we compute the gradients $D_Q^0(\theta, Q)$ for each θ , and use it to derive the price signal log-likelihood ratio and the public posterior odds ratio. Third, we use full posterior beliefs—including private information—to compute the new aggregate demand function $D^1(\theta, Q)$ according to (28) and (31). We continue iterating if the distance between the two aggregate demand functions is larger than an error criterion, and stop otherwise. Once convergence is reached, we can back out the equilibrium price density $f_Q(q)$, the mapping from the fundamental and net supply shocks to the price $Q(\theta, \tilde{S} - b)$, and all our other objects of interest.

F.2.2 Model Characterization

Figure F.1 displays further aspects of the model, complementing Figure 6 in the main text. The left panel shows informed and uninformed demand as a function of the equilibrium price. When Q is close to its bounds, informed demand is constrained regardless of the value of the fundamental, and the price is fully determined by uninformed demand. The central panel shows the equilibrium price as a function of gross supply $\tilde{S}$ and of the fundamental θ . The right panel plots the marginal density of the equilibrium price.

To compute the unconditional moments illustrated in Figure 7, we proceed as follows.

Market price and fundamental valuation. The average market price is computed as $\int q f_Q(q) dq$. To compute the public valuation, we consider a modification of informed traders' demand that is given by (31) where $\sigma_x \rightarrow \infty$, i.e., where unconstrained demand is given by $b^u(Q) = [\ell_0 + \ell(Q) - u(Q)]/(\rho(\theta_H - \theta_L))$. The public valuation, which we denote with $\widehat{Q}(Q)$, is thus the hypothetical price that clears the market by solving (29) with the modified version of informed traders' demand we just described. Note that, in this model, the public valuation contains a risk premium and is affected by the presence of uninformed demand. Below we will discuss how this differs from the *risk-neutral* public valuation.

Risk-neutral price. To compute the risk-neutral price, we do the following. We use investor i 's posterior beliefs from (F.1). To proceed, we need to choose the appropriate value of x_i . In the unconstrained case, we consider an informed investor whose private signal coincides with the price signal. The demand of such an investor is given by

$$b^u(x_m(Q), Q) = \mu_s - b - N(Q), \quad (\text{F.4})$$

that is, the investor's demand in any state would coincide with the average supply, net of uninformed demand, at such state. The investor's expected payoff would be

$$\begin{aligned} Q^{\text{rn}}(x_m) &= \theta_L + (\theta_H - \theta_L) P(\theta_H | x_i = x_m, x_m) \\ &= \theta_L + (\theta_H - \theta_L) \left[1 + \frac{1 - \gamma}{\gamma} \exp\left\{ -(\theta_H - \theta_L)(\tau_x + \tau_q)(x_m - x^\dagger) \right\} \right]^{-1}, \end{aligned} \quad (\text{F.5})$$

which coincides with the terms in (F.3) that are not multiplied by ρ .

In the constrained case we cannot follow these steps, since there does not exist a transformation of the price that is an unbiased, additive signal of θ . We thus select the private signal $x_i = z(Q)$ of the *hypothetical* informed investor whose unconstrained demand equals the average supply net of the demand of the central bank and of uninformed traders, that is, $b^u(z(Q), Q) = \mu_s - b - N(Q)$. We thus have

$$Q^{\text{rn}}(Q) = \theta_L + (\theta_H - \theta_L) P(\theta_H | x_i = z(Q), Q). \quad (\text{F.6})$$

Note that, in the presence of position bounds, such informed investor may not exist for all states (e.g., for extreme supply shocks and prices, *all* informed agents are constrained), but we follow this approach nonetheless because it delivers the same risk-neutral prices of the unconstrained case when position constraints are slack.

Computing the risk-neutral public valuation is instead straightforward, as it is simply

given by public posterior beliefs, that is, $\widehat{Q}^{\text{rn}}(x_m(Q)) = \theta_L + (\theta_H - \theta_L) P(\theta_H | x_m(Q))$ in the unconstrained case, and $\widehat{Q}^{\text{rn}}(Q) = \theta_L + (\theta_H - \theta_L) P(\theta_H | Q)$ in the constrained case, where $P(\theta_H | Q) = [1 + \exp \{-\ell_0 - \ell(Q)\}]^{-1}$.

Price sensitivity and informativeness. Both price sensitivity and informativeness are computed adapting the definitions of (19) and (20) to the current model. Mutual information is given by

$$\begin{aligned} I(Q; \Theta)(b) &= H(\Theta) - H(\Theta | Q) \\ &= H_b(\gamma) - \int H_b(P(\theta_H | q)) f_Q(q) dq. \end{aligned} \tag{F.7}$$

The sensitivity of a generic function $G(Q)$ of the equilibrium price Q is given by

$$\begin{aligned} \mathcal{S}_G(b) &= \frac{\mathbb{E}[G(Q) | \theta_H] - \mathbb{E}[G(Q) | \theta_L]}{\theta_H - \theta_L} \\ &= \frac{1}{\theta_H - \theta_L} \int G(q) (f_{Q|\Theta}(q | \theta_H) - f_{Q|\Theta}(q | \theta_L)) dq, \end{aligned}$$

where G can be $\widehat{Q}(Q)$, $Q^{\text{rn}}(Q)$, $\widehat{Q}^{\text{rn}}(Q)$, or the identity function for the equilibrium price itself.

F.3 Models Satisfying Proposition 5

This subsection shows that the case in which aggregate demand is additively separable is common to other models that are well-known in the literature.

CARA utility and normal-linear payoff. Consider the same model of Section 3 without position bounds, with an asset payoff given by a normally distributed fundamental $\theta \sim N(\mu_\theta, 1/\tau_\theta)$, and with no uninformed agents' demand.⁴² Investors choose b_i to maximize (26). Their asset demand is given by

$$b_i = \frac{\mathbb{E}[\theta | x_i, Q] - Q}{\rho \mathbb{V}(\theta | x_i, Q)}.$$

As is standard, we guess and later verify that the price is a linear function of $(\theta, \tilde{S}, b)$, so that there exists a transformation of the price $x_m(Q, b)$ which is a linear, unbiased signal of

⁴²This assumption is just for simplicity. The conclusion would not change if we kept some price-elastic demand from uninformed agents

θ with precision τ_q . Bayesian normal updating implies $\mathbb{E}[\theta | x_i, Q] = \frac{\tau_\theta \mu_\theta + \tau_x x_i + \tau_q x_m}{\tau_\theta + \tau_x + \tau_q}$. Applying a version of the law of large numbers, the market clearing condition becomes

$$\frac{\tau_\theta \mu_\theta + \tau_x \theta + \tau_q x_m(Q, b)}{\rho} - \frac{\tau_\theta + \tau_x + \tau_q}{\rho} Q = \tilde{S} - b,$$

which can be rearranged to show that aggregate demand is linear in both θ and Q , satisfying the requirements of Proposition 5.

Hellwig et al. (2006). Consider a variant of the risk-neutral model of Section 2 where supply is given by $\Phi(\tilde{S} - b)$, the fundamental follows an exogenous distribution F_Θ over the entire real line, while aggregate demand is unchanged and equal to $\Phi((\theta - x_m(Q, b))/\sigma_x)$, where x_m denotes the private signal of the agent that is marginal. For $b = 0$, this coincides with the framework of Hellwig et al. (2006) when supply is not elastic to the equilibrium price. The market clearing condition can be rearranged in the linear form

$$\frac{\theta - x_m(Q, b)}{\sigma_x} = \tilde{S} - b, \quad (\text{F.8})$$

from which one can show that aggregate demand satisfies the requirements of Proposition 5, being additively separable in θ and Q , increasing in the former, and decreasing in the latter.

G Baseline Model Extensions

G.1 Wider Bounds, Binary Fundamental

This section considers a more general version of the model of Section 2 with position bounds outside the unit interval, i.e., with generic long and short position bounds. Unless otherwise specified, the environment is the same as in the main text.

Position bounds are $[-\underline{b}, \bar{b}]$ with $\underline{b} \geq 0$ and $\bar{b} \geq 1$. Aggregate demand by the unit continuum of informed, risk-neutral agents is

$$D^{\text{inf}}(\theta, Q) = \bar{b} \Phi\left(\frac{\theta - x_m}{\sigma_x}\right) - \underline{b} \left[1 - \Phi\left(\frac{\theta - x_m}{\sigma_x}\right)\right], \quad (\text{G.1})$$

where x_m denotes the signal of the marginal investor who is indifferent between buying $\bar{b}$

and selling $\underline{b}$ units short. Market clearing $D^{\text{inf}}(\theta, Q) = \tilde{S} - b$ yields

$$\Phi\left(\frac{\theta - x_m}{\sigma_x}\right) = \frac{\max\{\tilde{S} + \underline{b} - b, 0\}}{\underline{b} + \bar{b}} =: S. \quad (\text{G.2})$$

Net supply S is uniformly distributed on $[\underline{S}, \bar{S}]$ where

$$\underline{S} := \frac{\max\{0, \underline{b} - b\}}{\underline{b} + \bar{b}}, \quad \bar{S} := \frac{1 + \underline{b} - b}{\underline{b} + \bar{b}}. \quad (\text{G.3})$$

The market is active (a clearing price exists) if and only if $\tilde{S} \geq b - \underline{b}$, which occurs with probability

$$P_A := 1 - \max(b - \underline{b}, 0). \quad (\text{G.4})$$

We assume uncontingent APs are such that $b \in [0, 1 + \underline{b})$, that is, the market is active with positive probability.

The price signal $x_m = \theta - \sigma_x \Phi^{-1}(S)$ has the following support for $S \in [\underline{S}, \bar{S}]$

$$x_m \mid (\theta, b) \in [\underline{x}(\theta), \bar{x}(\theta)] \quad (\text{G.5})$$

where $\underline{x}(\theta) = \theta - \sigma_x \Phi^{-1}(\bar{S})$ and $\bar{x}(\theta) = \theta - \sigma_x \Phi^{-1}(\underline{S})$ if $\underline{S} > 0$ and $\bar{x}(\theta) = +\infty$ otherwise. We assume throughout that the non-revelation region is non-empty:

$$\underline{x}(\theta_H) < \bar{x}(\theta_L),$$

which implies that the model primitives are such that $\frac{\theta_H - \theta_L}{\sigma_x} < \Phi^{-1}\left(\frac{1 + \underline{b} - b}{\underline{b} + \bar{b}}\right) - \Phi^{-1}\left(\frac{\max\{0, \underline{b} - b\}}{\underline{b} + \bar{b}}\right)$.

We denote by $\mathcal{M}_b \mid \theta$ the likelihood of the price signal conditional on $\tilde{S} \geq b - \underline{b}$, i.e., when the equilibrium bond price is determined by the market. The p.d.f. is given by $\frac{1}{P_A} f_{\mathcal{M}_b \mid \Theta}(x_m \mid \theta)$, where

$$f_{\mathcal{M}_b \mid \Theta}(x_m \mid \theta) = \frac{\underline{b} + \bar{b}}{\sigma_x} \phi\left(\frac{x_m - \theta}{\sigma_x}\right), \quad x_m \in [\underline{x}(\theta), \bar{x}(\theta)]. \quad (\text{G.6})$$

The unconditional distribution of the price signal is $\frac{1}{P_A} f_{\mathcal{M}_b}(x_m)$, where $f_{\mathcal{M}_b}(x_m) = \gamma f_{\mathcal{M}_b \mid \Theta}(x_m \mid \theta_H) + (1 - \gamma) f_{\mathcal{M}_b \mid \Theta}(x_m \mid \theta_L)$ is the mixture density analogous to (11).

The price-signal log-likelihood ratio is

$$\ell(x_m) = \begin{cases} -\infty, & x_m < \underline{x}(\theta_H), \\ \frac{\theta_H - \theta_L}{\sigma_x^2}(x_m - x^\dagger), & \underline{x}(\theta_H) \leq x_m \leq \bar{x}(\theta_L), \\ +\infty, & x_m > \bar{x}(\theta_L) \end{cases} \quad (\text{G.7})$$

where $\underline{x}, \bar{x}$ are defined in (G.5).

Posterior beliefs and prices. Since θ is binary, let $(\theta_H - \theta_L) := \theta_H - \theta_L > 0$, and consider posterior beliefs $p(x_m), \hat{p}(x_m)$ as defined in (14) and (12). Three regions arise:

- If $x_m < \underline{x}(\theta_H)$: the price signal is incompatible with θ_H , so $p = \hat{p} = 0$ and $Q = \theta_L$.
- If $\bar{x}(\theta_L) < x_m < \infty$: the price signal is incompatible with θ_L , so $p = \hat{p} = 1$ and $Q = \theta_H$.
- Otherwise, standard Bayesian updating gives

$$\frac{\tilde{p}(x_m; \sigma)}{1 - \tilde{p}(x_m; \sigma)} = \frac{\gamma}{1 - \gamma} \exp \left\{ \frac{(\theta_H - \theta_L)}{\sigma^2} (x_m - x^\dagger) \right\}, \quad (\text{G.8})$$

where $\tilde{p}(x_m; \sigma_x) = \hat{p}(x_m)$ is the public posterior, and $\tilde{p}(x_m; \sigma_x/\sqrt{2}) = p(x_m)$ is the market posterior, i.e., the posterior for the marginal investor who uses the price signal as a second observation.

G.1.1 Short Selling and Support Shifts

We now consider three useful cases that clarify how position bounds and APs reshape the support of the price signal. Figure G.1 illustrates these three parameter configurations.

The left panel represents the baseline model without interventions $(\underline{b}, \bar{b}, b) = (0, 1, 0)$, where $\underline{S} = 0$ and $\bar{S} = 1$, so the support of x_m is $(-\infty, +\infty)$. The center panel shows the case with short selling, long position bounds above one, and no APs $(\underline{b} > 0, \bar{b} > 1, b = 0)$. In this case $\underline{S} > 0$ and $\bar{S} < 1$, so both tails are truncated: very optimistic and very pessimistic investors are never marginal. Finally, the right panel shows that with short selling and sufficiently large APs $(\underline{b} > 0, b > \underline{b})$, APs push the lower supply bound to zero, eliminating the upper truncation while leaving the lower truncation in place. The result is the asymmetric support structure of the baseline one-sided model, with a truncation and a revealing region only in the left tail of x_m .

Because the posterior probability formula in (G.8) depends on $(\underline{b}, \bar{b}, b)$ only through the

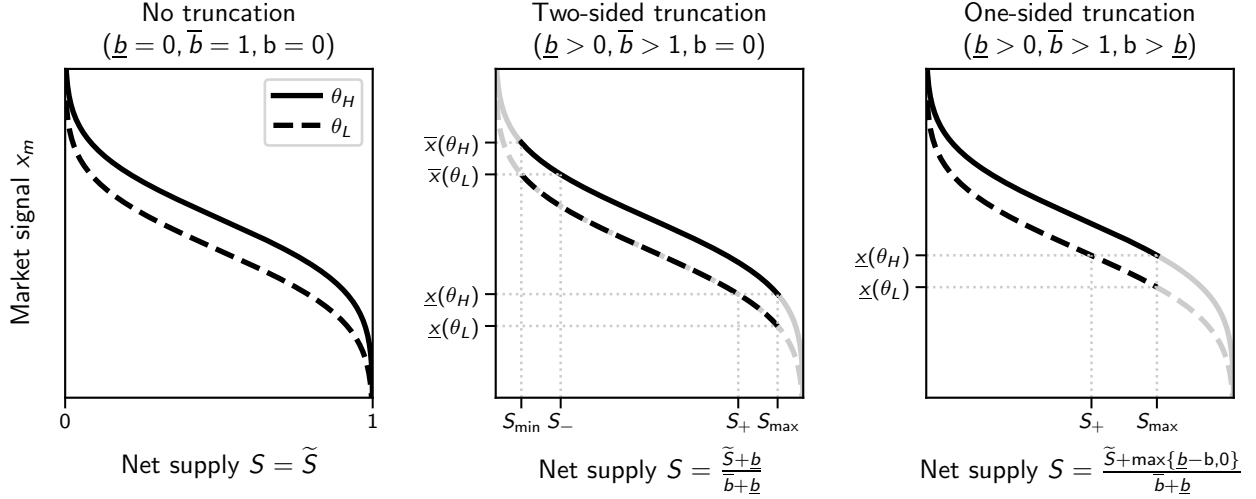


Figure G.1: The figure plots the price signal x_m against transformed net supply S for θ_H (solid) and θ_L (dashed) for three different parametrizations: $(\underline{b}, \bar{b}, b) = (0, 1, 0)$ (left panel); $(\underline{b}, \bar{b}, b) = (0.1, 1.1, 0)$ (center panel); $(\underline{b}, \bar{b}, b) = (0.1, 1.1, 0.2)$ (right panel). The remaining parameters are the same as Section 2.

support endpoints $\underline{x}(\theta_H)$ and $\bar{x}(\theta_L)$, changing position bounds or APs only re-partitions the price signal support into revealing and non-revealing regions. Figure G.2 illustrates this. The left panel reproduces the one-sided baseline. The central panel shows how short-selling introduces bilateral support truncations and creates revealing regions at both tails of the support. Holding these position bounds fixed, the right panel shows that APs shift the entire posterior structure rightward, widening the lower revealing region, and eliminating the upper one. Within the non-revelation region, however, the posterior curves are unchanged across panels: position bounds and APs change which x_m realizations are feasible or correspond with full revelation, but not the functional form of posterior beliefs conditional on a given non-revealing signal realization. Proposition G.1 formalizes this reasoning.

Proposition G.1 (Equivalence of APs and position bounds). *Any parameter triplet $(b, \underline{b}, \bar{b})$ delivering the same pair $(\underline{S}, \bar{S})$ generates the same active-market distribution of x_m , the same public and market posteriors as functions of x_m , and hence the same market price, public valuation, and active-market price-informativeness objects.*

Proof. The net supply bounds $(\underline{S}, \bar{S})$ determine both the price signal distribution inside the non-revealing region $[\underline{x}(\theta), \bar{x}(\theta)]$, and the value of such support endpoints. The active-market distribution of x_m depends only on $(\underline{S}, \bar{S})$. It follows that the net supply bounds $(\underline{S}, \bar{S})$ alone determine the public posterior $\hat{p}(x_m)$, the market posterior $p(x_m)$, the price signal log-likelihood in (G.7), and all other conditional and unconditional objects that depend on

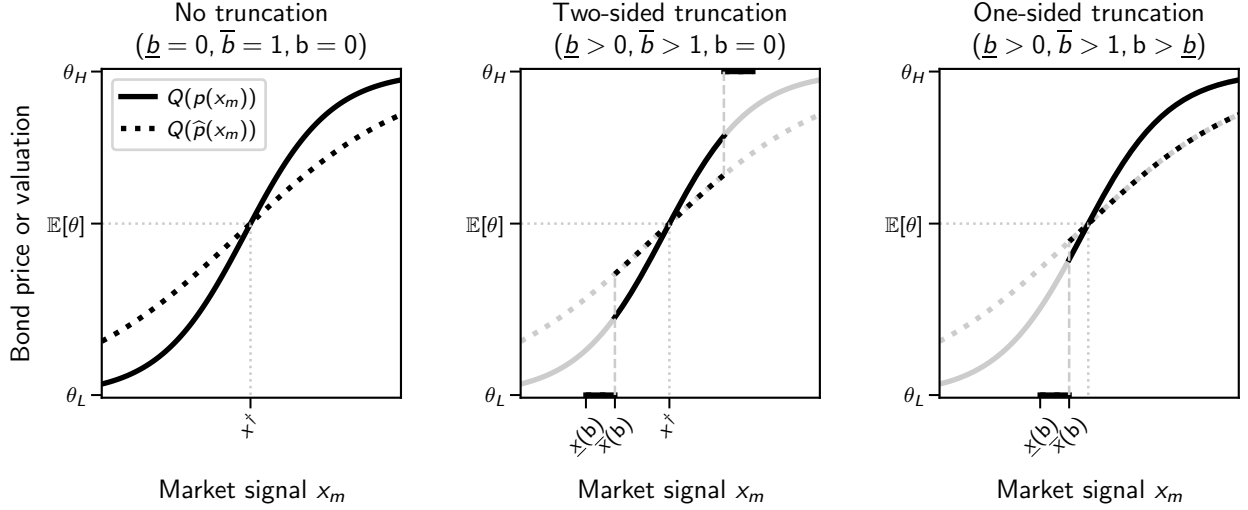


Figure G.2: Equilibrium bond price and fundamental valuation as a function of the price signal. The parametrization is the same as in Figure G.1. The x-axis is $x^\dagger \pm 3\sigma_x$.

these functions and on the price signal distribution. ■

The proposition shows that what matters for equilibrium prices, their informational content, and their sensitivity to fundamentals are not b , $\underline{b}$, and $\bar{b}$ separately, but how they combine to determine the support $[\underline{S}, \bar{S}]$ of transformed net supply. APs shift such interval leftward, while changes in position bounds can both shift it and alter its width.

G.1.2 Price Informativeness, Sensitivity, and Average Wedge

This subsection extends the results of Section 2.6 to the case where position bounds lie outside the unit interval. The three active-market moments are the same as in (19), (20), and (23), except that the non-revealing region is now $[x(\theta_H), \bar{x}(\theta_L)]$ and the active-market normalization is $1/P_A$:

$$I(Q; \Theta)(b) = H_b(\gamma) - \int_{\underline{x}(\theta_H)}^{\bar{x}(\theta_L)} H_b(\hat{p}(x_m)) \frac{1}{P_A} f_{\mathcal{M}_b}(x_m) dx_m \quad (G.9)$$

$$W(b) = \frac{\theta_H - \theta_L}{P_A} \int_{\underline{x}(\theta_H)}^{\bar{x}(\theta_L)} [p(x_m) - \hat{p}(x_m)] f_{\mathcal{M}_b}(x_m) dx_m. \quad (G.10)$$

The sensitivity expression also includes the upper revealing region, when this region exists:

$$\begin{aligned} \mathcal{S}(b) = \frac{1}{P_A} & \left[\int_{\underline{x}(\theta_H)}^{\bar{x}(\theta_L)} p(x_m) [f_{\mathcal{M}_b|\Theta}(x_m | \theta_H) - f_{\mathcal{M}_b|\Theta}(x_m | \theta_L)] dx_m \right. \\ & \left. + \int_{\bar{x}(\theta_L)}^{\bar{x}(\theta_H)} f_{\mathcal{M}_b|\Theta}(x_m | \theta_H) dx_m \right]. \end{aligned} \quad (G.11)$$

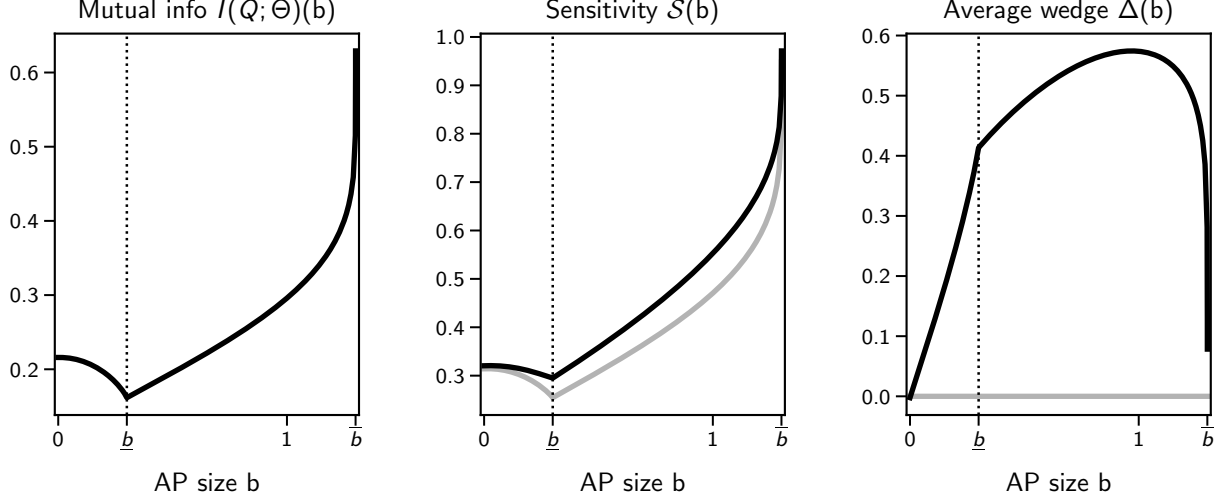


Figure G.3: Average informativeness, sensitivity, and information wedge for the equilibrium price. Where present, shaded lines represent the fundamental valuation. The parameter values are the same as Section 2, except for $\underline{b} = 0.3$, $\bar{b} = 1.3$.

Proposition G.2 (The Average Effects of APs with Wider Bounds). *(i) If $b \in [0, \underline{b}]$, then $P_A = 1$ and both net supply bounds depend on APs. Let $x_- := \underline{x}(\theta_H)$ and $x_+ := \bar{x}(\theta_L)$. Then*

$$\frac{\partial I(Q; \Theta)(b)}{\partial b} = \frac{\gamma}{\hat{p}(x_-)} H_b(\hat{p}(x_-)) - \frac{1 - \gamma}{1 - \hat{p}(x_+)} H_b(\hat{p}(x_+)), \quad (\text{G.12})$$

$$\frac{\partial \mathcal{S}(b)}{\partial b} \propto p(x_-) \left(\frac{\phi\left(\frac{x_- - \theta_L}{\sigma_x}\right)}{\phi\left(\frac{x_- - \theta_H}{\sigma_x}\right)} - 1 \right) + (1 - p(x_+)) \left(1 - \frac{\phi\left(\frac{x_+ - \theta_H}{\sigma_x}\right)}{\phi\left(\frac{x_+ - \theta_L}{\sigma_x}\right)} \right), \quad (\text{G.13})$$

$$\frac{\partial W(b)}{\partial b} \propto \left((p(x_+) - \hat{p}(x_+)) \frac{1 - \gamma}{1 - \hat{p}(x_+)} - (p(x_-) - \hat{p}(x_-)) \frac{\gamma}{\hat{p}(x_-)} \right). \quad (\text{G.14})$$

Both $\partial I / \partial b$ and $\partial \mathcal{S} / \partial b$ are positive if $x_+ \leq x^\dagger$, negative if $x_- \geq x^\dagger$, and ambiguous in the straddling case $x_- < x^\dagger < x_+$. In the straddling case, $\partial I / \partial b$ is strictly decreasing in b , so it has at most one zero. Finally, $\partial W / \partial b$ is positive in the straddling case and otherwise ambiguous. The average valuation and price satisfy

$$\hat{\mathcal{Q}}_{\text{qt}}(b) = \mathbb{E}[\theta], \quad (\text{G.15})$$

$$\mathcal{Q}_{\text{qt}}(b) = \mathbb{E}[\theta] + W(b), \quad (\text{G.16})$$

where W is defined in (G.10).

(ii) The case where $b \in (b, 1 + \underline{b})$ has the same properties of the one-sided model of Proposition 2 for the transformed AP variable $\tilde{b} := 1 - \bar{S} = 1 - \frac{1+b-\underline{b}}{b+\underline{b}}$ under the restricted

interval $\tilde{b} \in [1 - 1/(\underline{b} + \bar{b}), 1)$.

Proof.

Case (i): $b \in [0, \underline{b})$ and $P_A = 1$. First, consider $\partial I / \partial b$. Then

$$x'_- := \frac{\partial x_-}{\partial b} = \frac{\sigma_x}{(\underline{b} + \bar{b})\phi\left(\frac{x_- - \theta_H}{\sigma_x}\right)}, \quad x'_+ := \frac{\partial x_+}{\partial b} = \frac{\sigma_x}{(\underline{b} + \bar{b})\phi\left(\frac{x_+ - \theta_L}{\sigma_x}\right)}.$$

Bayes' rule at the two boundaries gives

$$f_{\mathcal{M}_b}(x_-)x'_- = \frac{\gamma}{\hat{p}(x_-)}, \quad f_{\mathcal{M}_b}(x_+)x'_+ = \frac{1 - \gamma}{1 - \hat{p}(x_+)}. \quad (\text{G.17})$$

Since only the integration limits in (G.9) move,

$$\frac{\partial H(\Theta | Q)}{\partial b} = H_b(\hat{p}(x_+))f_{\mathcal{M}_b}(x_+)x'_+ - H_b(\hat{p}(x_-))f_{\mathcal{M}_b}(x_-)x'_-,$$

and therefore (G.12) follows from (G.17). For the sign of (G.12), let $\eta_H(p) := \frac{\gamma}{p}H_b(p)$ and $\eta_L(p) := \frac{1-\gamma}{1-p}H_b(p)$, so the one-sided function $\eta(r)$ in Proposition 2 is $\eta_H(r)$. Note that

$$\eta'_H(r) = \frac{\gamma \log(1-r)}{r^2} < 0, \quad \eta'_L(r) = -\frac{(1-\gamma) \log r}{(1-r)^2} > 0,$$

and

$$\eta_H(r) - \eta_L(r) = H_b(r) \frac{\gamma - r}{r(1-r)}.$$

If $x_+ \leq x^\dagger$, then $\hat{p}(x_+) \leq \gamma$ and $\hat{p}(x_-) < \hat{p}(x_+)$, so

$$\eta_H(\hat{p}(x_-)) - \eta_L(\hat{p}(x_+)) > \eta_H(\hat{p}(x_+)) - \eta_L(\hat{p}(x_+)) \geq 0.$$

If $x_- \geq x^\dagger$, then $\hat{p}(x_-) \geq \gamma$ and $\hat{p}(x_+) > \hat{p}(x_-)$, so

$$\eta_H(\hat{p}(x_-)) - \eta_L(\hat{p}(x_+)) < \eta_H(\hat{p}(x_-)) - \eta_L(\hat{p}(x_-)) \leq 0.$$

If $x_- < x^\dagger < x_+$, both boundary posteriors increase with b , so

$$\frac{\partial}{\partial b} \left(\frac{\partial I(Q; \Theta)(b)}{\partial b} \right) = \eta'_H(\hat{p}(x_-)) \frac{\partial \hat{p}(x_-)}{\partial b} - \eta'_L(\hat{p}(x_+)) \frac{\partial \hat{p}(x_+)}{\partial b} < 0.$$

Second, consider $\partial\mathcal{S}/\partial b$. Differentiating (G.11) gives

$$\begin{aligned}\frac{\partial\mathcal{S}}{\partial b} &\propto -p(x_-) \left[f_{\mathcal{M}_b|\Theta}(x_-|\theta_H) - f_{\mathcal{M}_b|\Theta}(x_-|\theta_L) \right] x'_- \\ &\quad + p(x_+) \left[f_{\mathcal{M}_b|\Theta}(x_+|\theta_H) - f_{\mathcal{M}_b|\Theta}(x_+|\theta_L) \right] x'_+ \\ &\quad - f_{\mathcal{M}_b|\Theta}(x_+|\theta_H) x'_+ + f_{\mathcal{M}_b|\Theta}(\bar{x}(\theta_H)|\theta_H) \frac{\partial\bar{x}(\theta_H)}{\partial b}.\end{aligned}$$

Using

$$\begin{aligned}f_{\mathcal{M}_b|\Theta}(x_-|\theta_H) x'_- &= 1, & f_{\mathcal{M}_b|\Theta}(x_-|\theta_L) x'_- &= \frac{\phi\left(\frac{x_- - \theta_L}{\sigma_x}\right)}{\phi\left(\frac{x_- - \theta_H}{\sigma_x}\right)}, \\ f_{\mathcal{M}_b|\Theta}(x_+|\theta_L) x'_+ &= 1, & f_{\mathcal{M}_b|\Theta}(x_+|\theta_H) x'_+ &= \frac{\phi\left(\frac{x_+ - \theta_H}{\sigma_x}\right)}{\phi\left(\frac{x_+ - \theta_L}{\sigma_x}\right)},\end{aligned}$$

and

$$f_{\mathcal{M}_b|\Theta}(\bar{x}(\theta_H)|\theta_H) \frac{\partial\bar{x}(\theta_H)}{\partial b} = 1,$$

we obtain (G.13). If $x_+ \leq x^\dagger$, then

$$\frac{\phi\left(\frac{x_- - \theta_L}{\sigma_x}\right)}{\phi\left(\frac{x_- - \theta_H}{\sigma_x}\right)} \geq 1, \quad \frac{\phi\left(\frac{x_+ - \theta_H}{\sigma_x}\right)}{\phi\left(\frac{x_+ - \theta_L}{\sigma_x}\right)} \leq 1,$$

so both terms in (G.13) are nonnegative and at least one is strictly positive. If $x_- \geq x^\dagger$, the two inequalities are reversed, so both terms are nonpositive and at least one is strictly negative. If $x_- < x^\dagger < x_+$, the two terms have opposite signs in general.

Third, consider the average price, valuation, and wedge. The equalities (G.15)–(G.16) follow from iterated expectations and from $p = \hat{p}$ in revealing regions. Finally, differentiating (G.10) gives

$$\frac{\partial W}{\partial b} = (\theta_H - \theta_L) \left[(p(x_+) - \hat{p}(x_+)) f_{\mathcal{M}_b}(x_+) x'_+ - (p(x_-) - \hat{p}(x_-)) f_{\mathcal{M}_b}(x_-) x'_- \right],$$

which becomes (G.14) by (G.17). Since $p(x_m) - \hat{p}(x_m)$ is negative below $x^\dagger$ and positive above it, the two terms are both positive in the straddling case; outside it, their signs oppose each other.

Case (ii): $b \geq \underline{b}$ and $P_A \leq 1$. Now we have $P_A = 1 + \underline{b} - b$, $\bar{S} = P_A/(\underline{b} + \bar{b})$, and $\bar{x}(\theta_L) =$

$+\infty$. After the recoding $\tilde{b} = 1 - \bar{S}$, the active-market density is

$$\frac{1}{P_A} f_{\mathcal{M}_b|\Theta}(x_m | \theta) = \frac{1}{(1 - \tilde{b})\sigma_x} \phi\left(\frac{x_m - \theta}{\sigma_x}\right),$$

with lower boundary $\theta - \sigma_x \Phi^{-1}(1 - \tilde{b})$. Thus Proposition 2 and its proof apply after the affine change of the AP policy variable: price informativeness and sensitivity are strictly increasing, the average valuation and price satisfy (G.15)–(G.16), and the average wedge W has the same properties of Proposition 2. $\blacksquare$

G.2 Wider Bounds, Continuous Fundamental

This section makes one change to Online Appendix G.1: the fundamental is continuous, $\theta \sim N(\mu_\theta, \sigma_\theta^2)$. We continue to assume that the bond payoff is equal to θ : this is a symmetric payoff that is tractable, and confirms that the effect of APs does not depend on whether the asset payoff is symmetric. The remaining assumptions on the private signal noise, the supply shock distribution, investors' position bounds, the AP policy rule are all unchanged.

Price signal and posterior support. The market clearing condition (G.2), the net supply bounds (G.3), and the probability that the market is active (G.4) all carry over from the previous subsection. The price signal expression $x_m = \theta - \sigma_x \Phi^{-1}(S)$ is unchanged, although its implications are now different. First, observing x_m restricts the continuous fundamental to

$$\theta \in [\underline{\theta}(x_m), \bar{\theta}(x_m)], \quad \underline{\theta}(x_m) := x_m + \sigma_x \Phi^{-1}(\underline{S}), \quad \bar{\theta}(x_m) := x_m + \sigma_x \Phi^{-1}(\bar{S}). \quad (\text{G.18})$$

The binary model has fully revealing tails; here the same mechanism appears as a truncation of the posterior support of θ . Second, since the fundamental is distributed on the real line, the unconditional support of the price signal x_m is also given by the real line, and is independent of the support of net supply.

Posteriors and prices. The posterior distribution of θ conditional on either public information (the price signal) or market information (private and price signals) follows standard Bayesian Normal updating, and is given by truncated normal distributions. Public posterior beliefs are

$$\theta | x_m \sim N(\mu_{\text{pub}}, \sigma_{\text{pub}}) \quad \text{for } \theta \in [\underline{\theta}(x_m), \bar{\theta}(x_m)],$$

where

$$\mu_{\text{pub}}(x_m) = \frac{\tau_\theta \mu_\theta + \tau_x x_m}{\tau_{\text{pub}}}, \quad \tau_{\text{pub}} = \tau_\theta + \tau_x, \quad \sigma_{\text{pub}} = 1/\sqrt{\tau_{\text{pub}}}.$$

Analogously, investors' posterior beliefs are

$$\theta \mid x_m, x_i \sim N(\mu_{\text{mkt}}, \sigma_{\text{mkt}}) \quad \text{for } \theta \in [\underline{\theta}(x_m), \bar{\theta}(x_m)],$$

where

$$\mu_{\text{mkt}}(x_m) = \frac{\tau_\theta \mu_\theta + 2\tau_x x_m}{\tau_{\text{mkt}}}, \quad \tau_{\text{mkt}} = \tau_\theta + 2\tau_x, \quad \sigma_{\text{mkt}} = 1/\sqrt{\tau_{\text{mkt}}}.$$

It follows that both the public valuation and the market price are given by

$$Q_j(x_m) = \mu_j(x_m) - \sigma_j \frac{\phi(\beta_j(x_m)) - \phi(\alpha_j(x_m))}{\Phi(\beta_j(x_m)) - \Phi(\alpha_j(x_m))} \quad (\text{G.19})$$

where $j \in \{\text{pub}, \text{mkt}\}$, $\beta_j(x_m) := \frac{\bar{\theta}(x_m) - \mu_j(x_m)}{\sigma_j}$, and $\alpha_j(x_m) := \frac{\underline{\theta}(x_m) - \mu_j(x_m)}{\sigma_j}$. We get simple truncated means because of the assumption of payoff linearity. We write $Q(x_m) := Q_{\text{mkt}}(x_m)$ and $\hat{Q}(x_m) := Q_{\text{pub}}(x_m)$.⁴³

Figure G.4 is the analog of Figure 2 and shows the equilibrium price and the fundamental valuation as a function of the price signal x_m . The conditional wedge is still given by the difference between price $Q(x_m)$ and valuation $\hat{Q}(x_m)$, but unlike in the binary case, APs can change the posterior maps themselves at a fixed x_m because they move the truncation interval in (G.18).

⁴³For a generic payoff $\pi(\theta)$, we would have

$$Q_j(x_m) = \int_{\underline{\theta}(x_m)}^{\bar{\theta}(x_m)} \pi(\theta) \frac{\frac{1}{\sigma_j} \phi\left(\frac{\theta - \mu_j(x_m)}{\sigma_j}\right)}{\Phi\left(\frac{\bar{\theta}(x_m) - \mu_j(x_m)}{\sigma_j}\right) - \Phi\left(\frac{\underline{\theta}(x_m) - \mu_j(x_m)}{\sigma_j}\right)} d\theta, \quad j \in \{\text{pub}, \text{mkt}\}.$$

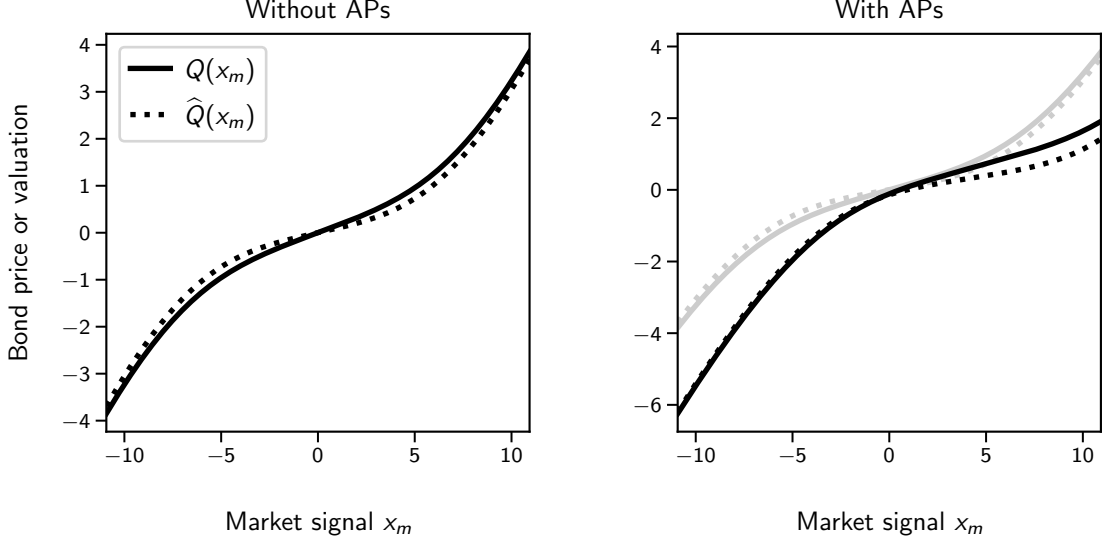


Figure G.4: Equilibrium price and public valuation as functions of the price signal, without and with central bank APs. We set parameters as close as possible to our previous choices: $\sigma_x = 10.5$ is the same as in Section 2, while $\mu_\theta = 0$ and $\sigma_\theta = 3.04$ are set to match the first and second moment of the binary distribution; the bounds $\underline{b}, \bar{b}$ are the same as in Figure G.3; and $b = 0.15$ in the right panel is chosen for illustrative purposes.

Average effects. The marginal distribution of the price signal is $\frac{1}{P_A} f_{\mathcal{M}_b}(x_m)$, where

$$f_{\mathcal{M}_b}(x_m) = \frac{\underline{b} + \bar{b}}{\sqrt{\sigma_\theta^2 + \sigma_x^2}} \phi\left(\frac{x_m - \mu_\theta}{\sqrt{\sigma_\theta^2 + \sigma_x^2}}\right) \left[\Phi\left(\frac{\bar{\theta}(x_m) - \mu_{\text{pub}}(x_m)}{\sigma_{\text{pub}}}\right) - \Phi\left(\frac{\underline{\theta}(x_m) - \mu_{\text{pub}}(x_m)}{\sigma_{\text{pub}}}\right) \right], \quad (\text{G.20})$$

and again we normalize by the probability that the market is active. The analogues of the key objects in Section 2.6 are given by

$$\mathcal{Q}(b) := \int_{-\infty}^{+\infty} Q(x_m) \frac{f_{\mathcal{M}_b}(x_m)}{P_A} dx_m, \quad (\text{G.21})$$

$$\hat{\mathcal{Q}}(b) := \int_{-\infty}^{+\infty} \hat{Q}(x_m) \frac{f_{\mathcal{M}_b}(x_m)}{P_A} dx_m, \quad (\text{G.22})$$

$$I(Q; \Theta) := h(\Theta) - \int_{-\infty}^{+\infty} h(\Theta | x_m) \frac{f_{\mathcal{M}_b}(x_m)}{P_A} dx_m, \quad (\text{G.23})$$

$$\mathcal{S}(b) := \mathbb{E}\left[\frac{\partial Q(x_m(\theta, S))}{\partial \theta}\right] = \mathbb{E}[Q'(x_m(\theta, S))] \quad (\text{G.24})$$

where h denotes differential entropy, and are plotted in Figure G.5. The figure illustrates the same support-shift mechanism of Online Appendix G.1. When APs move both supply bounds, they change the width and location of the posterior truncation interval; once $b \geq \bar{b}$,

the lower bound is zero and only the upper supply bound keeps moving. Qualitatively, the continuous model thus generates the same unconditional results of its binary counterpart (see Figure G.3), while not requiring literal revelation regions.⁴⁴

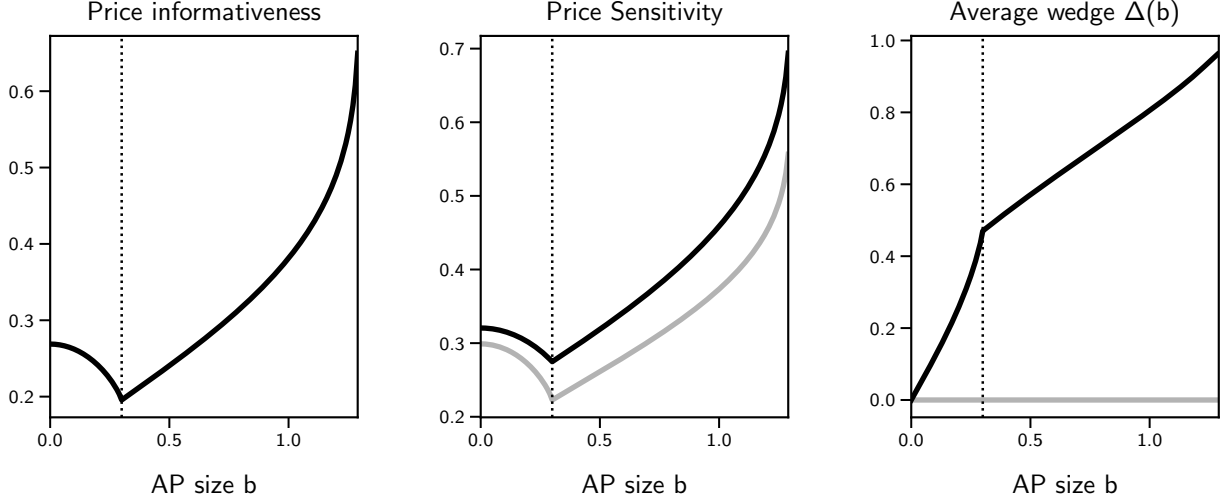


Figure G.5: Average price, price informativeness, and price sensitivity as functions of APs. The dashed vertical line marks $b = \bar{b}$; parameters are as in Figure G.4.

H Price-Targeting Asset Purchases

In this section, we explore the workings of price-targeting AP policies, where the central bank buys bonds only at or below a specific price target.

Definition H.1 (Price-targeting APs). *Under price-targeting APs, the central bank submits, simultaneously to investors, a limit order to buy up to a quantity $\bar{b}_n$ of bonds if the price is below a target Q_n , and nothing otherwise, that is*

$$b_{cb} \begin{cases} = \bar{b}_n & \text{if } Q < Q_n, \\ \in [0, \bar{b}_n] & \text{if } Q = Q_n, \\ = 0 & \text{if } Q > Q_n. \end{cases} \quad (\text{H.1})$$

with $Q_n \in [\theta_L, \theta_H]$ being the announced price target.⁴⁵

⁴⁴The only difference appears in the behavior of the average wedge in the limit for $b \rightarrow \bar{b}$, which is due to the fact that the binary model has a bounded payoff support while the continuous model does not.

⁴⁵We treat Q_n as a policy parameter, though it could just as well reflect the central bank's private valuation of the asset, in which case it should be interpreted as a public signal of the fundamental. What matters is that the central bank is ex-ante committed to it.

The central bank has no information about θ and, in fact, does not need to observe the state $(\theta, \tilde{S})$ to implement its price-targeting policy. It only needs to be able to condition its demand on the bond price Q . As we will see, the quantity b_{cb} that the central bank effectively purchases to achieve the target will be an equilibrium outcome and, as such, provide information to investors.

Most of our derivations for the case of quantity-targeting policies carry on to this section, so we organize the following exposition around the key differences.

H.1 State Space Partitions

We have seen above that a quantity-targeting policy can be understood as partitioning the state space of shocks $(\theta, \tilde{S})$ into three distinct regions: one of full revelation, one where the market operates under uncertainty, and one where the market remains passive. Price-targeting policies perform a similar partition of the state space into three regions: one of full revelation, one where the market operates under uncertainty, and one where the market price equals the central bank target. In both cases, the last region is one where the equilibrium price formation mechanism is disrupted. With quantity-targeting, this disruption arises as the central bank buys all the supply. On the other hand, with price targeting, the equilibrium price is artificially held at its target value as a result of the central bank intervention. We proceed by characterizing such regions.

To start, we recover from equation (16) the mapping from the equilibrium price Q to the signal of the marginal investor

$$x(Q, \sigma) = x^\dagger + \frac{\sigma^2}{\theta_H - \theta_L} \log \left(\frac{1 - \gamma}{\gamma} \frac{Q - \theta_L}{\theta_H - Q} \right) \quad (\text{H.2})$$

where $\sigma \in \{\sigma_x, \sigma_x/\sqrt{2}\}$ depending on whether the relevant beliefs are those of the investors or of the public. We define two values of interest:

- $\tilde{x}_n := x(Q_n, \sigma_x/\sqrt{2})$, which defines the price signal at which the equilibrium bond price is equal to the target in the absence of APs, i.e., $Q_n = \mathbb{E}[\theta \mid \tilde{x}_n \sim \mathcal{N}, \tilde{x}_n \sim \mathcal{M}_0]$;
- $x_n := x(Q_n, \sigma_x)$, which defines the price signal at which the *marginal private valuation* is equal to the price target in the absence of APs, i.e., $Q_n = \mathbb{E}[\theta \mid x_n \sim \mathcal{N}]$.

We will focus our discussion on price targets such that $Q_n \geq \mathbb{E}[\theta]$, which implies that $x_n \geq \tilde{x}_n \geq x^\dagger$. The case $Q_n \leq \mathbb{E}[\theta]$ requires a more nuanced analysis and will be briefly discussed later.

The pair $(\tilde{x}_n, x_n)$ fully characterizes the regions in which price-targeting APs partition the state space.

No-intervention region. This region is characterized by all combinations of fundamental and supply shocks that deliver a price signal $x_m \geq \tilde{x}_n$ in the absence of APs, which implies an equilibrium price larger than the target ($Q \geq Q_n$), and indeed no central bank intervention ($b_{cb} = 0$) as a result. It occurs with probability

$$P(Q \geq Q_n) = \gamma \Phi\left(\frac{\theta_H - \tilde{x}_n}{\sigma_x}\right) + (1 - \gamma) \Phi\left(\frac{\theta_L - \tilde{x}_n}{\sigma_x}\right).$$

In these states, the equilibrium bond price is determined by the market in the way described in Section 2, i.e., according to (16), with the posterior beliefs in (14), such that investors learn from the price and the scale parameter in their beliefs is $\sigma_x/\sqrt{2}$.

Price target region. This region is characterized by the equilibrium price being equal to the policy target ($Q = Q_n$) and an AP quantity $b_{cb} \in [\underline{b}_n, \bar{b}_n]$, where $(\underline{b}_n, \bar{b}_n)$ are bounds that we derive below. It occurs with probability

$$P(Q = Q_n) = 1 - \Phi\left(\frac{\theta_H - \tilde{x}_n}{\sigma_x}\right). \quad (\text{H.3})$$

In this region, investors learn from both the equilibrium price Q_n and the AP program size b_{cb} —which is endogenous to the equilibrium—that is needed to keep the price at the target. The following proposition establishes this mapping.

Proposition H.1 (Price-targeting APs). *Let*

$$b_n(\theta, \tilde{S}) = \tilde{S} - \Phi\left(\frac{\theta - x_n}{\sigma_x}\right) \quad (\text{H.4})$$

be the AP size that corresponds to price signal x_n and state $(\theta, \tilde{S})$. When the equilibrium price is at the target Q_n , the size of central bank APs is given by

$$b_{cb} = \begin{cases} b_n(\theta, \tilde{S}) & \text{if } b_n(\theta, \tilde{S}) \in [\underline{b}_n, \bar{b}_n] \\ 0 & \text{otherwise,} \end{cases} \quad (\text{H.5})$$

where $\underline{b}_n := b_n(\theta_H, \Phi(\frac{\theta_H - \tilde{x}_n}{\sigma_x}))$ and $\bar{b}_n := b_n(\theta_H, 1)$. We can state the following properties:

- *conditional on $Q = Q_n$, b_{cb} is distributed as a Uniform $[\underline{b}_n, \bar{b}_n]$, independently of θ ;*

- for all $b_{cb} \in [\underline{b}_n, \bar{b}_n]$, the market signals $(Q = Q_n, b_{cb})$ are uninformative, i.e.
 $\mathbb{E}[\theta \mid Q = Q_n, b_{cb}] = \mathbb{E}[\theta]$.
- x_n is such that $Q_n = \mathbb{E}[\theta \mid x_n \sim \mathcal{N}]$,

Proof. Postponed to Section H.3. ■

The proposition establishes several properties of price-targeting APs. Let us just sketch the main intuition here. Equation (H.4) derives the size of the central bank intervention needed to make net supply consistent with a given x_n . It shows that, conditional on θ , the distribution of $b_n(\theta, \tilde{S})$ is uniform because it follows that of gross supply. For APs to not be fully revealing, any realization of b_n must be compatible with all values of the fundamental θ . The set of AP quantities that satisfy this property is given by the interval $[\underline{b}_n, \bar{b}_n]$. It follows that the likelihood of observing any $b_{cb} \in [\underline{b}_n, \bar{b}_n]$ is constant and independent of the fundamental θ . Standard Bayesian updating then implies that

$$\mathbb{E}[\theta \mid x_i \sim \mathcal{N}, Q = Q_n, b_{cb}] = \mathbb{E}[\theta \mid x_i \sim \mathcal{N}] \quad \text{for all } b_{cb} \in [\underline{b}_n, \bar{b}_n],$$

that is, the expected bond payoff for investor i , conditional on observing the public signals $Q = Q_n$ and any b_{cb} inside $[\underline{b}_n, \bar{b}_n]$, is uninformative about θ . The indifference condition $Q_n = \mathbb{E}[\theta \mid x_n \sim \mathcal{N}]$ finally pins down the marginal investor's (or price) signal $x_n = x(Q_n, \sigma_x)$ that is consistent with the equilibrium price being equal to target *and* investors learning nothing from public information. The intuition behind this uninformativeness result is that, when the price target is achieved, the central bank becomes the marginal investor in the market. Since the likelihood of its purchases is independent of θ , investors do not learn anything from observing them.

The upper bound of the equilibrium APs is given by $\bar{b}_n = b_n(\theta_H, 1)$.⁴⁶ In fact, (H.4) implies that the range of AP interventions corresponding to θ_L is wider than that for θ_H , so the latter defines the range of AP quantities that achieve the price target, while its complement is fully revealing as uniquely determined when $\theta = \theta_L$ and consists of the fully revealing region discussed next.

It is worth noting that, in this case, the lower bound $\underline{b}_n$ is generally greater than zero, which means a discontinuity exists at the margin when APs are implemented. To see why, consider a situation where the equilibrium price in the absence of APs is just equal to the target, i.e., the market signal is equal to $\tilde{x}_n$ and we are at the lower bound of the non-

⁴⁶One could, of course, exogenously set a stricter upper bound; $\bar{b}_n = b_n(\theta_H, 1)$ entails the maximal upper bound possible.

intervention region. Consider now a marginal increase in gross supply, which makes the equilibrium price drop below target, triggering central bank intervention. Such intervention implies a discontinuous drop in the precision of investors' beliefs, since the equilibrium outcome (the target price and central bank purchases) becomes uninformative; in this case, x_n has only an allocational role as it still identifies the mass of investors buying. Such discontinuity in information manifests itself as a discontinuity in investors' demand. In particular, given that the wedge is positive whenever $Q_n \geq \mathbb{E}[\theta]$, investors' demand shrinks, which must be compensated by further purchases by the central bank for the target to be achieved.

Restricting our analysis to the cases where $Q_n \geq \mathbb{E}[\theta]$ ensures that the target can always be met without the need for central bank asset sales. In case $Q_n < \mathbb{E}[\theta]$, the value of the wedge at the non-intervention limit is negative, and the discontinuity logic flips sign: investors' demand jumps up, and the central bank must implement short sales to reach the price target—a form of quantitative tightening. The case $Q_n \leq \mathbb{E}[\theta]$ thus requires an explicit assumption about the possibility of short selling.

Fully revealing region. This region is characterized by a price signal $x_m < \tilde{x}_n$, an equilibrium price equal to the lower bound ($Q = \theta_L$), and no central bank intervention ($b_{cb} = 0$). It occurs with probability

$$P(Q = \theta_L) := (1 - \gamma) \left[\Phi \left(\frac{\theta_H - \tilde{x}_n}{\sigma_x} \right) - \Phi \left(\frac{\theta_L - \tilde{x}_n}{\sigma_x} \right) \right].$$

As discussed in the previous paragraph, the presence of bounds to b_{cb} generates cases where investors have common knowledge that $\theta = \theta_L$. This happens when gross supply takes values such that an intervention $b_{cb} \notin [\underline{b}_n, \bar{b}_n]$ would be needed to sustain the price target. We slightly modify rule (H.1) and assume that, in these instances, the central bank refrains from intervening, and the market signal fully reveals the state. This assumption is innocuous, since these interventions would happen at the perfect information price $Q = \theta_L$ and the central bank would make zero gains.

An illustration. Figure H.1 illustrates the equilibrium with a given price target level $Q_n > \mathbb{E}[\theta]$. The left panel plots the equilibrium price as a function of the price signal with black solid lines, the mass point where the price is equal to the target with a circle marker, and the equilibrium price and the marginal private valuation in the absence of APs with thick and thin gray lines, respectively. The center and right panels plot the price signal and central bank purchases respectively, as a function of gross bond supply (on the x-axis) and the value of the fundamental (solid and dashed lines represent θ_H and θ_L , respectively).

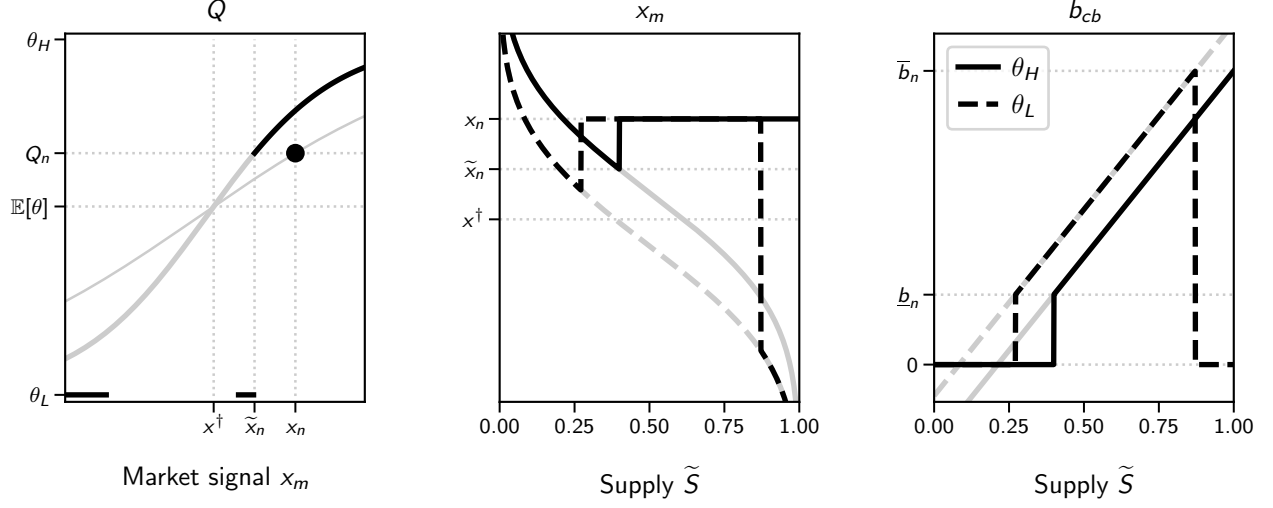


Figure H.1: Illustration of the workings of price-targeting APs. The price target for this example is set at $Q_n = 0.01$, which corresponds to a target repayment probability of 68%, 15 p.p. above its prior value. The remaining parameters are the same as Section 2.6.

The gray lines represent the price signal in the absence of APs in the center panel,⁴⁷ and $b_n(\theta, \tilde{S})$, which is the “unconstrained” version of APs as per (H.4), in the right panel. The no-intervention region is characterized by an equilibrium price above both the target and the marginal private valuation for all price signals above $\tilde{x}_n$, and APs equal to zero. When the equilibrium price equals the target, it lies on the marginal private valuation line because the public price signal is uninformative. In this region, the central bank expands APs in response to changes in gross supply to sustain the target price as a market outcome, b_{cb} is inside the $[\underline{b}_n, \bar{b}_n]$ interval, and the price signal is equal to x_n . The extent of the full revelation region is best understood by looking at the center and right panels. Full revelation never takes place when $\theta = \theta_H$, and occurs in two instances when $\theta = \theta_L$: first, when gross supply is large enough that the target price could only be sustained with APs larger than $\bar{b}_n$; second, for all states where the marginal agent’s signal without APs drops below $\tilde{x}_n$, and the target price could only be sustained with APs smaller than $\underline{b}_n$.

H.2 The Distribution of Gains

We now characterize gains and losses under this class of AP policies. In the case of the central bank, we only need to consider states where APs are positive, i.e., when $Q = Q_n$, in which case ex-post gains or losses are given by $\pi_{cb}(\theta) = \theta - Q_n$. Since the distribution of APs is uniformly distributed and uncorrelated with the fundamental θ (see Proposition H.1),

⁴⁷Note that these curves coincide with the functions depicted in Figure 1.

average central bank gains are

$$\mathbb{E}[\pi_{cb}] = \mathbb{E}[b_{cb}(\theta - Q_n)] = P(Q = Q_n) \frac{\bar{b}_n + \underline{b}_n}{2} (\mathbb{E}[\theta] - Q_n). \quad (\text{H.6})$$

This is the simple product of the unconditional probability of hitting the target, the average purchased size $\frac{\bar{b}_n + \underline{b}_n}{2}$, and the per-unit gain or loss $\mathbb{E}[\theta] - Q_n$. In fact, since in equilibrium APs occur independently from the realization of the fundamental, the value of the purchased asset evaluated at those states is its risk-neutral valuation $\mathbb{E}[\theta]$. Thus, when $Q_n = \mathbb{E}[\theta]$, average gains are exactly zero; when $Q_n > \mathbb{E}[\theta]$, the central bank makes losses; when $Q_n < \mathbb{E}[\theta]$, the sign is ambiguous because so is the average intervention size, since $\underline{b}_n$ is negative.

Investors' average gains are given by

$$\begin{aligned} \mathbb{E}[\pi_{inv}] = & \int_{\tilde{x}_n}^{\infty} \sum_{j \in \{H, L\}} \Phi\left(\frac{\theta_j - x}{\sigma_x}\right) (\theta_j - Q(p(x))) P(\theta_j | x \sim \mathcal{M}_0) f_{\mathcal{M}_0}(x) dx \\ & + P(Q = Q_n) \sum_{j \in \{H, L\}} \Phi\left(\frac{\theta_j - x_n}{\sigma_x}\right) (\theta_j - Q_n) \varpi_j, \end{aligned} \quad (\text{H.7})$$

where $\varpi_H = \gamma, \varpi_L = 1 - \gamma$, and the first and second lines account for states in the no intervention and price target regions, respectively. Two similarities with the case of quantity-targeting APs are worth noting. First, the probability that investors enter the market is positively correlated with bond payoffs, which implies investors' gains are always non-negative. Second, the first line of (H.7) coincides with average gains under quantity-targeting if $\bar{x}(b) = \tilde{x}_n$.

The government's gains are simply given by the difference between the average price and the prior mean of the fundamental, while noise traders' gains are determined residually from the zero-sum game of the financial market. The following proposition characterizes how price-targeting APs shape the unconditional price and the distribution of average gains in the market.

Proposition H.2 (Average properties of price-targeting APs). *For $Q_n \geq \mathbb{E}[\theta]$, price-targeting APs have the following properties:*

- (i) *The average size of APs is strictly increasing in Q_n ;*
- (ii) *Central bank average gains are weakly negative and decreasing in Q_n ;*
- (iii) *Investors' average gains are positive, decreasing in Q_n , and $\lim_{Q_n \rightarrow \theta_H} \mathbb{E}[\pi_{inv}] = 0$;*

(iv) Noise traders' average gains are strictly negative, and government gains are strictly positive;

(v) The average bond price as a function of the price target Q_n is given by

$$\mathcal{Q}_{\text{pt}}(Q_n) = \left(1 - P(Q = Q_n)\right) \mathcal{Q}_{\text{qt}}(\tilde{b}_n) + P(Q = Q_n) Q_n \quad (\text{H.8})$$

where $\tilde{b}_n = \Phi((\tilde{x}_n - \theta_H)/\sigma_x)$ is the level of quantity-targeting APs that delivers $\bar{x}(b) = \tilde{x}_n$. The average price $\mathcal{Q}_{\text{pt}}(Q_n)$ is such that $\mathcal{Q}_{\text{pt}}(\mathbb{E}[\theta]) > \mathbb{E}[\theta]$, $\mathcal{Q}_{\text{pt}}(\theta_H) = \theta_H$, and $\mathcal{Q}_{\text{pt}}(Q_n) \leq Q_n$ for Q_n large enough.

Proof. See Section H.3. ■

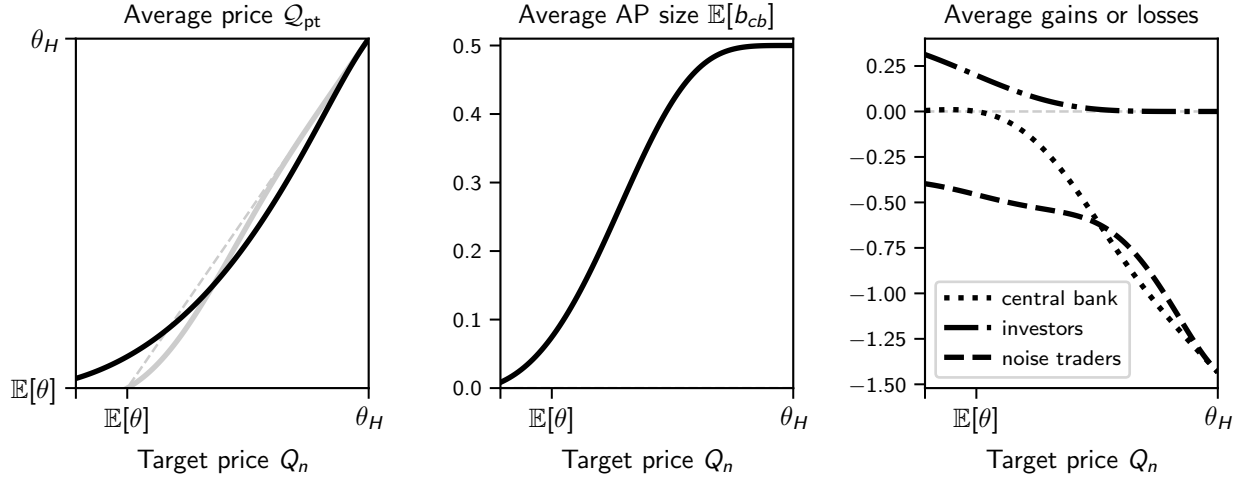


Figure H.2: Average bond price, asset purchases, and gains with price-targeting APs.

Statement (v) of the proposition characterizes the average price under price-targeting as a weighted average of the target, and the average price under a quantity-targeting program that delivers the same partitions of the state space. This is useful since we already derived the property of $\mathcal{Q}_{\text{qt}}$. Figure H.2 illustrates the proposition. The left, center and right panel plot the average equilibrium bond price, the average size of the price-targeting AP program, and average gains, respectively, all as a function of the central bank price target Q_n .⁴⁸ The shape of the average equilibrium price $\mathcal{Q}_{\text{pt}}$ as a function of Q_n is the result of two contrasting forces, which can be seen from equation (H.8). First, the average price in the absence of the learning-from-prices wedge (represented by a gray solid line in the left panel) is never larger

⁴⁸Government gains are not plotted in the right panel because they can be seen in the left panel, as the difference between the average price and the prior mean of θ .

than the target ($Q_{\text{pt}}(Q_n) \leq Q_n$), and is equal to it at $Q_n = \{\mathbb{E}[\theta], \theta_H\}$, since we know that $Q_{\text{qt}}(\tilde{b}_n) = \mathbb{E}[\theta]$.⁴⁹ Second, the learning-from-prices wedge is strictly positive in all states in which the central bank does not intervene, i.e. $W(\tilde{b}_n) > 0$, and its average value is decreasing in Q_n and converges to zero as $Q_n \rightarrow \theta_H$, because so does the measure $P(Q > Q_n)$ of the no-intervention region. The combined effects of these two forces implies the single-crossing result in the last statement of the proposition.

Statement (ii) of the proposition establishes that central bank gains are zero at a target of $\mathbb{E}[\theta]$, and negative above it. The former part of this statement implies that price-targeting APs can inflate average bond prices to some extent, at no balance sheet cost. The latter part states that the higher the price target, the larger the required central bank intervention, and the larger the associated losses. As equation (H.6) shows, losses are inevitable because the central bank is, by construction, buying bonds only when they are priced above the fundamental valuation.

Statement (iii) of the proposition states that investors' gains qualitatively behave as in the case of quantity-targeting APs, and converge to zero as $Q_n \rightarrow \theta_H$, because so does the extent of investors' participation in the market. Statement (iv) says that noise traders lose and the government gains in the absence of APs, and these gains and losses become bigger, the larger the AP price target. This implies that the government benefits from the intervention of the central bank, because it gets positive resource transfers from it. In summary, the key difference from the quantity-targeting case is that the central bank must always pay the marginal investor's valuation to intervene, leading to significant losses when attempting to purchase the entire market.

Finally, we extend figure H.2 to show the case where $Q_n < \mathbb{E}[\theta]$. The central panel shows that the average AP size converges to zero in these cases, since $\underline{b}_n$ slides towards negative territory. Although targeting a price below the prior mean delivers positive gains per unit purchased, maintaining the target forces the bank to implement sales because of the information discontinuity described earlier. Thus, overall gains are smaller in absolute value, and potentially ambiguous in sign.

⁴⁹Note that, when APs target the highest possible price ($Q_n = \theta_H$), the central bank must intervene and buy all net supply left after noise-trader demand at all times ($P(Q = Q_n) = 1$), which implies its average size equals average net supply ($\mathbb{E}[b_{\text{cb}}] = \mathbb{E}[\tilde{S}] = 1/2$).

H.3 Proofs

H.3.1 Proof of Proposition H.1 (Price-targeting APs)

For convenience and to lighten the notation, define

$$\tilde{B}_i := \Phi\left(\frac{\theta_i - \tilde{x}_n}{\sigma_x}\right) \quad \text{and} \quad B_i := \Phi\left(\frac{\theta_i - x_n}{\sigma_x}\right) \quad (\text{H.9})$$

as the mass of investors buying bonds when the fundamental is θ_i for $i \in \{H, L\}$ and the marginal agent's signal is $\tilde{x}_n$ as defined in the paragraph of equation (H.2). We consider states where the central bank intervenes, which requires

$$x_m \in (-\infty, \tilde{x}_n] \quad \Leftrightarrow \quad \tilde{S} \in [\tilde{B}_i, 1].$$

For now, suppose that the AP policy must deliver a marginal agent's signal equal to some value x_n to achieve the price target, which results in the central bank purchasing a quantity $b_n(\theta, \tilde{S})$ as defined in (H.4). It follows that $b_n | \theta$ is uniformly distributed over the interval $[b_n(\theta, \tilde{B}_i), b_n(\theta, 1)]$.

If investors were to observe an AP quantity that is only compatible with one value of the fundamental, then APs would be fully revealing. This implies that the interval of non-revealing purchases must be the intersection of the supports of $b_n | \theta_H$ and $b_n | \theta_L$. This coincides with the support of θ_H . The fact that $b_n(\theta_H, 1) < b_n(\theta_L, 1)$ is straightforward to show. To see why $b_n(\theta_H, \tilde{B}_H) > b_n(\theta_L, \tilde{B}_L)$, we plug in the definitions of $\tilde{x}_n$ and x_n , and after a bit of algebra we get to the expression $\int_{L_n/\sqrt{2}}^{L_n} \phi(A+y)dy < \int_{L_n/\sqrt{2}}^{L_n} \phi(A-y)dy$, where $L_n := \frac{\sigma_x}{\theta_H - \theta_L} \log\left(\frac{1-\gamma}{\gamma} \frac{Q_n - \theta_L}{\theta_H - Q_n}\right) > 0$ and $A := \frac{\theta_H - \theta_L}{2\sigma_x} > 0$, which is satisfied for all $Q_n > \mathbb{E}[\theta]$ because $\phi(A+y) < \phi(A-y)$ for any y , due to the symmetry of the standard normal density.

We have thus shown that the price-targeting, non-revealing AP policy $b_n | \theta \sim \text{Uniform}[\underline{b}_n, \bar{b}_n]$. Crucially, this distribution is independent of θ , so it is also the marginal distribution of b_n , whose c.d.f. and p.d.f. are given by

$$P(b_n < y | Q_n) = \frac{y - \underline{b}_n}{\bar{b}_n - \underline{b}_n} \quad \text{and} \quad f(b_n = y | Q_n) = \frac{1}{\bar{b}_n - \underline{b}_n} \quad \text{for all } y \in [\underline{b}_n, \bar{b}_n].$$

We now have all the elements to show that, when $b_{cb} \in [\underline{b}_n, \bar{b}_n]$, observing $(Q = Q_n, b_{cb})$ does not convey any new information. Using Bayes' law

$$P(\theta_H | Q_n, b_{cb}) = \frac{f(Q_n, b_{cb} | \theta_H) P(\theta_H)}{\sum_{j \in \{H, L\}} f(Q_n, b_{cb} | \theta_j) P(\theta_j)} = \frac{P(\theta_H)}{\sum_{j \in \{H, L\}} P(\theta_j)} = \gamma$$

where $P(\theta_H) = \gamma, P(\theta_L) = 1 - \gamma$, and the second equality uses the fact we derived above that the distribution of APs is independent of the fundamental.

As a result, the expected bond payoff for investor i conditional on observing $(Q = Q_n, b_{cb})$ is given by

$$\mathbb{E}[\theta | x_i \sim \mathcal{N}, Q_n, b_n > 0] = \mathbb{E}[\theta | x_i \sim \mathcal{N}],$$

so that $Q_n = \mathbb{E}[\theta | x_n \sim \mathcal{N}]$ pins down the price (or marginal agent's) signal $x_n = x(Q_n, \sigma_x)$ that needs to obtain for the price to equal the target when the central bank is intervening in the bond market.

H.3.2 Proof of Prop. H.2 (Average properties of price-targeting APs)

Proof of statement (i) We continue to use the notation introduced in (H.9) for brevity. The average size of APs is

$$\mathbb{E}[b_{cb}] = P(Q_n) \mathbb{E}[b_{cb} | Q_n] = (1 - \tilde{B}_H) \frac{1}{2} [1 - 2B_H + \tilde{B}_H]$$

where

$$\begin{aligned} \mathbb{E}[b_{cb} | Q_n] &= \frac{\bar{b}_n + b_n}{2} = \frac{1}{2} [1 - 2B_H + \tilde{B}_H] \\ P(Q_n) &= 1 - \tilde{B}_H. \end{aligned}$$

To show that the average size is increasing in Q_n , let $\tilde{B}'_H := \frac{\partial \tilde{B}_H}{\partial x_n} \frac{\partial x_n}{\partial Q_n}$ and $B'_H := \frac{\partial B_H}{\partial x_n} \frac{\partial x_n}{\partial Q_n}$ denote derivatives with respect to the price target, both of which can be shown to be negative. We have

$$\begin{aligned} \frac{\partial \mathbb{E}[b_{cb}]}{\partial Q_n} &= \frac{1}{2} [-\tilde{B}'_H(1 - 2B_H + \tilde{B}_H) + (1 - \tilde{B}_H)(-2B'_H + \tilde{B}'_H)] \\ &= -B'_H(1 - \tilde{B}_H) - \tilde{B}'_H(\tilde{B}_H - B_H). \end{aligned}$$

which is positive since $\tilde{B}_H > B_H$ when $Q_n > \mathbb{E}[\theta]$.

Proof of statement (ii) Since average APs are positive, the sign of central bank gains follows directly from the sign of $\mathbb{E}[\theta] - Q_n$, and is thus either zero or negative when $Q_n > \mathbb{E}[\theta]$. Deriving with respect to Q_n we get

$$\frac{\partial \mathbb{E}[\pi_{cb}]}{\partial Q_n} = \frac{\partial \mathbb{E}[b_{cb}]}{\partial Q_n} (\mathbb{E}[\theta] - Q_n) - \mathbb{E}[b_{cb}]$$

which is strictly negative, since average APs are increasing in Q_n as per statement (i).

Proof of statement (iii) This requires taking the derivative of equation (H.7) with respect to the price target Q_n . Before doing so, we rewrite (H.7) replacing $P(Q = Q_n)$ with $\Phi\left(\frac{\tilde{x}_n - \theta_H}{\sigma_x}\right)$ and $P(\theta_j | x \sim \mathcal{M}_0) f_{\mathcal{M}_0}(x)$ with $\frac{1}{\sigma_x} \phi\left(\frac{\theta_j - x}{\sigma_x}\right) \varpi_j$, where $j \in \{H, L\}$ and $\varpi_H = \gamma, \varpi_L = 1 - \gamma$. Finally, since x_n and Q_n are one-to-one and $\tilde{x}_n = (x_n + x^\dagger)/2$, we can take derivatives with respect to x_n rather than Q_n , since we are only concerned with the sign of derivatives. We compute

$$\frac{\partial}{\partial x_n} \left[\int_{\tilde{x}_n}^{\infty} \sum_j \Phi\left(\frac{\theta_j - x}{\sigma_x}\right) (\theta_j - Q(p(x))) \frac{1}{\sigma_x} \phi\left(\frac{\theta_j - x}{\sigma_x}\right) \varpi_j dx + \Phi\left(\frac{\tilde{x}_n - \theta_H}{\sigma_x}\right) \sum_j \Phi\left(\frac{\theta_j - x_n}{\sigma_x}\right) (\theta_j - Q_n) \varpi_j \right]$$

which yields the following derivative (that we explain below)

$$\begin{aligned} &= -\frac{1}{2} \sum_j \Phi\left(\frac{\theta_j - \tilde{x}_n}{\sigma_x}\right) (\theta_j - Q_n) \frac{1}{\sigma_x} \phi\left(\frac{\theta_j - \tilde{x}_n}{\sigma_x}\right) \varpi_j + \\ &+ \frac{1}{2} \frac{1}{\sigma_x} \phi\left(\frac{\theta_H - \tilde{x}_n}{\sigma_x}\right) \sum_j \Phi\left(\frac{\theta_j - x_n}{\sigma_x}\right) (\theta_j - Q_n) \varpi_j + \\ &- \Phi\left(\frac{\tilde{x}_n - \theta_H}{\sigma_x}\right) \sum_j \frac{1}{\sigma_x} \phi\left(\frac{\theta_j - x_n}{\sigma_x}\right) (\theta_j - Q_n) \varpi_j \\ &- \Phi\left(\frac{\tilde{x}_n - \theta_H}{\sigma_x}\right) \sum_j \Phi\left(\frac{\theta_j - x_n}{\sigma_x}\right) \frac{\partial Q_n}{\partial x_n} \varpi_j. \end{aligned} \tag{H.10}$$

In words, a higher price target has the following effects, each corresponding to one line of the derivative in (H.10): first, it shrinks the set of states in which there are no interventions (first line); second, it expands the set of states in which there are interventions and $Q = Q_n$ (second line); third, it lowers the amount of bonds absorbed by investors in such states (third line); fourth, it increases the price and shrinks the gains margin $\theta - Q$ in such states (fourth line).

To show that this derivative is negative for all $Q_n \geq \mathbb{E}[\theta]$, we proceed as follows. The fourth line is strictly negative since $\frac{\partial Q_n}{\partial x_n} > 0$. The third line is equal to zero: to see that, we first rearrange the statement as

$$\frac{\phi\left(\frac{\theta_L - x_n}{\sigma_x}\right)}{\phi\left(\frac{\theta_H - x_n}{\sigma_x}\right)} \frac{1 - \gamma}{\gamma} \frac{Q_n - \theta_L}{\theta_H - Q_n} = 1,$$

and then define $T := \frac{1-\gamma}{\gamma} \frac{Q_n - \theta_L}{\theta_H - Q_n}$ and note that

$$\frac{\phi\left(\frac{\theta_L - x_n}{\sigma_x}\right)}{\phi\left(\frac{\theta_H - x_n}{\sigma_x}\right)} = \exp\left\{\frac{(x^\dagger - x_n)(\theta_H - \theta_L)}{\sigma_x^2}\right\} = T^{-1}, \quad (\text{H.11})$$

according to (H.2), which verifies that the third line of (H.10) is indeed zero.

To complete the proof, it is thus sufficient to compare the first two lines of the derivative in (H.10). Collecting some terms, simplifying, and opening up the summation over θ we get

$$\begin{aligned} & \gamma(\theta_H - Q_n) \phi\left(\frac{\theta_H - \tilde{x}_n}{\sigma_x}\right) \left[\Phi\left(\frac{\theta_H - \tilde{x}_n}{\sigma_x}\right) - \Phi\left(\frac{\theta_H - x_n}{\sigma_x}\right) \right] \geq \\ & \geq (1 - \gamma)(Q_n - \theta_L) \left[\phi\left(\frac{\theta_L - \tilde{x}_n}{\sigma_x}\right) \Phi\left(\frac{\theta_L - \tilde{x}_n}{\sigma_x}\right) - \phi\left(\frac{\theta_H - \tilde{x}_n}{\sigma_x}\right) \Phi\left(\frac{\theta_L - x_n}{\sigma_x}\right) \right]. \end{aligned}$$

Dividing through by $\gamma(\theta_H - Q_n) \phi\left(\frac{\theta_H - \tilde{x}_n}{\sigma_x}\right)$ we get

$$\left[\Phi\left(\frac{\theta_H - \tilde{x}_n}{\sigma_x}\right) - \Phi\left(\frac{\theta_H - x_n}{\sigma_x}\right) \right] \geq T \left[\frac{\phi\left(\frac{\theta_L - \tilde{x}_n}{\sigma_x}\right)}{\phi\left(\frac{\theta_H - \tilde{x}_n}{\sigma_x}\right)} \Phi\left(\frac{\theta_L - \tilde{x}_n}{\sigma_x}\right) - \Phi\left(\frac{\theta_L - x_n}{\sigma_x}\right) \right]$$

where T is defined in the previous paragraph, and we know $T \geq 1$ when $Q_n \geq \mathbb{E}[\theta]$. Derivations similar to (H.11) yield

$$\frac{\phi\left(\frac{\theta_L - \tilde{x}_n}{\sigma_x}\right)}{\phi\left(\frac{\theta_H - \tilde{x}_n}{\sigma_x}\right)} = e^{\frac{(x^\dagger - \tilde{x}_n)(\theta_H - \theta_L)}{2\sigma_x^2}} = T^{-1/2}.$$

We thus have

$$\left[\Phi\left(\frac{\theta_H - \tilde{x}_n}{\sigma_x}\right) - \Phi\left(\frac{\theta_H - x_n}{\sigma_x}\right) \right] \geq \sqrt{T} \Phi\left(\frac{\theta_L - \tilde{x}_n}{\sigma_x}\right) - T \Phi\left(\frac{\theta_L - x_n}{\sigma_x}\right). \quad (\text{H.12})$$

Given that with $Q_n > \mathbb{E}[\theta]$ we have $T > 1$, a sufficient condition for (H.12) to hold is

$$\left[\Phi\left(\frac{\theta_H - \tilde{x}_n}{\sigma_x}\right) - \Phi\left(\frac{\theta_H - x_n}{\sigma_x}\right) \right] \geq \sqrt{T} \left(\Phi\left(\frac{\theta_L - \tilde{x}_n}{\sigma_x}\right) - \Phi\left(\frac{\theta_L - x_n}{\sigma_x}\right) \right). \quad (\text{H.13})$$

The proof of this is tedious and proceeds as follows. To lighten the notation, let $\chi := \frac{\theta_H - \theta_L}{2\sigma_x}$ and $t := \frac{1}{2} \frac{\sigma_x}{\theta_H - \theta_L} \log(T) = \frac{\tilde{x}_n - x^\dagger}{\sigma_x} = \frac{x_n - x^\dagger}{2\sigma_x}$, where we note $\chi > 0$ and $t > 0$, and recall

$x^\dagger = \frac{\theta_H + \theta_L}{2}$. The sufficient condition (H.13) can be rewritten as

$$\begin{aligned}
\Phi(\chi - t) - \Phi(\chi - 2t) &\geq e^{2\chi t} [\Phi(-\chi - t) - \Phi(-\chi - 2t)] \\
\int_{\chi-2t}^{\chi-t} \phi(x) dx &\geq e^{2\chi t} \int_{\chi+t}^{\chi+2t} \phi(x) dx \\
t \int_1^2 \phi(\chi - ty) dy &\geq e^{2\chi t} t \int_1^2 \phi(\chi + ty) dy \\
\int_1^2 e^{-\frac{1}{2}(\chi-ty)^2} dy &\geq \int_1^2 e^{-\frac{1}{2}(\chi+ty)^2} e^{2\chi t} dy \\
\int_1^2 e^{-\frac{1}{2}(\chi+ty)^2} e^{2\chi ty} dy &\geq \int_1^2 e^{-\frac{1}{2}(\chi+ty)^2} e^{2\chi t} dy \\
\int_1^2 e^{-\frac{1}{2}(\chi+ty)^2} (e^{2\chi ty} - e^{2\chi t}) dy &\geq 0
\end{aligned}$$

which concludes the proof since $2\chi ty > 2\chi t$ for $y \in [1, 2]$.

Proof of statement (iv) We first prove the result for noise traders. In the no-intervention region, noise-trader gains coincide with the baseline no-AP gains conditional on the same non-revealing price signal, and Proposition 3 gives that they are strictly negative for every such signal. In the fully revealing region, gains are zero for all players. It remains to consider the target region. Conditional on $Q = Q_n$ and a realized intervention $b_{cb} = b$, noise-trader gains are

$$\mathbb{E}[\pi_{nt} | Q = Q_n, b] = \sum_{j \in \{H, L\}} \varpi_j \left(1 - \Phi \left(\frac{\theta_j - x_n}{\sigma_x} \right) - b \right) (\theta_j - Q_n).$$

Rearranging,

$$\mathbb{E}[\pi_{nt} | Q = Q_n, b] = (1 - b) (\mathbb{E}[\theta] - Q_n) - \sum_{j \in \{H, L\}} \varpi_j \Phi \left(\frac{\theta_j - x_n}{\sigma_x} \right) (\theta_j - Q_n).$$

For $Q_n \geq \mathbb{E}[\theta]$ and $b \leq \bar{b}_n = 1 - \Phi((\theta_H - x_n)/\sigma_x) \leq 1$, the first term is weakly negative. For $Q_n < \theta_H$, the second term is strictly negative because

$$\sum_{j \in \{H, L\}} \varpi_j \Phi \left(\frac{\theta_j - x_n}{\sigma_x} \right) (\theta_j - Q_n) = \mathbb{E}[\mathbf{1}\{x_i \geq x_n\} (\theta - Q_n)] > 0,$$

where the inequality follows from the fact that the investor with $x_i = x_n$ is indifferent and investors with $x_i > x_n$ have strictly positive expected gains. Thus target-region noise-trader gains are weakly negative for each b . If $Q_n < \theta_H$, the second term makes them strictly negative for each b . If $Q_n = \theta_H$, then $P(Q = Q_n) = 1$, b is not identically one, and

$\mathbb{E}[\theta] - \theta_H < 0$, so target-region gains are strictly negative on average. Averaging across regions gives $\mathbb{E}[\pi_{\text{nt}}] < 0$.

We next prove the result for the government. Since the government issues one unit, its average gains are

$$\mathbb{E}[\pi_{\text{gov}}] = \mathbb{E}[Q - \theta] = \mathcal{Q}_{\text{pt}}(Q_n) - \mathbb{E}[\theta].$$

Using the average-price formula proved below,

$$\mathbb{E}[\pi_{\text{gov}}] = (1 - P(Q = Q_n)) W(\tilde{b}_n) + P(Q = Q_n) (Q_n - \mathbb{E}[\theta]).$$

For $Q_n \geq \mathbb{E}[\theta]$, we have $\tilde{b}_n \geq b(x^\dagger)$, so Appendix A.2.3 implies $W(\tilde{b}_n) > 0$ whenever $Q_n < \theta_H$. The second term is weakly positive. At $Q_n = \theta_H$, $P(Q = Q_n) = 1$ and $Q_n - \mathbb{E}[\theta] > 0$. Therefore $\mathbb{E}[\pi_{\text{gov}}] > 0$ for all $Q_n \geq \mathbb{E}[\theta]$.

Proof of statement (v) We compute the average price in three different regions of the state space.

First, with probability $P(Q > Q_n) = \gamma \Phi\left(\frac{\theta_H - \tilde{x}_n}{\sigma_x}\right) + (1 - \gamma) \Phi\left(\frac{\theta_L - \tilde{x}_n}{\sigma_x}\right)$, the price signal is above $\tilde{x}_n$, there are no APs and $Q > Q_n$, and the average price can be written as:

$$\frac{\int_{\tilde{x}_n}^{\infty} Q(p(x_m)) dF_{\mathcal{M}_b}(x_m)}{P(Q > Q_n)}.$$

With some algebra, the numerator of this expression can be rewritten as

$$\begin{aligned} & \int_{\tilde{x}_n}^{\infty} Q(p(x_m)) dF_{\mathcal{M}_b}(x_m) \\ &= \int_{\tilde{x}_n}^{\infty} Q(\hat{p}(x_m)) dF_{\mathcal{M}_b}(x_m) + \int_{\tilde{x}_n}^{\infty} (Q(p(x_m)) - Q(\hat{p}(x_m))) dF_{\mathcal{M}_b}(x_m) \\ &= P(Q > Q_n) \theta_L + (\theta_H - \theta_L) \left[\int_{\tilde{x}_n}^{\infty} \hat{p}(x_m) dF_{\mathcal{M}_b}(x_m) + \int_{\tilde{x}_n}^{\infty} (p(x_m) - \hat{p}(x_m)) dF_{\mathcal{M}_b}(x_m) \right] \\ &= P(Q > Q_n) \theta_L + \Phi\left(\frac{\theta_H - \tilde{x}_n}{\sigma_x}\right) \left[\gamma(\theta_H - \theta_L) + W(\tilde{b}_n) \right] \end{aligned}$$

where the last step uses (23) to rewrite the third term of the summation as $W(\tilde{b}_n)(1 - \tilde{b}_n)/(\theta_H - \theta_L)$ with $\tilde{b}_n = \Phi((\tilde{x}_n - \theta_H)/\sigma_x)$.

Second, with probability $P(Q = Q_n) = 1 - \Phi\left(\frac{\theta_H - \tilde{x}_n}{\sigma_x}\right) - \Phi\left(\frac{\theta_L - \tilde{x}_n}{\sigma_x}\right)$ there are APs and $Q = Q_n$; the average price in this case is Q_n .

Third, with the remaining probability $P(Q = \theta_L) = (1 - \gamma) \left[\Phi\left(\frac{\theta_H - \tilde{x}_n}{\sigma_x}\right) - \Phi\left(\frac{\theta_L - \tilde{x}_n}{\sigma_x}\right) \right]$ there are no APs and there is full revelation that $\theta = \theta_L$; the average price in this case is θ_L .

Finally, we write the average price as

$$\begin{aligned}
\mathcal{Q}_{\text{pt}} &= \int_{\tilde{x}_n}^{\infty} Q(p(x_m)) dF_{\mathcal{M}_b}(x_m) + P(Q = Q_n)Q_n + P(Q = \theta_L)\theta_L = \\
&= \Phi\left(\frac{\theta_H - \tilde{x}_n}{\sigma_x}\right) (\mathbb{E}[\theta] + W(\tilde{b}_n)) + P(Q = Q_n)Q_n \\
&= (1 - P(Q = Q_n)) (\mathbb{E}[\theta] + W(\tilde{b}_n)) + P(Q = Q_n)Q_n \\
&= (1 - P(Q = Q_n))\mathcal{Q}_{\text{qt}}(\tilde{b}_n) + P(Q = Q_n)Q_n
\end{aligned}$$

where we used $P(Q > Q_n) + P(Q = \theta_L) = \Phi\left(\frac{\theta_H - \tilde{x}_n}{\sigma_x}\right) = 1 - P(Q = Q_n)$.

We now move on to prove how the average price compares with the price target. First, consider $Q_n = \mathbb{E}[\theta]$. Note that $\mathcal{Q}_{\text{pt}}(\mathbb{E}[\theta]) > \mathbb{E}[\theta]$ if and only if $W(\tilde{b}_n) > 0$. In this case $\tilde{b}_n = \Phi\left(\frac{-(\theta_H - \theta_L)}{2\sigma_x}\right)$, and from Appendix A.2.3 we know that $\tilde{b}_n = b(x^\dagger)$ and $W(b(x^\dagger)) > 0$.

Second, when $Q_n = \theta_H$, we have that $P(Q = Q_n) = 1$ and thus $\mathcal{Q}_{\text{pt}} = \theta_H$. In this case, the central bank must crowd out all investors, thus paying the valuation of the most optimistic one.

Third, with some algebra we can show that $\mathcal{Q}_{\text{pt}}(Q_n) < Q_n$ if and only if $\tilde{W}(\tilde{b}_n) < \frac{p_n}{\gamma}$, where $p_n = (Q_n - \theta_L)/(\theta_H - \theta_L)$. We have shown in Appendix A.2.3 that $\tilde{W}'(b) < 0$ for all $b \geq b^*$, which implies $\tilde{W}(b) < w(\bar{x}(b))$. It follows that, for all price targets such that $\tilde{b}_n > b^*$,

$$\tilde{W}(\tilde{b}_n) < w(\bar{x}(\tilde{b}_n)) = w(\tilde{x}_n) = \frac{p_n}{\hat{p}(\tilde{x}_n)} < \frac{p_n}{\gamma}$$

which must hold since $\tilde{b}_n > b^*$ implies $\tilde{x}_n > x^\dagger$ and in turn $\hat{p}(\tilde{x}_n) > \hat{p}(x^\dagger) = \gamma$.

This concludes the proof of all statements of Proposition H.2.

I Asset Purchases at Market Value

We study a variant of the quantity-targeting rule in which the central bank fixes an expenditure—or market value—target $Q \times b_{\text{cb}} = M > 0$ rather than a face value target. The purpose of this online appendix is to isolate the one structural change that nominal targeting introduces—the equilibrium price enters the central bank's own purchase quantity—and to show how far the logic of Section 2 carries over. We work throughout under the assumptions that $\theta_L > 0$ and $M < \theta_H$, so that the purchase rule below is always well defined.

Definition I.1 (Expenditure-targeting APs). *Under expenditure-targeting APs the central*

bank spends up to M at the market-clearing price Q , purchasing

$$b_{\text{cb}} = \begin{cases} M/Q & \text{if } \tilde{S} \geq M/\theta_H, \\ \tilde{S} & \text{if } \tilde{S} < M/\theta_H, \end{cases} \quad (\text{I.1})$$

with the passive-market convention that the price equals the realized fundamental, $Q_{\text{pas}} = \theta$, when $\tilde{S} < M/\theta_H$.

The announced parameter M is observed by all investors, and the face value of bonds actually purchased when the market is active, $b_{\text{cb}} = M/Q$, is endogenous, decreasing in Q , and common knowledge.

I.1 Market Clearing and Price Signals

We refer throughout to $\tilde{S}$ —the government bond issuance net of noise traders’ demand—simply as supply. In an active market ($\tilde{S} \geq M/\theta_H$), the supply absorbed by private investors net of central bank APs is $S = \tilde{S} - M/Q$, and the market-clearing condition (9) of the baseline model becomes

$$\Phi\left(\frac{\theta - x_m}{\sigma_x}\right) = \tilde{S} - \frac{M}{Q(x_m)}. \quad (\text{I.2})$$

As in the baseline, the equilibrium price is given by the marginal investor’s expected payoff $Q(x_m) = p(x_m)\theta_H + (1 - p(x_m))\theta_L$, where $p(x_m)$ denotes her posterior probability of a high payoff, and depends on the realized fundamental and supply shocks only through the price signal x_m . Rearranging (I.2) gives

$$x_m = \theta + \sigma_x \Phi^{-1}\left(1 - \tilde{S} + \frac{M}{Q(x_m)}\right). \quad (\text{I.3})$$

The contrast with quantity targeting is immediate. Under a fixed quantity b , the analogue of (I.3) carries b in place of $M/Q(x_m)$; its right-hand side does not contain x_m , so the price signal is the closed-form object of equation (10). Under expenditure targeting, x_m appears on both sides through $Q(x_m)$, so the price signal solves a fixed-point equation.

From now on, we focus on equilibrium price functions that are non-decreasing in the price signal. The following result shows that, taking the equilibrium price function as given, this fixed point is unique whenever the market is active.

Proposition I.1 (Price-signal mapping). *Let $Q : \mathbb{R} \rightarrow [\theta_L, \theta_H]$ be continuous and non-decreasing, with $\lim_{x \rightarrow -\infty} Q(x) = \theta_L$ and $\lim_{x \rightarrow +\infty} Q(x) = \theta_H$. Fix $\theta \in \{\theta_L, \theta_H\}$ and a*

realization $\tilde{S} \in (0, 1]$.

(i) If $\tilde{S} > M/\theta_H$, equation (I.3) has a unique solution $x_m^*(\theta, \tilde{S})$.

(ii) If $\tilde{S} \leq M/\theta_H$, equation (I.3) has no solution: the central bank would absorb at least the entire supply at every price, so the market is passive.

Proof. Define $F(x) \equiv x - \theta - \sigma_x \Phi^{-1}(a(x))$ with $a(x) := 1 - \tilde{S} + M/Q(x)$, on the set where $a(x) \in (0, 1)$.

Monotonicity. As x increases, $Q(x)$ is non-decreasing, so $M/Q(x)$ and hence $a(x)$ are non-increasing; since Φ^{-1} is increasing, $\sigma_x \Phi^{-1}(a(x))$ is non-increasing. Therefore F , the sum of the strictly increasing term x and a non-decreasing term, is strictly increasing wherever it is defined.

Part (ii). If $\tilde{S} \leq M/\theta_H$ then, since $Q(x) \leq \theta_H$, we have $M/Q(x) \geq M/\theta_H \geq \tilde{S}$, so $a(x) \geq 1$ for every x and $\Phi^{-1}(a(x))$ is never defined and no solution exists.

Part (i). Let $\tilde{S} > M/\theta_H$. Since $Q(x) \in [\theta_L, \theta_H]$ and $\tilde{S} \leq 1$, the argument $a(x) = 1 - \tilde{S} + M/Q(x)$ is always strictly positive. It is strictly below 1 when $Q(x) > M/\tilde{S}$, which holds for x large enough, so the domain $\mathcal{D} = \{x : a(x) \in (0, 1)\}$ is a non-empty interval that is unbounded from above. On $\mathcal{D}$, F is continuous and strictly increasing. As $x \rightarrow +\infty$, $a(x) \rightarrow 1 - \tilde{S} + M/\theta_H \in (0, 1)$, so $\Phi^{-1}(a(x))$ stays bounded and $F(x) \rightarrow +\infty$. At the left edge of $\mathcal{D}$, either $a(x) \uparrow 1$ and $\Phi^{-1}(a(x)) \rightarrow +\infty$, or $x \rightarrow -\infty$; in either case $F(x) \rightarrow -\infty$. It follows that $F(x)$ is a continuous, strictly increasing function running from $-\infty$ to $+\infty$, and so it has a unique root $x_m^*(\theta, \tilde{S})$. ■

Proposition I.1 pins down the price signal as a function of the realized state *given* the price function $Q(\cdot)$. It does not by itself characterize the price function: in equilibrium Q must equal the marginal investor's posterior expectation, which is itself shaped by the distribution of x_m that (I.3) induces. We turn to this consistency requirement next, where the one genuine departure from the baseline quantity-targeting model appears.

I.2 From Price Signal to Investors' Beliefs

In the baseline model, the price signal likelihood that is behind the marginal density in (11) is exactly Gaussian: with uniform supply and constant APs, the change of variable from $\tilde{S}$ to x_m has a Jacobian term, which is why investors' posterior collapses to the closed form in (13). Under expenditure targeting, the purchase quantity $M/Q(x_m)$ varies with the signal, and the likelihood of the price signal acquires an extra term, derived in the following lemma.

Lemma I.1 (Conditional density of the price signal). *Let $Q(x_m)$ be the continuously differentiable, non-decreasing equilibrium price function. The density of the price signal conditional on θ is*

$$f(x_m | \theta) = \frac{1}{\sigma_x} \phi\left(\frac{\theta - x_m}{\sigma_x}\right) + \frac{M Q'(x_m)}{Q(x_m)^2} \quad (\text{I.4})$$

for all x_m in the support $\{x_m^*(\theta, \tilde{S}) \mid \forall \tilde{S} \in [M/\theta_H, 1]\}$.

Proof. Consider the market clearing condition (I.2) conditional on θ . Let $D(x_m) := \Phi\left(\frac{\theta - x_m}{\sigma_x}\right) + \frac{M}{Q(x_m)}$ denote aggregate bond demand. If $Q(x_m)$ is non-decreasing, then D is a decreasing function and $P(x_m < y | \theta) = P(\tilde{S} > D(y) | \theta) = 1 - D(y)$. Differentiating this conditional c.d.f. we readily obtain the likelihood function of x_m . ■

Marginalizing (I.4) over θ we get the density of the price signal, *unconditional* on active market trading. This is the counterpart of $f_{\mathcal{M}}(x_m)$ in (11), and indeed integrates to the probability that the market is active, $1 - M/\theta_H$. This constant cancels out in the posterior beliefs, to which we now turn.

Writing $g_\theta(x_m) := \frac{1}{\sigma_x} \phi\left(\frac{\theta - x_m}{\sigma_x}\right)$ for the private-signal likelihood evaluated at $x_i = x_m$, and using (I.4) for the price signal, Bayes' rule gives the following posterior probability that $\theta = \theta_H$ for the marginal agent

$$p(x_m) = \frac{\gamma g_{\theta_H} [g_{\theta_H} + J]}{\gamma g_{\theta_H} [g_{\theta_H} + J] + (1 - \gamma) g_{\theta_L} [g_{\theta_L} + J]}, \quad (\text{I.5})$$

where $J(x_m) := \frac{M Q'(x_m)}{Q(x_m)^2}$ and all functions are evaluated at x_m . Each fundamental state contributes the product of the private signal likelihood g_θ and the price signal likelihood $g_\theta + J$. The tilt J is where nominal targeting enters: it is common to both states (it depends on the price signal, not the payoff) but it does not factor out. Equivalently, the price-signal likelihood ratio is $(g_{\theta_H} + J)/(g_{\theta_L} + J)$ rather than $g_{\theta_H}/g_{\theta_L}$; adding a common positive J pulls it toward one, so the central bank's price-contingent policy makes the price a *locally* less informative signal of θ than under quantity targeting.

Setting $J \equiv 0$ —a constant AP quantity—collapses each bracket in (I.5) to g_θ and returns the symmetric Gaussian posterior of (13). For $M > 0$ instead the posterior has no closed form, and $Q(\cdot)$ solves a functional fixed point: a given price function Q pins down the price signal mapping $x_m^*(\theta, \tilde{S})$ and its likelihood function, these determine posterior beliefs (I.5), and the posterior yields the equilibrium price function Q . We do not solve this fixed point; existence of a continuous increasing price function is a standard but non-trivial question in noisy rational-expectations models, and a full characterization of the equilibrium price

distribution is beyond our scope.

I.3 Equilibrium Characterization

Some conclusions of the baseline model of Section 2 are robust to the complications we just highlighted, in the sense that they hold for any equilibrium price function in the class we consider.

First, since $b_{cb} = M/Q(x_m)$ and Q is non-decreasing in x_m , the central bank buys more bonds precisely when the price and the corresponding public signal are low. Since the price signal is increasing in the fundamental, APs are larger when the fundamental is low. Nominal targeting therefore tilts the intervention countercyclically: a fixed budget M translates into larger bond purchases in bad states. This is qualitatively similar to the crowding-out channel of Section 2, which operates asymmetrically across states, and only reveals the fundamental when its value is low.

Second, by the law of iterated expectations, the unconditional mean of the fundamental valuation $\widehat{Q}(x_m) = \mathbb{E}[\theta \mid x_m]$ equals the prior mean of the fundamental

$$\mathbb{E}[\widehat{Q}(x_m)] = \mathbb{E}[\theta] \quad \text{for all } M > 0,$$

regardless of the equilibrium distribution of x_m . Hence, as in the case of quantity-targeting (Proposition 2), expenditure-targeting leaves the average fundamental valuation unchanged; any effect on the average market price operates through the wedge between price and valuation.

Third, there may exist regions where the value of the fundamental is fully revealed. In the baseline model, the truncation of net supply from above creates an interval of low price signals that occur only in the low-payoff state and so reveal it. Under expenditure-targeting such a region can exist only if the central bank targets a level of expenditures that is small enough, that is, $M < \theta_L$.

In this case, the lowest signal observable when the low state is fully revealed is the solution of (I.3) evaluated at $\tilde{S} = 1$ and $Q = \theta_L$, which is

$$x^*(\theta_L, 1) = \theta_L + \sigma_x \Phi^{-1}\left(\frac{M}{\theta_L}\right).$$

This is strictly smaller than $x^*(\theta_H, 1)$ —the lowest signal observable under θ_H —so any signal between them fully reveals that the fundamental is low. When instead $M \geq \theta_L$, the central bank's AP rule would require purchasing the entire supply when $Q = \theta_L$, so the left bound

of the price signal support would be $-\infty$ regardless of θ , and no fully-revealing region would exist.

J Heterogeneous Preferences: a Comparison

This section compares our mechanism to more traditional models, where the elasticity of aggregate asset demand depends on the distribution of required rates of return across the investor population, and investor heterogeneity is structural (e.g. Vayanos and Vila (2021) among others). To explore this, we consider a modification of our model where investors have heterogeneous preferences. Our findings reveal a stark contrast in predictions: this different form of heterogeneity implies that asset prices increase monotonically with the size of APs, while the expected impact on the central bank's balance sheet is strictly positive.

Let us first show how our framework extends to include heterogeneity in required term premia. As before, bonds pay according to the lottery specified in (1). Each investor i is risk-neutral, can buy up to one unit of the asset, and has an expected payoff given by $\mathbb{E}[\theta | Q, x_i]$. We assume that investor i discounts the payoff at a discount rate given by an idiosyncratic required term premium λ_i , and thus buys bonds if $\frac{\mathbb{E}[\theta | Q, x_i]}{1 + \lambda_i} > Q$.⁵⁰ That is, investor i buys bonds if her subjective required premium is smaller than the equilibrium one. The case investigated in the main text obtains in the case of a homogeneous required term premium ($\lambda_i = \lambda$) through a simple rescaling of θ . Here we explore the case of homogeneous expectations ($\mathbb{E}[\theta | Q, x_i] = \mathbb{E}[\theta]$ for all i) and heterogeneous term premia.

The distribution of required premia across investors is assumed to be exogenous, with a cumulative distribution function $\tilde{F}$ on the support $[\underline{\lambda}, \bar{\lambda}]$, with $\underline{\lambda} > 0$. We denote by F the c.d.f. of reservation prices $\mathbb{E}[\theta]/(1 + \lambda)$ that is induced by $\tilde{F}$. The marginal agent m is the one whose reservation price is equal to the equilibrium one, so we write her subjective premium as a function of the equilibrium price: $\lambda_m(Q) = \mathbb{E}[\theta]/Q - 1$.

The bond market is assumed to work as in the previous sections: the supply of bonds is given by $\tilde{S}$ and distributed as in (2), and the central bank implements APs according to the quantity-targeting rule (5).⁵¹ Market clearing in the bond market requires

$$1 - F(Q) = \max\{\tilde{S} - b, 0\}$$

⁵⁰This can be easily extended to also include a common component r (e.g., the short-term risk-free rate). Heterogeneity in required term premia can be thought as differences in discounting or aversion to risk across investors, for example.

⁵¹We consider quantity-targeting APs for simplicity, but our results here would also go through with price-targeting policies.

where aggregate demand $1 - F(Q) = \tilde{F}(\lambda_m(Q))$ denotes the mass of investors with a reservation price larger than Q , or equivalently a subjective premium smaller than $\lambda_m(Q)$. The equilibrium price is given by

$$Q(\tilde{S}, b) = \begin{cases} F^{-1}(1 - \tilde{S} + b) & \text{if } \tilde{S} \geq b \\ Q_{\text{pas}} & \text{if } \tilde{S} < b \end{cases} \quad (\text{J.1})$$

where we set $Q_{\text{pas}} = \theta$ to rule out direct transfers between the central bank and government, as we do in our baseline model. Equation (J.1) shows that the equilibrium bond price is independent of the realization of θ . It follows that the equilibrium value of the marginal agent's subjective premium is given by $\lambda_m(Q) = \tilde{F}^{-1}(\tilde{S} - b)$.

Assuming that subjective premia are uniformly distributed, we obtain simple results stated in the following proposition, including an equilibrium price—or yield—expression conceptually close to that typically assumed in the finance literature.

Proposition J.1 (APs with structural heterogeneity). *Let $\tilde{F} = \text{Uniform}[\underline{\lambda}, \bar{\lambda}]$. Approximating to a first order, bond demand is given by*

$$\tilde{F}(\lambda_m(Q)) = \frac{\lambda_m(Q) - \underline{\lambda}}{\bar{\lambda} - \underline{\lambda}} \approx \beta - \alpha \log(Q) \quad (\text{J.2})$$

where $\alpha := 1/(\bar{\lambda} - \underline{\lambda})$ and $\beta := \alpha \left[\log(\mathbb{E}[\theta]) - \underline{\lambda} \right]$. The equilibrium yield (or term premium) is given by

$$\lambda_m(\tilde{S}, b) = \underline{\lambda} + (\tilde{S} - b)(\bar{\lambda} - \underline{\lambda}).$$

A quantity-targeting intervention of size b leads to a drop in the average equilibrium yield

$$\mathbb{E}_b[\lambda_m] - \mathbb{E}_0[\lambda_m] = -b \frac{\bar{\lambda} - \underline{\lambda}}{2},$$

and to central bank average gains

$$\mathbb{E}[\pi_{cb}] = \frac{b}{1 - b} \mathbb{E}[\theta] \int_b^1 \frac{\lambda_m(\tilde{S}, b)}{1 + \lambda_m(\tilde{S}, b)} d\tilde{S} > 0.$$

Proof. To derive (J.2), we take the equilibrium expression for the subjective premium of the marginal agent $1 + \lambda_m(Q) = \mathbb{E}[\theta]/Q$, take logs and use $\log(1 + x) \approx x$ to get

$$\lambda_m(Q) \approx \log(\mathbb{E}[\theta]) - \log(Q).$$

Plugging this into $\tilde{F}(\lambda_m(Q))$ yields expression (J.2).

The average effect of APs on yields conditional on $\tilde{S} \geq b$ is

$$\begin{aligned}
\mathbb{E}_b[\lambda_m] - \mathbb{E}_0[\lambda_m] &= \frac{1}{1-b} \int_b^1 \lambda_m(\tilde{S}, b) d\tilde{S} - \int_0^1 \lambda_m(\tilde{S}, 0) d\tilde{S} \\
&= \frac{1}{1-b} \int_0^{1-b} \lambda_m(S, 0) dS - \int_0^{1-b} \lambda_m(\tilde{S}, 0) d\tilde{S} - \int_{1-b}^1 \lambda_m(\tilde{S}, 0) d\tilde{S} \\
&= \frac{b}{1-b} \int_0^{1-b} \lambda_m(\tilde{S}, 0) d\tilde{S} - \int_{1-b}^1 \lambda_m(\tilde{S}, 0) d\tilde{S} \\
&= \frac{b}{1-b} \left((1-b)\underline{\lambda} + (1-b)^2(\bar{\lambda} - \underline{\lambda})/2 \right) - \left(b\bar{\lambda} + b(2-b)(\bar{\lambda} - \underline{\lambda})/2 \right) \\
&= -b(\bar{\lambda} - \underline{\lambda})/2.
\end{aligned}$$

The average gains of the central bank are

$$\begin{aligned}
\mathbb{E}[\pi_{cb}] &= \frac{b}{1-b} \int_b^1 \left(\mathbb{E}[\theta] - Q(\tilde{S}, b) \right) d\tilde{S} \\
&= \frac{b}{1-b} \int_b^1 \left(\mathbb{E}[\theta] - \mathbb{E}[\theta]/(1 + \lambda_m) \right) d\tilde{S} \\
&= \frac{b}{1-b} \mathbb{E}[\theta] \int_b^1 \frac{\lambda_m(\tilde{S}, b)}{1 + \lambda_m(\tilde{S}, b)} d\tilde{S} \\
&= \frac{b}{1-b} \mathbb{E}[\theta] \int_b^1 \frac{\underline{\lambda} + (\bar{\lambda} - \underline{\lambda})(\tilde{S} - b)}{1 + \underline{\lambda} + (\bar{\lambda} - \underline{\lambda})(\tilde{S} - b)} d\tilde{S} \\
&= b \mathbb{E}[\theta] \left[1 - \frac{1}{(1-b)(\bar{\lambda} - \underline{\lambda})} \log \left(1 + \frac{(\bar{\lambda} - \underline{\lambda})(1-b)}{1 + \underline{\lambda}} \right) \right],
\end{aligned}$$

which is strictly positive for any $b \in (0, 1)$. ■

The proposition characterizes a simple example where aggregate asset demand is downward-sloping and microfounded by heterogeneity in preferences. This is useful to highlight some key differences with our model of belief heterogeneity. First, the effect of APs on yields is strictly negative and is a monotonic function of the intervention size. Second, the effect on the central bank balance sheet is strictly positive. These conclusions go beyond the specific functional form explored in this example. APs have monotonic effects because they do not affect the fundamental characteristics of investors, so their only effect is to crowd out those less willing to enter the market. Since investors require positive term premia, the central bank always intervenes at a price that is below the fundamental value of the asset, generating positive average gains. These results imply that APs have no downsides, and any reason to

tame their size, if any, is external to their functioning. In our model, in contrast, investors respond to central bank intervention by adjusting their beliefs. This has two important implications: first, APs can backfire and depress asset prices in low payoff states; second, central banks may incur losses if they purchase assets when they are overpriced by the market.